



ABSTRACT ALGEBRA – (MMT522J1)

Written by:

Dr. Aijaz

ASSISTANT PROFESSOR OF MATHEMATICS

JK HIGHER EDUCATION DEPARTMENT

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Chapter 1

Groups and Subgroups

1.1 Binary Operations and Equivalence Relations

The study of algebraic structures begins with sets equipped with operations that combine their elements according to prescribed rules. Familiar examples include addition and multiplication of numbers, multiplication of matrices, and composition of functions. Although these operations arise in different settings, they often satisfy common algebraic properties. The systematic study of these properties leads to important algebraic structures such as groups, rings, and fields.

Before introducing groups, we examine two fundamental concepts: binary operations and equivalence relations. Binary operations provide the algebraic framework for combining elements, whereas equivalence relations allow us to classify elements into disjoint classes. Both concepts play a central role in the subsequent development of abstract algebra.

1.1.1 Binary Operations

Definition 1.1.1 (Binary Operation). Let S be a nonempty set. A *binary operation* on S is a function

$$* : S \times S \longrightarrow S.$$

For each ordered pair $(a, b) \in S \times S$, the image under $*$ is denoted by $a * b$.

Thus, a binary operation assigns to every ordered pair of elements of S a unique element of S . In particular, the result of combining any two elements must again belong to S . This requirement is called the *closure property*.

It is important to observe that an operation is defined relative to a particular set. A rule may define a binary operation on one set but fail to do so on another.

Example 1.1.2. The following illustrate binary operations and the importance of closure.

- (i) Addition, subtraction, and multiplication are binary operations on $\mathbb{Z}$, since the sum, difference, and product of any two integers are again integers.
- (ii) Subtraction is not a binary operation on $\mathbb{N} = \{1, 2, 3, \dots\}$. Indeed, $2, 5 \in \mathbb{N}$, but $2 - 5 = -3 \notin \mathbb{N}$.
- (iii) Multiplication is a binary operation on $M_2(\mathbb{R})$, the set of all 2×2 matrices with real entries, because the product of two such matrices is again a 2×2 real matrix.
- (iv) The rule $a * b = a + b + ab$ defines a binary operation on $\mathbb{Z}$. For example, $2 * 3 = 2 + 3 + 6 = 11$.
- (v) Division does not define a binary operation on $\mathbb{R}$, since division by zero is undefined. Moreover, division does not define a binary operation on $\mathbb{Z} \setminus \{0\}$, since the quotient of two nonzero integers need not be an integer.

Remark 1.1.3. Closure is determined by both the operation and the underlying set. For instance, subtraction is closed on $\mathbb{Z}$ but not on $\mathbb{N}$, while division is closed on $\mathbb{R} \setminus \{0\}$ but not on $\mathbb{Z} \setminus \{0\}$.

Properties of Binary Operations

Certain properties of binary operations occur frequently in algebra. The two most important are associativity and commutativity.

Definition 1.1.4. A binary operation $*$ on a set S is said to be

- (i) *associative* if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$;
- (ii) *commutative* if $a * b = b * a$ for all $a, b \in S$.

Associativity permits us to change the grouping of elements without changing their order. Thus, for an associative operation, the expression $a * b * c$ is unambiguous. Commutativity, on the other hand, permits us to interchange the order of the elements.

These properties are independent: an operation may satisfy either one, both, or neither.

Example 1.1.5. (i) Addition and multiplication on $\mathbb{Z}$ are both associative and commutative.

(ii) Subtraction on $\mathbb{Z}$ is neither associative nor commutative. For example,

$$(8 - 5) - 2 = 1, \quad 8 - (5 - 2) = 5,$$

while $8 - 5 = 3$ and $5 - 8 = -3$.

(iii) Matrix multiplication is associative but is not commutative in general. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Hence $AB \neq BA$.

(iv) On $\mathbb{Z}$, define $a * b = 2a + 2b$. This operation is commutative, but it is not associative. Indeed,

$$(1 * 2) * 3 = 18, \quad 1 * (2 * 3) = 22.$$

Identity and Inverse Elements

Besides associativity and commutativity, the existence of identity and inverse elements is fundamental in the study of groups.

Definition 1.1.6 (Identity Element). Let $*$ be a binary operation on a nonempty set S . An element $e \in S$ is called an *identity element* if

$$a * e = e * a = a$$

for every $a \in S$.

The identity element leaves every element unchanged when combined with it, regardless of the order in which the operation is performed.

For example, 0 is the identity for addition on $\mathbb{Z}$, while 1 is the identity for multiplication on $\mathbb{R}$.

Proposition 1.1.7. If a binary operation on a nonempty set has an identity element, then that identity element is unique.

Proof. Suppose e and f are identity elements of S . Since e is an identity, $e * f = f$. Similarly, since f is an identity, $e * f = e$. Therefore $e = f$, proving uniqueness. ■

Definition 1.1.8 (Inverse Element). Suppose a binary operation $*$ on S has an identity element e . An element $b \in S$ is called an *inverse* of $a \in S$ if

$$a * b = b * a = e.$$

The inverse of an element therefore reverses its effect under the given operation. For addition on $\mathbb{Z}$, the inverse of a is $-a$, whereas for multiplication on $\mathbb{R} \setminus \{0\}$, the inverse of a is $1/a$.

Remark 1.1.9. The existence of an identity does not guarantee that every element has an inverse. For example, multiplication on $\mathbb{Z}$ has identity 1, but only 1 and -1 possess multiplicative inverses in $\mathbb{Z}$.

Example 1.1.10 (An Operation Defined by a Formula). On $\mathbb{Z}$, define

$$a * b = a + b + 1.$$

We examine its algebraic properties.

First, $*$ is a binary operation, since $a + b + 1 \in \mathbb{Z}$ whenever $a, b \in \mathbb{Z}$.

To verify associativity, let $a, b, c \in \mathbb{Z}$. Then

$$(a * b) * c = (a + b + 1) + c + 1 = a + b + c + 2,$$

and

$$a * (b * c) = a + (b + c + 1) + 1 = a + b + c + 2.$$

Therefore $(a * b) * c = a * (b * c)$, so the operation is associative.

Also, $a * b = a + b + 1 = b + a + 1 = b * a$, showing that $*$ is commutative.

To determine the identity, suppose $e \in \mathbb{Z}$ satisfies $a * e = a$. Then $a + e + 1 = a$, which gives $e = -1$. Since $(-1) * a = a * (-1) = a$, the element -1 is the identity.

Finally, the inverse of a is an element b satisfying $a * b = -1$. Thus,

$$a + b + 1 = -1,$$

which gives $b = -a - 2$. Hence every element $a \in \mathbb{Z}$ has inverse $-a - 2$.

Consequently, $(\mathbb{Z}, *)$ satisfies the group axioms. In fact, it is an abelian group.

Remark 1.1.11. The preceding example illustrates an important principle of abstract algebra: the algebraic structure of a set depends on the operation under consideration, not merely on the nature of its elements.

1.1.2 Equivalence Relations

In mathematics, two objects may be regarded as equivalent even when they are not literally equal. For example, two integers may leave the same remainder upon division by a fixed positive integer. Equivalence relations provide a precise mathematical framework for expressing such situations.

Definition 1.1.12 (Equivalence Relation). Let S be a nonempty set. A relation $\sim$ on S is called an *equivalence relation* if the following conditions hold for all $a, b, c \in S$:

- (i) *Reflexivity*: $a \sim a$.
- (ii) *Symmetry*: If $a \sim b$, then $b \sim a$.
- (iii) *Transitivity*: If $a \sim b$ and $b \sim c$, then $a \sim c$.

Reflexivity means that every element is equivalent to itself. Symmetry means that equivalence works in both directions, while transitivity ensures that equivalence is preserved through intermediate elements.

Example 1.1.13 (Congruence Modulo n). Let n be a fixed positive integer. Define a relation on $\mathbb{Z}$ by

$$a \sim b \iff n \mid (a - b).$$

This relation is called *congruence modulo n* and is written $a \equiv b \pmod{n}$.

We verify that it is an equivalence relation.

Reflexivity. Since $a - a = 0$ and $n \mid 0$, we have $a \equiv a \pmod{n}$.

Symmetry. If $a \equiv b \pmod{n}$, then $n \mid (a - b)$. Since $b - a = -(a - b)$, it follows that $n \mid (b - a)$, so $b \equiv a \pmod{n}$.

Transitivity. Suppose $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. Then $n \mid (a - b)$ and $n \mid (b - c)$. Consequently,

$$n \mid ((a - b) + (b - c)) = a - c,$$

and hence $a \equiv c \pmod{n}$.

Therefore congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Example 1.1.14. (i) Equality on any set is an equivalence relation.

(ii) On $\mathbb{R}$, define $a \sim b$ if $|a| = |b|$. This is an equivalence relation. For instance, $3 \sim -3$.

(iii) On the set of all people, the relation “has the same month of birth as” is an equivalence relation.

(iv) The usual relation $\leq$ on $\mathbb{R}$ is not an equivalence relation. Although it is reflexive and transitive, it is not symmetric. For example, $2 \leq 5$, but $5 \not\leq 2$.

(v) On $\mathbb{Z}$, define $a \sim b$ if $a - b$ is odd. This relation is symmetric, but it is neither reflexive nor transitive. Indeed, $a - a = 0$ is not odd, and although $1 \sim 2$ and $2 \sim 3$, we have $1 \not\sim 3$.

1.1.3 Equivalence Classes

An equivalence relation groups together elements that are equivalent to one another. Each such collection is called an equivalence class.

Definition 1.1.15 (Equivalence Class). Let $\sim$ be an equivalence relation on a set S . For $a \in S$, the *equivalence class* of a is the subset

$$[a] = \{x \in S : x \sim a\}.$$

The element a is called a *representative* of the class $[a]$.

Since $a \sim a$, every equivalence class contains its representative and is therefore nonempty. Different representatives may determine the same equivalence class.

Example 1.1.16 (Residue Classes Modulo 3). Consider congruence modulo 3 on $\mathbb{Z}$. Its equivalence classes are

$$\begin{aligned}[0] &= \{3k : k \in \mathbb{Z}\}, \\ [1] &= \{3k + 1 : k \in \mathbb{Z}\}, \\ [2] &= \{3k + 2 : k \in \mathbb{Z}\}.\end{aligned}$$

By the division algorithm, every integer belongs to exactly one of these three classes.

For example, $8 \equiv 2 \pmod{3}$ and $-4 \equiv 2 \pmod{3}$. Hence $[8] = [-4] = [2]$.

Similarly, $[6] = [0]$ and $[10] = [1]$. Thus there are exactly three distinct equivalence classes, even though $\mathbb{Z}$ contains infinitely many elements.

Example 1.1.17. On $\mathbb{R}$, define $a \sim b$ if $|a| = |b|$. The equivalence class of a is

$$[a] = \{a, -a\}.$$

In particular, $[0] = \{0\}$, while $[5] = \{5, -5\}$. This shows that equivalence classes need not contain the same number of elements.

The following result provides a useful criterion for determining when two elements represent the same equivalence class.

Proposition 1.1.18. Let $\sim$ be an equivalence relation on S . For any $a, b \in S$,

$$[a] = [b] \iff a \sim b.$$

Proof. Suppose $[a] = [b]$. Since $a \in [a]$, we also have $a \in [b]$. By the definition of $[b]$, this gives $a \sim b$.

Conversely, suppose $a \sim b$. Let $x \in [a]$. Then $x \sim a$, and transitivity gives $x \sim b$. Thus $x \in [b]$, proving $[a] \subseteq [b]$.

By symmetry, $b \sim a$, and the same argument gives $[b] \subseteq [a]$. Hence $[a] = [b]$. ■

1.1.4 Partitions and Equivalence Relations

Definition 1.1.19 (Partition). A *partition* of a nonempty set S is a collection $\{S_i : i \in I\}$ of nonempty subsets of S such that

- (i) $S = \bigcup_{i \in I} S_i$;
- (ii) $S_i \cap S_j = \emptyset$ whenever $i \neq j$.

A partition divides a set into nonempty, pairwise disjoint subsets whose union is the entire set. Every element therefore belongs to exactly one part of the partition.

Example 1.1.20. Let $S = \{1, 2, 3, 4, 5, 6\}$. The collection

$$\{\{1, 3, 5\}, \{2, 4, 6\}\}$$

is a partition of S into odd and even elements.

However, $\{\{1, 2, 3\}, \{3, 4, 5, 6\}\}$ is not a partition because the two subsets overlap.

Theorem 1.1.21 (Equivalence Classes Form a Partition). Let $\sim$ be an equivalence relation on a nonempty set S . Then the collection of all distinct equivalence classes of S forms a partition of S .

Proof. We verify the defining properties of a partition.

First, every equivalence class is nonempty. Indeed, for each $a \in S$, reflexivity gives $a \sim a$, so $a \in [a]$.

Next, every element of S belongs to an equivalence class, namely its own. Consequently, the union of all equivalence classes is S .

It remains to show that two distinct equivalence classes are disjoint. Suppose $[a] \cap [b] \neq \emptyset$. Choose $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$, and hence transitivity gives $a \sim b$.

By the preceding proposition, $[a] = [b]$. Therefore any two equivalence classes are either equal or disjoint.

Thus the collection of all distinct equivalence classes forms a partition of S . ■

The converse is equally important: every partition determines an equivalence relation.

Theorem 1.1.22 (A Partition Determines an Equivalence Relation). *Let $\{S_i : i \in I\}$ be a partition of a nonempty set S . Define a relation $\sim$ on S by declaring $a \sim b$ if a and b belong to the same member of the partition. Then $\sim$ is an equivalence relation.*

Proof. Since the subsets S_i form a partition, every element $a \in S$ belongs to some S_i . Thus a and itself belong to the same subset, giving $a \sim a$. Hence the relation is reflexive.

If $a \sim b$, then a and b belong to the same subset S_i . Consequently, b and a belong to that same subset, so $b \sim a$. Thus the relation is symmetric.

Finally, suppose $a \sim b$ and $b \sim c$. Then $a, b \in S_i$ and $b, c \in S_j$ for some $i, j \in I$. Since $b \in S_i \cap S_j$ and the members of a partition are pairwise disjoint, we must have $S_i = S_j$. Therefore a and c belong to the same subset, so $a \sim c$.

Hence $\sim$ is an equivalence relation. ■

Remark 1.1.23. Equivalence relations and partitions are two descriptions of the same mathematical phenomenon. An equivalence relation specifies which elements are regarded as equivalent, while the corresponding partition collects those elements into disjoint classes.

This connection is fundamental in abstract algebra. For example, congruence modulo n partitions $\mathbb{Z}$ into n residue classes, which form the underlying set of the ring $\mathbb{Z}_n$. Similarly, the cosets of a subgroup partition a group and provide the basis for the construction of quotient groups.

1.2 Groups and Their Basic Properties

A group is one of the fundamental structures of abstract algebra. It consists of a set equipped with a binary operation satisfying certain conditions that make it possible to combine elements, identify an element that leaves others unchanged, and reverse the effect of an operation.

Many familiar mathematical systems possess a group structure. The integers under addition, the nonzero real numbers under multiplication, invertible matrices under matrix multiplication, and permutations under composition are important examples. Although these systems differ considerably in their nature, they satisfy the same fundamental algebraic laws.

The concept of a group provides a common framework for studying these systems and the properties they share.

Definition 1.2.1 (Group). *A group is a nonempty set G equipped with a binary operation $*$ satisfying the following conditions:*

(G1) *Associativity:* $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.

(G2) *Identity element:* There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.

(G3) *Inverse element*: For every $a \in G$, there exists $b \in G$ such that $a * b = b * a = e$.

The element e is called an *identity element*, and b is called an *inverse* of a .

Since $*$ is a binary operation on G , the closure condition $a * b \in G$ is already included in the definition. Nevertheless, closure is often stated explicitly when verifying whether a particular set and operation form a group.

A group is generally denoted by $(G, *)$. When the operation is understood, we simply write G . Two common notational conventions are used:

- (i) In *additive notation*, the operation is written $+$, the identity is denoted by 0 , and the inverse of a is written $-a$.
- (ii) In *multiplicative notation*, the operation is written by juxtaposition, the identity is denoted by e or 1 , and the inverse of a is written a^{-1} .

These are not different definitions of a group; they are simply different ways of writing the same algebraic structure.

Example 1.2.2 (The Integers under Addition). The set $\mathbb{Z}$ forms a group under ordinary addition. Indeed, the sum of two integers is an integer, addition is associative, 0 is the identity element, and every integer a has additive inverse $-a$. Thus $(\mathbb{Z}, +)$ is a group.

The same argument shows that $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are groups.

Example 1.2.3 (Nonzero Rational Numbers under Multiplication). Let $G = \mathbb{Q} \setminus \{0\}$. Then $(G, \cdot)$ is a group under ordinary multiplication.

The product of two nonzero rational numbers is again a nonzero rational number, and multiplication is associative. The identity element is 1 , while the inverse of $a \in G$ is $1/a$, which also belongs to G .

Similarly, $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$ are groups under multiplication.

Example 1.2.4 (Integers Modulo n). Let $n \geq 2$ and consider the set of residue classes

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}.$$

Define addition by $[a] + [b] = [a + b]$, where the class on the right is taken modulo n .

This operation is well-defined. Indeed, if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then n divides both $a - a'$ and $b - b'$. Consequently,

$$n \mid ((a + b) - (a' + b')),$$

so $[a + b] = [a' + b']$.

Associativity follows from integer addition. The identity element is $[0]$, and the additive inverse of $[a]$ is $[-a]$. Thus $(\mathbb{Z}_n, +)$ is a group.

For example, in $\mathbb{Z}_5$, we have $[3] + [4] = [2]$ and $[3] + [2] = [0]$. Hence $[2]$ is the additive inverse of $[3]$.

Example 1.2.5 (The General Linear Group). Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries, and define

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}.$$

Then $GL_n(\mathbb{R})$ is a group under matrix multiplication.

If $A, B \in GL_n(\mathbb{R})$, then $\det(AB) = \det(A)\det(B) \neq 0$, so $AB \in GL_n(\mathbb{R})$. Matrix multiplication is associative, and the identity matrix I_n is the identity element.

Furthermore, every matrix $A \in GL_n(\mathbb{R})$ is invertible, and its inverse A^{-1} also has nonzero determinant. Therefore $GL_n(\mathbb{R})$ satisfies all the group axioms.

Not every set with a binary operation forms a group. The failure of even one group axiom is sufficient to show that a given structure is not a group.

- Example 1.2.6** (Nonexamples of Groups). (i) $(\mathbb{N}, +)$, where $\mathbb{N} = \{1, 2, 3, \dots\}$, is not a group because it does not contain the additive identity 0.
- (ii) $(\mathbb{Z}, \cdot)$ is not a group because most integers do not have multiplicative inverses in $\mathbb{Z}$. In particular, there is no integer b satisfying $2b = 1$.
- (iii) $(\mathbb{Z}, -)$ is not a group because subtraction is not associative.
- (iv) $M_n(\mathbb{R})$ under matrix multiplication is not a group, since singular matrices do not possess multiplicative inverses.
- (v) $(\mathbb{N}_0, +)$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, has an identity but is not a group, since positive integers have no additive inverses in $\mathbb{N}_0$.

1.2.1 Finite, Infinite, Abelian and Nonabelian Groups

Groups may be classified according to the number of elements they contain and whether their operation is commutative.

Definition 1.2.7 (Order of a Group). The *order* of a group G , denoted by $|G|$, is the number of elements in G . If G has finitely many elements, it is called a *finite group*; otherwise, it is called an *infinite group*.

For example, the additive group $\mathbb{Z}_n$ has order n , whereas the additive groups $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are infinite.

Definition 1.2.8 (Abelian Group). A group G is called *abelian* or *commutative* if

$$ab = ba$$

for every $a, b \in G$. A group that is not abelian is called *nonabelian*.

To establish that a group is abelian, one must show that every pair of its elements commutes. To establish that it is nonabelian, it is sufficient to find a single pair of elements that does not commute.

Example 1.2.9. The groups $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{Z}_n, +)$ are abelian because addition is commutative.

The multiplicative groups $\mathbb{Q} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$ are also abelian.

Example 1.2.10 (A Nonabelian Matrix Group). Consider the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both belong to $GL_2(\mathbb{R})$, since their determinants are nonzero. However,

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus $AB \neq BA$, proving that $GL_2(\mathbb{R})$ is nonabelian.

1.2.2 Permutation Groups and Symmetries

An important source of groups arises from permutations and geometric symmetries.

Definition 1.2.11 (Permutation). A *permutation* of a nonempty set S is a bijective mapping from S onto itself.

The composition of two permutations is again a permutation. Since function composition is associative, the identity mapping is an identity element, and every permutation has an inverse, the set of all permutations of S forms a group under composition.

When $S = \{1, 2, \dots, n\}$, this group is called the *symmetric group of degree n* , denoted by S_n .

Proposition 1.2.12. The symmetric group S_n has order $n!$.

Proof. A permutation is completely determined by the images of $1, 2, \dots, n$. There are n possible images for 1, followed by $n - 1$ choices for the image of 2, and so on. Since distinct elements must have distinct images, the number of permutations is

$$n(n - 1) \cdots 2 \cdot 1 = n!.$$

Therefore $|S_n| = n!$. ■

Example 1.2.13 (The Symmetric Group S_3). The symmetric group on $\{1, 2, 3\}$ consists of six permutations:

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

Here (12) denotes the permutation interchanging 1 and 2 while fixing 3, and (123) sends 1 to 2, 2 to 3, and 3 to 1.

Let $\sigma = (12)$ and $\tau = (23)$. Using the convention that the rightmost permutation is applied first, we obtain

$$\sigma\tau = (123), \quad \tau\sigma = (132).$$

Since $\sigma\tau \neq \tau\sigma$, the group S_3 is nonabelian.

Example 1.2.14 (Symmetries of an Equilateral Triangle). The symmetries of an equilateral triangle consist of three rotations, including the identity, and three reflections.

Composition of symmetries is associative, the identity symmetry leaves the triangle unchanged, and every symmetry has an inverse. Consequently, these six symmetries form a group.

If the three vertices are labelled 1, 2, 3, every symmetry determines a permutation of the vertices. In fact, every permutation of the three vertices is induced by exactly one symmetry. Thus the symmetry group of an equilateral triangle is isomorphic to S_3 .

1.2.3 Elementary Properties of Groups

The group axioms have several important consequences. We now establish the elementary results that will be used repeatedly in the study of subgroups, cosets, and homomorphisms.

Theorem 1.2.15 (Uniqueness of the Identity). *Every group has a unique identity element.*

Proof. Suppose e and f are both identity elements of a group G . Since e is an identity, $ef = f$. Since f is also an identity, $ef = e$. Hence $e = f$, proving that the identity element is unique. ■

Theorem 1.2.16 (Uniqueness of Inverses). *Every element of a group has a unique inverse.*

Proof. Let $a \in G$, and suppose b and c are both inverses of a . Then $ab = ba = e$ and $ac = ca = e$.

By associativity,

$$b = be = b(ac) = (ba)c = ec = c.$$

Therefore $b = c$, proving uniqueness. ■

Since the inverse of each element is unique, it may be denoted unambiguously by a^{-1} .

Theorem 1.2.17 (Cancellation Laws). *Let G be a group and let $a, b, c \in G$. Then:*

- (i) $ab = ac$ implies $b = c$;
- (ii) $ba = ca$ implies $b = c$.

Proof. Suppose $ab = ac$. Multiplying on the left by a^{-1} and using associativity, we obtain

$$a^{-1}(ab) = a^{-1}(ac),$$

so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$. Hence $b = c$.

Similarly, if $ba = ca$, multiplication on the right by a^{-1} yields

$$(ba)a^{-1} = (ca)a^{-1},$$

which simplifies to $be = ce$. Thus $b = c$. ■

Remark 1.2.18. In a nonabelian group, cancellation is valid only when the common factor occurs on the same side. In particular, the equation $ab = ca$ does not generally imply $b = c$.

Theorem 1.2.19 (Inverse Laws). *For every $a, b \in G$,*

- (i) $(a^{-1})^{-1} = a$;
- (ii) $(ab)^{-1} = b^{-1}a^{-1}$;
- (iii) $e^{-1} = e$.

Proof. Since $aa^{-1} = a^{-1}a = e$, the element a is an inverse of a^{-1} . Uniqueness of inverses gives $(a^{-1})^{-1} = a$.

Next, consider $b^{-1}a^{-1}$. By associativity,

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aa^{-1} = e.$$

Similarly,

$$(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = b^{-1}b = e.$$

Thus $b^{-1}a^{-1}$ is the inverse of ab , proving the second assertion.

Finally, since $ee = e$, the identity element is its own inverse. ■

Corollary 1.2.20. *For any $a_1, a_2, \dots, a_n \in G$,*

$$(a_1a_2 \cdots a_n)^{-1} = a_n^{-1} \cdots a_2^{-1}a_1^{-1}.$$

Proof. The result follows by repeated application of the identity $(ab)^{-1} = b^{-1}a^{-1}$. ■

Proposition 1.2.21 (Solutions of Group Equations). Let $a, b \in G$. Then:

- (i) the equation $ax = b$ has the unique solution $x = a^{-1}b$;
- (ii) the equation $ya = b$ has the unique solution $y = ba^{-1}$.

Proof. If $ax = b$, multiplication on the left by a^{-1} gives

$$x = (a^{-1}a)x = a^{-1}b.$$

Conversely, $a(a^{-1}b) = (aa^{-1})b = b$, so this element is indeed a solution. Uniqueness follows from left cancellation.

Similarly, from $ya = b$, multiplication on the right by a^{-1} gives $y = ba^{-1}$. Substitution verifies the solution, and right cancellation establishes uniqueness. ■

Example 1.2.22 (Solving an Equation in a Nonabelian Group). Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

We solve $AX = B$ in $GL_2(\mathbb{R})$.

Since

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

the unique solution is

$$X = A^{-1}B = \begin{pmatrix} 2 & -3 \\ 0 & 3 \end{pmatrix}.$$

Indeed, direct multiplication gives $AX = B$.

Notice that the order of multiplication is essential: the solution is $A^{-1}B$, not BA^{-1} .

Proposition 1.2.23. A group G is abelian if and only if

$$(ab)^{-1} = a^{-1}b^{-1}$$

for all $a, b \in G$.

Proof. If G is abelian, then $b^{-1}a^{-1} = a^{-1}b^{-1}$. Hence the usual inverse law gives $(ab)^{-1} = a^{-1}b^{-1}$.

Conversely, suppose $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$. Since $(ab)^{-1} = b^{-1}a^{-1}$ in every group, we obtain

$$a^{-1}b^{-1} = b^{-1}a^{-1}.$$

Taking inverses of both sides gives $ba = ab$. Since a, b were arbitrary, G is abelian. ■

1.2.4 Powers of Group Elements

Let G be a group, and let $a \in G$. For a positive integer n , the power a^n denotes the product of n copies of a . We also define $a^0 = e$ and $a^{-n} = (a^{-1})^n$.

The familiar laws of exponents remain valid for powers of a fixed group element.

Proposition 1.2.24 (Laws of Exponents). For every $a \in G$ and $r, s \in \mathbb{Z}$,

- (i) $a^r a^s = a^{r+s}$;
- (ii) $(a^r)^s = a^{rs}$;

$$(iii) \quad (a^r)^{-1} = a^{-r}.$$

Proof. For positive integers r, s , the first identity follows from associativity by combining r copies of a with s additional copies. The identity remains valid when either exponent is zero.

For negative exponents, the relation $a^{-m} = (a^{-1})^m$, together with $aa^{-1} = a^{-1}a = e$, allows opposite powers to cancel. This establishes $a^r a^s = a^{r+s}$ for arbitrary integers r, s .

The second identity follows by repeated application of the first when $s > 0$, is immediate when $s = 0$, and follows from inverses when $s < 0$.

Finally, since $a^r a^{-r} = a^0 = e = a^{-r} a^r$, the uniqueness of inverses gives $(a^r)^{-1} = a^{-r}$. ■

Remark 1.2.25. The exponent laws above concern powers of the same element. For arbitrary $a, b \in G$, the identity $(ab)^n = a^n b^n$ need not hold.

However, if $ab = ba$, then $(ab)^n = a^n b^n$ for every integer n .

1.2.5 Order of an Element

The order of a group is the number of elements it contains. The order of an individual element is a different, though closely related, concept.

Definition 1.2.26 (Order of an Element). Let a be an element of a group G . If there exists a positive integer n such that $a^n = e$, the least such integer is called the *order of a* , denoted by $\text{ord}(a)$.

If no positive integer n satisfies $a^n = e$, the element a is said to have *infinite order*.

For additive groups, the condition $a^n = e$ is replaced by $na = 0$, where na denotes the sum of n copies of a .

Example 1.2.27. (i) In $(\mathbb{Z}, +)$, the identity 0 has order 1, while every nonzero integer has infinite order.

(ii) In $(\mathbb{Z}_8, +)$, the element $[2]$ has order 4, since

$$[2], \quad [4], \quad [6], \quad [0]$$

are its successive positive multiples.

(iii) In $(\mathbb{Z}_6, +)$, the element $[2]$ has order 3, while $[3]$ has order 2.

(iv) In $GL_2(\mathbb{R})$, the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has order 4. Indeed, $A^2 = -I_2$, $A^3 = -A$, and $A^4 = I_2$, with no smaller positive power equal to I_2 .

Theorem 1.2.28 (Order and Powers of an Element). Let $a \in G$ have finite order n . Then, for every integer k ,

$$a^k = e \iff n \mid k.$$

Proof. Suppose $a^k = e$. By the division algorithm, there exist integers q, r such that $k = qn + r$, where $0 \leq r < n$. Therefore,

$$e = a^k = a^{qn+r} = (a^n)^q a^r = a^r.$$

Since n is the least positive integer for which $a^n = e$, the only possibility is $r = 0$. Hence $k = qn$, showing that $n \mid k$.

Conversely, if $n \mid k$, then $k = qn$ for some integer q . Consequently, $a^k = (a^n)^q = e$, as required. ■

Corollary 1.2.29. *If $\text{ord}(a) = n$, then*

$$a^r = a^s \iff r \equiv s \pmod{n}$$

for all integers r, s .

Proof. We have $a^r = a^s$ if and only if $a^{r-s} = e$. By the preceding theorem, this holds precisely when $n \mid (r - s)$, which is equivalent to $r \equiv s \pmod{n}$. ■

Proposition 1.2.30. An element a has infinite order if and only if all the powers a^k , $k \in \mathbb{Z}$, are distinct.

Proof. Suppose a has infinite order. If $a^r = a^s$ for integers $r > s$, then $a^{r-s} = e$, contradicting the infinite order of a . Hence distinct integers give distinct powers.

Conversely, if a has finite order n , then $a^n = e = a^0$, so its powers are not all distinct. ■

Theorem 1.2.31 (Order of a Power). *Suppose $a \in G$ has finite order n . Then, for every integer k ,*

$$\text{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

Here $\gcd(n, k)$ denotes the positive greatest common divisor.

Proof. Put $d = \gcd(n, k)$, and write $n = dn_1$ and $k = dk_1$, where $\gcd(n_1, k_1) = 1$.

Since

$$(a^k)^{n_1} = a^{kn_1} = a^{k_1n} = e,$$

the element a^k has finite order at most n_1 .

Now suppose $(a^k)^m = e$ for some positive integer m . Then $a^{km} = e$, so $n \mid km$. Substituting $n = dn_1$ and $k = dk_1$, we obtain $n_1 \mid k_1m$.

Since $\gcd(n_1, k_1) = 1$, Euclid's lemma implies $n_1 \mid m$. Thus every positive exponent giving $(a^k)^m = e$ is a multiple of n_1 .

Therefore the least such exponent is $n_1 = n/d$, proving the result. ■

Example 1.2.32. In $(\mathbb{Z}_{12}, +)$, the element $[1]$ has order 12. Consequently, the order of $[8] = 8[1]$ is

$$\text{ord}([8]) = \frac{12}{\gcd(12, 8)} = \frac{12}{4} = 3.$$

Indeed, $[8] + [8] = [4]$ and $[8] + [8] + [8] = [0]$.

Similarly, $\text{ord}([9]) = 12/\gcd(12, 9) = 4$.

1.3 Subgroups

A group may contain subsets that themselves form groups under the same operation. Such subsets are called *subgroups*. The study of subgroups provides a means of understanding the structure of a group through its smaller algebraic systems.

For example, the even integers form a group under addition and are contained in the additive group of all integers. Similarly, certain collections of invertible matrices form groups under matrix multiplication. These are instances of subgroups.

Definition 1.3.1 (Subgroup). Let $(G, *)$ be a group. A nonempty subset H of G is called a *subgroup* of G if $(H, *)$ is itself a group under the operation inherited from G . We write $H \leq G$.

Every group G contains at least two subgroups, namely $\{e\}$ and G itself, where e is the identity element of G . These are called the *trivial subgroups*. A subgroup H is called *proper* if $H \neq G$, and we write $H < G$.

Since the operation on H is inherited from G , associativity holds automatically in H . Thus, to determine whether a nonempty subset is a subgroup, it is sufficient to verify closure, the presence of the identity, and the existence of inverses.

The following theorem combines these requirements into a single condition.

Theorem 1.3.2 (Subgroup Test). A nonempty subset H of a group G is a subgroup of G if and only if

$$ab^{-1} \in H$$

for all $a, b \in H$.

Proof. Suppose first that H is a subgroup of G . Since H is a group, it contains the inverse b^{-1} of every $b \in H$. Closure under the group operation then gives $ab^{-1} \in H$ for all $a, b \in H$.

Conversely, suppose H is nonempty and satisfies the stated condition. We show that H is a group under the operation inherited from G .

Since H is nonempty, choose $a \in H$. Taking $b = a$ gives $aa^{-1} = e \in H$. Thus H contains the identity of G .

Next, let $b \in H$. Since $e \in H$, the assumed condition gives $eb^{-1} = b^{-1} \in H$. Hence every element of H has its inverse in H .

Finally, let $a, b \in H$. Since $b^{-1} \in H$, applying the condition to a and b^{-1} yields

$$a(b^{-1})^{-1} = ab \in H.$$

Therefore H is closed under multiplication.

Associativity is inherited from G , so H satisfies all the group axioms and is a subgroup of G . ■

Remark 1.3.3. For an additive group, the subgroup test takes the form

$$a - b \in H \quad \text{for all } a, b \in H.$$

This form is particularly convenient for studying subgroups of $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$.

Example 1.3.4 (Subgroups of the Integers). For any integer m , define $m\mathbb{Z} = \{mk : k \in \mathbb{Z}\}$. We show that $m\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Since $0 = m \cdot 0$, the set $m\mathbb{Z}$ is nonempty. If $ma, mb \in m\mathbb{Z}$, then

$$ma - mb = m(a - b) \in m\mathbb{Z}.$$

The subgroup test therefore gives $m\mathbb{Z} \leq \mathbb{Z}$.

In particular, $2\mathbb{Z}$ is the subgroup of even integers, while $3\mathbb{Z}$ consists of all multiples of 3. For $m = 0$, we obtain the trivial subgroup $\{0\}$.

Example 1.3.5 (A Subgroup of $\mathbb{Z}_{12}$). Consider the additive group $\mathbb{Z}_{12}$ and the subset

$$H = \{[0], [3], [6], [9]\}.$$

The sum or difference modulo 12 of any two elements of H again belongs to H . Hence H is a subgroup of $\mathbb{Z}_{12}$.

On the other hand, $K = \{[0], [4], [6]\}$ is not a subgroup, since $[4] + [6] = [10] \notin K$. Thus K is not closed under addition.

Example 1.3.6 (The Special Linear Group). Let $GL_n(\mathbb{R})$ be the group of all invertible $n \times n$ real matrices, and define

$$SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det A = 1\}.$$

Then $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$.

Indeed, the identity matrix belongs to $SL_n(\mathbb{R})$. Moreover, if $A, B \in SL_n(\mathbb{R})$, then

$$\det(AB^{-1}) = \det(A) \det(B^{-1}) = \frac{\det A}{\det B} = 1.$$

Consequently $AB^{-1} \in SL_n(\mathbb{R})$, and the subgroup test proves the assertion.

1.3.1 Subgroup Test for Finite Subsets

For a finite subset of a group, the subgroup test can be simplified considerably.

Theorem 1.3.7 (Finite Subgroup Test). *A nonempty finite subset H of a group G is a subgroup of G if and only if H is closed under the group operation.*

Proof. If H is a subgroup, closure follows immediately.

Conversely, suppose H is a nonempty finite subset of G that is closed under multiplication. Let $a \in H$. By closure, every positive power a^k belongs to H .

Since H is finite, two positive powers must be equal. Thus $a^r = a^s$ for some integers $r > s \geq 1$. Multiplying by a^{-s} in G , we obtain $a^{r-s} = e$. Since $r - s \geq 1$, closure implies $e \in H$.

Moreover, $a^{r-s-1} \in H$, where $a^0 = e$, and

$$aa^{r-s-1} = a^{r-s} = e.$$

Thus $a^{r-s-1} = a^{-1} \in H$.

Therefore H contains the identity and the inverse of each of its elements. Together with closure and associativity inherited from G , this shows that H is a subgroup. ■

Remark 1.3.8. The finiteness assumption is essential. For example, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ is closed under addition but is not a subgroup of $(\mathbb{Z}, +)$, since it does not contain the additive inverses of positive integers.

1.3.2 Intersections and Unions of Subgroups

The intersection of subgroups always retains the group structure. This observation is useful in constructing new subgroups from known ones.

Theorem 1.3.9 (Intersection of Subgroups). *Let $\{H_i : i \in I\}$ be a nonempty family of subgroups of a group G . Then*

$$H = \bigcap_{i \in I} H_i$$

is a subgroup of G .

Proof. Since every subgroup contains the identity e of G , we have $e \in H_i$ for all $i \in I$. Hence $e \in H$, so H is nonempty.

Let $a, b \in H$. Then $a, b \in H_i$ for every $i \in I$. Since each H_i is a subgroup, $ab^{-1} \in H_i$ for every $i \in I$. Therefore $ab^{-1} \in H$, and the subgroup test shows that H is a subgroup of G . ■

Unlike intersections, unions of subgroups need not be subgroups.

Example 1.3.10. Consider the subgroups $2\mathbb{Z}$ and $3\mathbb{Z}$ of $(\mathbb{Z}, +)$. Their union contains 2 and 3, but it does not contain $2 + 3 = 5$. Thus $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subgroup of $\mathbb{Z}$.

The precise condition under which the union of two subgroups is again a subgroup is given by the following theorem.

Theorem 1.3.11. *Let H and K be subgroups of a group G . Then $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.*

Proof. If $H \subseteq K$, then $H \cup K = K$, which is a subgroup. Similarly, if $K \subseteq H$, then $H \cup K = H$ is a subgroup.

Conversely, suppose $H \cup K$ is a subgroup, but neither subgroup is contained in the other. Choose $h \in H \setminus K$ and $k \in K \setminus H$.

Since $h, k \in H \cup K$ and this union is a subgroup, we have $hk \in H \cup K$.

If $hk \in H$, then $k = h^{-1}(hk) \in H$, a contradiction.

If $hk \in K$, then $h = (hk)k^{-1} \in K$, again a contradiction.

Therefore one of the inclusions $H \subseteq K$ or $K \subseteq H$ must hold. ■

1.3.3 Cyclic Subgroups

An especially important subgroup is obtained by taking all integer powers of a single group element.

Definition 1.3.12 (Cyclic Subgroup). Let $a \in G$. The subset

$$\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$$

is called the *cyclic subgroup generated by a* . The element a is called a *generator* of $\langle a \rangle$.

Theorem 1.3.13. *For every $a \in G$, the set $\langle a \rangle$ is a subgroup of G . Moreover, it is the smallest subgroup of G containing a .*

Proof. Since $a^0 = e$, the set $\langle a \rangle$ is nonempty. If $a^r, a^s \in \langle a \rangle$, then

$$a^r(a^s)^{-1} = a^{r-s} \in \langle a \rangle.$$

Thus $\langle a \rangle$ is a subgroup by the subgroup test.

Now let H be any subgroup of G containing a . Since H is closed under multiplication and inverses, it contains every integer power of a . Hence $\langle a \rangle \subseteq H$.

Therefore $\langle a \rangle$ is the smallest subgroup of G containing a . ■

Proposition 1.3.14. For every $a \in G$,

$$|\langle a \rangle| = \text{ord}(a),$$

where both sides are understood to be infinite when a has infinite order.

Proof. Suppose $\text{ord}(a) = n < \infty$. Then $a^n = e$, and every integer power of a is equal to one of $e, a, a^2, \dots, a^{n-1}$.

These n elements are distinct. Indeed, if $a^r = a^s$ for $0 \leq s < r < n$, then $a^{r-s} = e$, contradicting the minimality of n .

Therefore $\langle a \rangle = \{e, a, \dots, a^{n-1}\}$ and $|\langle a \rangle| = n$.

If a has infinite order, then all its integer powers are distinct, so $\langle a \rangle$ is infinite. ■

Example 1.3.15. In the additive group $\mathbb{Z}_{12}$, the cyclic subgroup generated by $[4]$ is

$$\langle [4] \rangle = \{[0], [4], [8]\}.$$

Thus $\text{ord}([4]) = 3$. Similarly,

$$\langle [3] \rangle = \{[0], [3], [6], [9]\},$$

so $\text{ord}([3]) = 4$.

1.4 Cosets and Lagrange's Theorem

Subgroups allow us to organize the elements of a group into collections called *cosets*. These collections partition the group into disjoint subsets of equal size. This observation leads to Lagrange's theorem, which establishes a fundamental relationship between the orders of a finite group and its subgroups.

1.4.1 Left and Right Cosets

Definition 1.4.1 (Cosets). Let H be a subgroup of a group G , and let $a \in G$. The set

$$aH = \{ah : h \in H\}$$

is called the *left coset* of H determined by a . Similarly,

$$Ha = \{ha : h \in H\}$$

is called the *right coset* of H determined by a .

A coset is obtained by multiplying every element of a subgroup by a fixed element of the group. The distinction between left and right cosets is important in nonabelian groups, since the two sets need not be equal.

Remark 1.4.2. Since $e \in H$, every element $a \in G$ belongs to both aH and Ha . Moreover, if $a \in H$, then $aH = Ha = H$.

In an abelian group, left and right cosets coincide because $ah = ha$ for all $a \in G$ and $h \in H$.

Example 1.4.3 (Cosets in $\mathbb{Z}$). Consider the additive group $(\mathbb{Z}, +)$ and its subgroup $H = 4\mathbb{Z}$. Since the operation is addition, the cosets are written as $a + H$.

The distinct cosets are

$$\begin{aligned} H &= \{\dots, -8, -4, 0, 4, 8, \dots\}, \\ 1 + H &= \{\dots, -7, -3, 1, 5, 9, \dots\}, \\ 2 + H &= \{\dots, -6, -2, 2, 6, 10, \dots\}, \\ 3 + H &= \{\dots, -5, -1, 3, 7, 11, \dots\}. \end{aligned}$$

These are precisely the four residue classes modulo 4. Every integer belongs to exactly one of them.

Example 1.4.4 (Cosets in $\mathbb{Z}_6$). Let $G = \mathbb{Z}_6$ and $H = \{[0], [3]\}$. The distinct cosets of H are

$$H = \{[0], [3]\}, \quad [1] + H = \{[1], [4]\}, \quad [2] + H = \{[2], [5]\}.$$

Thus the six elements of $\mathbb{Z}_6$ are partitioned into three cosets, each containing two elements.

1.4.2 Equality and Partition of Cosets

We first establish a criterion for determining when two elements produce the same left coset.

Theorem 1.4.5 (Coset Equality Criterion). *Let $H \leq G$ and let $a, b \in G$. Then*

$$aH = bH \iff a^{-1}b \in H.$$

Proof. Suppose $aH = bH$. Since $b = be \in bH$, we have $b \in aH$. Thus $b = ah$ for some $h \in H$, giving $a^{-1}b = h \in H$.

Conversely, suppose $a^{-1}b \in H$. Then $b = ah$ for some $h \in H$. Consequently,

$$bH = (ah)H = a(hH).$$

Since $h \in H$ and H is a subgroup, $hH = H$. Hence $bH = aH$. ■

Corollary 1.4.6. *For $H \leq G$ and $a, b \in G$,*

$$Ha = Hb \iff ab^{-1} \in H.$$

Proof. The proof is analogous to that of the left coset criterion, with multiplication performed on the right. ■

Theorem 1.4.7 (Cosets Partition a Group). *Let H be a subgroup of G . Then any two left cosets of H are either equal or disjoint, and their union is G .*

Proof. Suppose $aH \cap bH \neq \emptyset$. Then there exists $x \in G$ such that $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$. It follows that

$$a^{-1}b = h_1h_2^{-1} \in H.$$

By the coset equality criterion, $aH = bH$.

Thus any two distinct left cosets are disjoint.

Furthermore, every $a \in G$ belongs to aH , since $a = ae$ and $e \in H$. Hence the union of all left cosets is G .

Therefore the distinct left cosets form a partition of G . ■

Remark 1.4.8. The same argument shows that the distinct right cosets of a subgroup also form a partition of the group. Although the left and right cosets need not coincide, the numbers of distinct left and right cosets are equal.

1.4.3 Cardinality of Cosets

A remarkable property of cosets is that every coset contains exactly as many elements as the subgroup from which it is formed.

Theorem 1.4.9 (Equal Cardinality of Cosets). *Let $H \leq G$ and $a \in G$. Then H and aH have the same cardinality. In particular, if H is finite, then*

$$|aH| = |H|.$$

Proof. Define $\phi : H \rightarrow aH$ by $\phi(h) = ah$.

The map is surjective by the definition of aH . If $\phi(h_1) = \phi(h_2)$, then $ah_1 = ah_2$. By left cancellation, $h_1 = h_2$, so ϕ is injective.

Thus ϕ is a bijection, proving that H and aH have the same cardinality. ■

Definition 1.4.10 (Index of a Subgroup). Let H be a subgroup of G . The number of distinct left cosets of H in G is called the *index* of H in G , denoted by $[G : H]$.

Example 1.4.11. In $(\mathbb{Z}, +)$, the subgroup $5\mathbb{Z}$ has exactly five distinct cosets:

$$5\mathbb{Z}, \quad 1 + 5\mathbb{Z}, \quad 2 + 5\mathbb{Z}, \quad 3 + 5\mathbb{Z}, \quad 4 + 5\mathbb{Z}.$$

Therefore $[\mathbb{Z} : 5\mathbb{Z}] = 5$.

1.4.4 Lagrange's Theorem

The partition of a group into cosets, together with their equal cardinality, yields one of the most important results in finite group theory.

Theorem 1.4.12 (Lagrange's Theorem). *Let G be a finite group and let H be a subgroup of G . Then*

$$|G| = [G : H] |H|.$$

Consequently, the order of every subgroup of a finite group divides the order of the group.

Proof. Let $[G : H] = k$, and let $a_1H, a_2H, \dots, a_kH$ be the distinct left cosets of H in G . Since the cosets form a partition of G ,

$$G = a_1H \cup a_2H \cup \dots \cup a_kH,$$

where the union is disjoint.

By the equal cardinality theorem, every coset contains exactly $|H|$ elements. Therefore,

$$|G| = \sum_{i=1}^k |a_iH| = k|H| = [G : H]|H|.$$

Since $[G : H]$ is a positive integer, $|H|$ divides $|G|$. ■

Example 1.4.13. Let $G = \mathbb{Z}_{12}$ and $H = \{[0], [4], [8]\}$. Then $|G| = 12$ and $|H| = 3$.

By Lagrange's theorem,

$$[G : H] = \frac{|G|}{|H|} = \frac{12}{3} = 4.$$

The four distinct cosets are

$$H, \quad [1] + H, \quad [2] + H, \quad [3] + H.$$

Each coset contains three elements, and together they contain all twelve elements of G .

1.4.5 Consequences of Lagrange's Theorem

Lagrange's theorem has several important consequences concerning the orders of elements and the structure of finite groups.

Corollary 1.4.14. *If G is a finite group and $a \in G$, then*

$$\text{ord}(a) \mid |G|.$$

Proof. The cyclic subgroup $\langle a \rangle$ has order $\text{ord}(a)$. Since $\langle a \rangle \leq G$, Lagrange's theorem gives

$$|\langle a \rangle| \mid |G|.$$

Therefore $\text{ord}(a) \mid |G|$. ■

Corollary 1.4.15. *If G is a finite group, then $a^{|G|} = e$ for every $a \in G$.*

Proof. Let $\text{ord}(a) = n$. By the preceding result, $n \mid |G|$. Hence $|G| = kn$ for some positive integer k , and

$$a^{|G|} = a^{kn} = (a^n)^k = e. \quad \blacksquare$$

Corollary 1.4.16. *Every group of prime order is cyclic.*

Proof. Let G be a group of prime order p . Choose $a \in G$ with $a \neq e$.

By Lagrange's theorem, $\text{ord}(a)$ divides p . Since $a \neq e$, its order is not 1. As p is prime, we must have $\text{ord}(a) = p$.

Thus the cyclic subgroup $\langle a \rangle$ has p elements. Since $|G| = p$, it follows that $\langle a \rangle = G$. Therefore G is cyclic. ■

Corollary 1.4.17. *A group of prime order has no proper nontrivial subgroups.*

Proof. Let $|G| = p$, where p is prime, and let $H \leq G$. By Lagrange's theorem, $|H|$ divides p . Therefore $|H| = 1$ or $|H| = p$.

In the first case, $H = \{e\}$; in the second case, $H = G$. Hence there are no proper nontrivial subgroups. ■

Corollary 1.4.18. *Every group of prime order is abelian.*

Proof. Every group of prime order is cyclic. Since every cyclic group is abelian, the result follows. ■

Example 1.4.19 (Application of Lagrange's Theorem). Can a group of order 18 contain an element of order 4?

Suppose such an element a exists. Then $\langle a \rangle$ is a subgroup of order 4. By Lagrange's theorem, 4 must divide 18, which is impossible.

Therefore no group of order 18 can contain an element of order 4.

Remark 1.4.20. Lagrange's theorem gives a necessary condition for the existence of a subgroup of a given order, but the converse is not true in general.

In other words, if d divides $|G|$, it does not follow that G must contain a subgroup of order d .

Example 1.4.21 (The Converse of Lagrange's Theorem). The alternating group A_4 has order 12, but it has no subgroup of order 6. Thus, although 6 divides 12, there need not exist a subgroup of the corresponding order.

This example shows that the converse of Lagrange's theorem fails in general.

1.5 Cyclic Groups and Their Structure

Among the various classes of groups, cyclic groups occupy a particularly important position because their entire structure is determined by a single element. They provide the simplest examples of groups and play a fundamental role in understanding more general algebraic structures.

We have already seen that every element a of a group G determines a cyclic subgroup $\langle a \rangle$, consisting of all integer powers of a . When this subgroup is the whole group, G is called cyclic.

Definition 1.5.1 (Cyclic Group). A group G is said to be *cyclic* if there exists an element $a \in G$ such that

$$G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}.$$

Such an element a is called a *generator* of G .

Thus, a cyclic group is one in which every element can be obtained by repeatedly applying the group operation to a single element and its inverse.

In additive notation, the cyclic subgroup generated by a is written

$$\langle a \rangle = \{ka : k \in \mathbb{Z}\},$$

where ka denotes the sum of k copies of a when $k > 0$, with the usual interpretation for $k = 0$ and negative integers.

A cyclic group may have more than one generator. The problem of determining all its generators is an important part of the study of cyclic groups.

Example 1.5.2. The additive group $(\mathbb{Z}, +)$ is cyclic because every integer is an integral multiple of 1. Thus $\mathbb{Z} = \langle 1 \rangle$. It is also generated by -1 , since every integer is an integral multiple of -1 .

The group $(\mathbb{Z}_6, +)$ is cyclic, with

$$\mathbb{Z}_6 = \langle [1] \rangle = \langle [5] \rangle.$$

However, $[2]$ is not a generator of $\mathbb{Z}_6$, because

$$\langle [2] \rangle = \{[0], [2], [4]\}.$$

This is a proper subgroup of $\mathbb{Z}_6$.

Example 1.5.3 (A Multiplicative Cyclic Group). Consider the group of units modulo 10,

$$U(10) = \{[1], [3], [7], [9]\},$$

under multiplication modulo 10.

The successive powers of $[3]$ are

$$[3]^1 = [3], \quad [3]^2 = [9], \quad [3]^3 = [7], \quad [3]^4 = [1].$$

Therefore $U(10) = \langle [3] \rangle$, and hence $U(10)$ is cyclic of order 4.

1.5.1 Elementary Properties of Cyclic Groups

The first important property of cyclic groups is that their operation is necessarily commutative.

Theorem 1.5.4. *Every cyclic group is abelian.*

Proof. Let $G = \langle a \rangle$ be a cyclic group. If $x, y \in G$, then there exist integers r, s such that $x = a^r$ and $y = a^s$. By the laws of exponents,

$$xy = a^r a^s = a^{r+s} = a^{s+r} = a^s a^r = yx.$$

Since x and y were arbitrary, G is abelian. ■

The converse of this theorem is not true. An abelian group need not be cyclic.

Example 1.5.5 (An Abelian Group Which Is Not Cyclic). Consider the direct product

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{([0], [0]), ([1], [0]), ([0], [1]), ([1], [1])\},$$

under componentwise addition modulo 2.

The group G is abelian because addition in each component is commutative. However, every nonidentity element has order 2. Thus no element generates all four elements of G , and consequently G is not cyclic.

Proposition 1.5.6. Every group of order at most 3 is cyclic.

Proof. The group of order 1 is generated by its identity element. If G has order 2 or 3, then its order is prime. By the consequence of Lagrange's theorem established earlier, every group of prime order is cyclic. ■

Remark 1.5.7. The preceding example of $\mathbb{Z}_2 \times \mathbb{Z}_2$ shows that a group of order 4 need not be cyclic. It is therefore the smallest possible order of a noncyclic group.

1.5.2 Generators of Finite Cyclic Groups

A cyclic group may possess several generators. For finite cyclic groups, the generators can be characterized completely in terms of the greatest common divisor.

Recall that if $\text{ord}(a) = n$, then

$$\text{ord}(a^k) = \frac{n}{\gcd(n, k)}$$

for every integer k .

This formula immediately gives a criterion for determining when a power of a generator is again a generator.

Theorem 1.5.8 (Generators of a Finite Cyclic Group). *Let $G = \langle a \rangle$ be a finite cyclic group of order n . Then a^k is a generator of G if and only if*

$$\gcd(k, n) = 1.$$

Proof. Since $G = \langle a \rangle$, the order of a is n . The element a^k generates G if and only if its order is n .

By the formula for the order of a power,

$$\text{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

Consequently, $\text{ord}(a^k) = n$ if and only if $\gcd(n, k) = 1$.

Therefore a^k generates G precisely when k and n are relatively prime. ■

Example 1.5.9 (Generators of $\mathbb{Z}_{12}$). The additive group $\mathbb{Z}_{12}$ is cyclic of order 12, generated by $[1]$.

An element $[k]$ generates $\mathbb{Z}_{12}$ if and only if $\gcd(k, 12) = 1$. The integers between 1 and 11 relatively prime to 12 are 1, 5, 7, 11.

Therefore the generators of $\mathbb{Z}_{12}$ are

$$[1], [5], [7], [11].$$

In contrast, the element $[8]$ has order

$$\text{ord}([8]) = \frac{12}{\gcd(12, 8)} = 3,$$

and generates only the subgroup $\{[0], [4], [8]\}$.

The number of generators of a finite cyclic group is also determined by the order of the group.

Definition 1.5.10 (Euler's Totient Function). For a positive integer n , the number of integers k satisfying $1 \leq k \leq n$ and $\gcd(k, n) = 1$ is denoted by $\varphi(n)$ and is called *Euler's totient function*.

Corollary 1.5.11. A cyclic group of order n has exactly $\varphi(n)$ generators.

Proof. Let $G = \langle a \rangle$ have order n . Its elements are $e, a, \dots, a^{n-1}$, and the preceding theorem shows that a^k is a generator precisely when $\gcd(k, n) = 1$.

There are exactly $\varphi(n)$ such residue classes modulo n . Hence G has $\varphi(n)$ generators. ■

Example 1.5.12. The group $\mathbb{Z}_8$ has $\varphi(8) = 4$ generators, namely $[1], [3], [5], [7]$. Similarly, $\mathbb{Z}_{10}$ has $\varphi(10) = 4$ generators, namely $[1], [3], [7], [9]$.

1.5.3 Generators of Infinite Cyclic Groups

Infinite cyclic groups have a simpler description: they possess exactly two generators.

Theorem 1.5.13. Let $G = \langle a \rangle$ be an infinite cyclic group. Then the only generators of G are a and a^{-1} .

Proof. Clearly a generates G . Also, every element a^k can be written as $(a^{-1})^{-k}$, so a^{-1} generates G .

Now suppose b is any generator of G . Since $G = \langle a \rangle$, we have $b = a^k$ for some integer k .

Since b generates G , there exists an integer m such that

$$a = b^m = (a^k)^m = a^{km}.$$

Because a has infinite order, distinct integer powers of a are distinct. Hence $km = 1$.

The only integers k for which this is possible are $k = 1$ and $k = -1$. Thus $b = a$ or $b = a^{-1}$, proving the result. ■

Example 1.5.14. The additive group $\mathbb{Z}$ is infinite cyclic, and its only generators are 1 and -1 .

For example, 2 is not a generator of $\mathbb{Z}$, since $\langle 2 \rangle = 2\mathbb{Z}$, the subgroup of even integers.

1.5.4 Subgroups of Cyclic Groups

One of the most useful properties of cyclic groups is that all their subgroups are themselves cyclic.

Theorem 1.5.15 (Subgroups of Cyclic Groups). *Every subgroup of a cyclic group is cyclic.*

Proof. Let $G = \langle a \rangle$ be a cyclic group, and let H be a subgroup of G .

If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic.

Suppose $H \neq \{e\}$. Since every element of G is an integer power of a , there exists a nonzero integer k such that $a^k \in H$. If $k < 0$, then $a^{-k} = (a^k)^{-1} \in H$. Hence H contains a positive power of a .

Let d be the least positive integer such that $a^d \in H$. Since H is a subgroup, every integer power of a^d lies in H . Therefore

$$\langle a^d \rangle \subseteq H.$$

Conversely, let $a^k \in H$. By the division algorithm, write $k = qd + r$, where $q, r \in \mathbb{Z}$ and $0 \leq r < d$. Then

$$a^r = a^k (a^d)^{-q}.$$

Since both factors on the right belong to H , we have $a^r \in H$.

By the minimality of d , we must have $r = 0$. Thus $k = qd$, so $a^k = (a^d)^q \in \langle a^d \rangle$.

Therefore $H \subseteq \langle a^d \rangle$, and hence $H = \langle a^d \rangle$. Thus H is cyclic. ■

Example 1.5.16 (Subgroups of $\mathbb{Z}$). Every subgroup of $(\mathbb{Z}, +)$ is cyclic. More precisely, every subgroup of $\mathbb{Z}$ has the form

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

for some integer $n \geq 0$.

For example, $4\mathbb{Z}$ is generated by 4, while $7\mathbb{Z}$ is generated by 7. The trivial subgroup $\{0\}$ corresponds to $n = 0$.

For finite cyclic groups, a stronger classification is possible.

Theorem 1.5.17 (Subgroups of a Finite Cyclic Group). *Let $G = \langle a \rangle$ be a cyclic group of order n . For every positive divisor d of n , there exists exactly one subgroup of G of order d , namely*

$$\langle a^{n/d} \rangle.$$

Proof. Let d be a positive divisor of n , and put $m = n/d$. Since m divides n , we have $\gcd(n, m) = m$. Therefore

$$\text{ord}(a^m) = \frac{n}{\gcd(n, m)} = \frac{n}{m} = d.$$

Thus $\langle a^m \rangle = \langle a^{n/d} \rangle$ is a subgroup of order d .

To prove uniqueness, let H be any subgroup of G with $|H| = d$. Since every subgroup of a cyclic group is cyclic, write $H = \langle a^k \rangle$ for some integer k .

Then

$$d = |H| = \text{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

Hence $\gcd(n, k) = n/d = m$. In particular, m divides k , so a^k belongs to $\langle a^m \rangle$.

Consequently, $H \subseteq \langle a^m \rangle$. Since both subgroups have order d , they must be equal.

Thus $\langle a^{n/d} \rangle$ is the unique subgroup of order d . ■

Corollary 1.5.18. *A finite cyclic group of order n has exactly as many subgroups as n has positive divisors.*

Proof. By the preceding theorem, each positive divisor d of n determines exactly one subgroup of order d . Conversely, by Lagrange's theorem, the order of every subgroup divides n . Hence there is a one-to-one correspondence between the subgroups of G and the positive divisors of n . ■

Example 1.5.19 (All Subgroups of $\mathbb{Z}_{12}$). The positive divisors of 12 are 1, 2, 3, 4, 6, 12. Therefore $\mathbb{Z}_{12}$ has exactly six subgroups.

They are

$$\begin{aligned}\langle [0] \rangle &= \{[0]\}, \\ \langle [6] \rangle &= \{[0], [6]\}, \\ \langle [4] \rangle &= \{[0], [4], [8]\}, \\ \langle [3] \rangle &= \{[0], [3], [6], [9]\}, \\ \langle [2] \rangle &= \{[0], [2], [4], [6], [8], [10]\}, \\ \langle [1] \rangle &= \mathbb{Z}_{12}.\end{aligned}$$

Their respective orders are 1, 2, 3, 4, 6, and 12.

Proposition 1.5.20. Let $G = \langle a \rangle$ be a finite cyclic group of order n . For integers r, s ,

$$\langle a^r \rangle \subseteq \langle a^s \rangle \iff \gcd(n, s) \mid \gcd(n, r).$$

Proof. Put $d_r = \gcd(n, r)$ and $d_s = \gcd(n, s)$. By the subgroup classification theorem, $\langle a^r \rangle = \langle a^{d_r} \rangle$ and $\langle a^s \rangle = \langle a^{d_s} \rangle$.

If $d_s \mid d_r$, then every power of a^{d_r} is a power of a^{d_s} , so $\langle a^r \rangle \subseteq \langle a^s \rangle$.

Conversely, suppose $\langle a^{d_r} \rangle \subseteq \langle a^{d_s} \rangle$. Then $a^{d_r} = (a^{d_s})^k$ for some integer k , and hence $n \mid (d_r - kd_s)$. Since $d_s \mid n$, it follows that $d_s \mid d_r$. ■

1.5.5 Classification of Cyclic Groups

Two groups may be constructed from entirely different sets and operations yet possess the same essential algebraic structure. The concept of an isomorphism makes this precise.

Definition 1.5.21 (Group Isomorphism). Let G and K be groups. A bijective mapping $\varphi : G \rightarrow K$ is called an *isomorphism* if

$$\varphi(xy) = \varphi(x)\varphi(y)$$

for all $x, y \in G$, where the operations on the two sides are those of G and K , respectively.

If such a mapping exists, the groups are said to be *isomorphic*, written $G \cong K$.

An isomorphism preserves the group operation and therefore the fundamental algebraic properties of the group.

The following theorem shows that, up to isomorphism, there is only one infinite cyclic group and exactly one cyclic group of each finite positive order.

Theorem 1.5.22 (Structure Theorem for Cyclic Groups). *Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$. Moreover, every finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.*

Proof. Let $G = \langle a \rangle$ be a cyclic group. We consider two cases.

Case 1: G is infinite.

Define $\varphi : \mathbb{Z} \rightarrow G$ by $\varphi(k) = a^k$. For any $r, s \in \mathbb{Z}$,

$$\varphi(r + s) = a^{r+s} = a^r a^s = \varphi(r)\varphi(s).$$

Thus φ preserves the group operation.

Since $G = \langle a \rangle$, every element of G has the form a^k , so φ is surjective.

If $\varphi(r) = \varphi(s)$, then $a^r = a^s$. As a has infinite order, distinct integer powers are distinct. Hence $r = s$, so φ is injective.

Therefore φ is an isomorphism and $G \cong \mathbb{Z}$.

Case 2: G is finite of order n .

Define $\psi : \mathbb{Z}_n \rightarrow G$ by

$$\psi([k]) = a^k.$$

We first show that ψ is well-defined. If $[k] = [r]$ in $\mathbb{Z}_n$, then $n \mid (k - r)$. Since $\text{ord}(a) = n$, this implies $a^k = a^r$. Thus the value of $\psi([k])$ does not depend on the chosen representative.

For $[r], [s] \in \mathbb{Z}_n$,

$$\psi([r] + [s]) = \psi([r + s]) = a^{r+s} = a^r a^s = \psi([r])\psi([s]).$$

Hence ψ preserves the operation.

Since a generates G , the map ψ is surjective. Furthermore, if $\psi([r]) = \psi([s])$, then $a^r = a^s$, which implies $n \mid (r - s)$ and hence $[r] = [s]$. Thus ψ is injective.

Therefore ψ is an isomorphism and $G \cong \mathbb{Z}_n$. ■

Corollary 1.5.23. Any two cyclic groups of the same finite order are isomorphic.

Proof. Let G and K be cyclic groups of order n . By the structure theorem, both are isomorphic to $\mathbb{Z}_n$. Consequently, $G \cong K$. ■

Remark 1.5.24. The structure theorem gives a complete classification of cyclic groups up to isomorphism. It shows that the essential structure of a cyclic group depends only on whether it is infinite or, in the finite case, on its order.

For example, the rotational symmetry group of a regular pentagon is cyclic of order 5 and is therefore isomorphic to $(\mathbb{Z}_5, +)$.

Example 1.5.25 (The Rotation Group of a Square). Let G be the group of rotations of a square through multiples of 90° , under composition. If r denotes counterclockwise rotation through 90° , then

$$G = \{e, r, r^2, r^3\}, \quad r^4 = e.$$

Thus $G = \langle r \rangle$ is cyclic of order 4.

The mapping $\varphi : \mathbb{Z}_4 \rightarrow G$ defined by $\varphi([k]) = r^k$ is an isomorphism. Hence $G \cong \mathbb{Z}_4$.

1.6 Exercises

Unless otherwise stated, all operations are the usual ones.

1. Determine whether each operation on $\mathbb{Z}$ is associative and commutative: (a) $a * b = a - b$; (b) $a * b = a + b + 2$; (c) $a * b = ab$.
2. Prove that the relation $a \sim b$ on $\mathbb{Z}$ defined by $5 \mid (a - b)$ is an equivalence relation. Write all its distinct equivalence classes.
3. Show that $a \sim b$ on $\mathbb{R}$ defined by $a^2 = b^2$ is an equivalence relation. Describe the equivalence class of 3 and of 0.
4. Verify directly that $\mathbb{Z}_5$ is a group under addition modulo 5. Construct its addition table.
5. Determine whether the positive rational numbers form a group under multiplication.

6. Give two noncommuting elements of S_3 and calculate their products in both orders.
7. Prove that every group of order 1 or 2 is abelian.
8. Show that in a group G , $(abc)^{-1} = c^{-1}b^{-1}a^{-1}$.
9. If $a^2 = e$ for every $a \in G$, prove that G is abelian.
10. Solve $axb = c$ for x in an arbitrary group, where $a, b, c \in G$.
11. Prove that $3\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subgroup?
12. Determine all subgroups of the additive group $\mathbb{Z}_6$.
13. Prove that the center $Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}$ is a subgroup of G .
14. List the distinct cosets of $\langle [4] \rangle$ in $\mathbb{Z}_{12}$ and determine their number.
15. Can a finite group of order 21 contain an element of order 6? Justify your answer.
16. Prove that a group of prime order is abelian.
17. Find the orders of $[4]$, $[6]$ and $[9]$ in $\mathbb{Z}_{18}$.
18. Find all generators of $\mathbb{Z}_{10}$ and $\mathbb{Z}_{15}$.
19. Show that every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$ for a unique nonnegative integer m .
20. Describe all subgroups of a cyclic group of order 18, giving their generators and orders.
21. Prove that $\langle a^r \rangle = \langle a^s \rangle$ in a cyclic group of order n if and only if $\gcd(r, n) = \gcd(s, n)$.
22. Explain why the rule $[k] \mapsto a^k$ requires a well-definedness check when its domain is $\mathbb{Z}_n$, but not when its domain is $\mathbb{Z}$.

Chapter 2

Normal Subgroups and Homomorphisms

The study of subgroups provides important information about the internal structure of a group. Among subgroups, those called *normal subgroups* play a particularly significant role because they allow us to construct new groups whose elements are cosets of the original group. These are known as *quotient groups*.

The notion of a normal subgroup is closely connected with that of a group homomorphism, which is a mapping between groups that preserves their operations. Together, these concepts provide a systematic method for studying relationships between groups and understanding their structure.

In this chapter, we develop the theory of normal subgroups, quotient groups, and homomorphisms. We also study permutation groups, Cayley's theorem, and the symmetry groups of regular polygons. Unless otherwise stated, group operations are written multiplicatively, and permutations are composed from right to left.

2.1 Normal Subgroups

Recall that if H is a subgroup of a group G and $g \in G$, then the *left coset* and *right coset* of H determined by g are, respectively,

$$gH = \{gh : h \in H\}, \quad Hg = \{hg : h \in H\}.$$

In an abelian group, these cosets are equal because $gh = hg$ for all $g \in G$ and $h \in H$. In a nonabelian group, however, left and right cosets need not coincide.

Subgroups for which these cosets are always equal are called normal subgroups.

Definition 2.1.1 (Normal Subgroup). A subgroup N of a group G is called a *normal subgroup* of G if

$$gN = Ng$$

for every $g \in G$. We denote this by $N \trianglelefteq G$.

Normality therefore means that multiplying the elements of a subgroup by any fixed element of G produces the same subset, whether multiplication is performed on the left or on the right.

Normality can also be expressed in terms of *conjugation*. For $g, x \in G$, the element gxg^{-1} is called the conjugate of x by g .

For a subset N of G , we write

$$gNg^{-1} = \{gng^{-1} : n \in N\}.$$

The following theorem gives several equivalent characterizations of normal subgroups.

Theorem 2.1.2 (Normal Subgroup Criterion). *Let N be a subgroup of G . The following conditions are equivalent:*

- (i) N is normal in G ;
- (ii) $gNg^{-1} = N$ for every $g \in G$;
- (iii) $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.

Proof. Suppose first that $N \trianglelefteq G$. Then $gN = Ng$ for every $g \in G$. Multiplying both sides on the right by g^{-1} gives

$$gNg^{-1} = Ng g^{-1} = N.$$

Thus (i) implies (ii).

Conversely, if $gNg^{-1} = N$, multiplying on the right by g gives $gN = Ng$. Hence (ii) implies (i).

Clearly, (ii) implies (iii), since every element gng^{-1} belongs to the set $gNg^{-1} = N$.

Finally, suppose (iii) holds. Then $gNg^{-1} \subseteq N$ for every $g \in G$.

Replacing g by g^{-1} gives $g^{-1}Ng \subseteq N$. Multiplying this inclusion on the left by g and on the right by g^{-1} yields

$$N \subseteq gNg^{-1}.$$

Combining the two inclusions gives $gNg^{-1} = N$. Therefore (iii) implies (ii), completing the proof. ■

Remark 2.1.3. The third condition is often the most convenient test for normality. To prove that a subgroup N is normal, it is sufficient to show that the conjugate of every element of N by every element of G again belongs to N .

2.1.1 Examples of Normal Subgroups

Example 2.1.4 (Trivial Normal Subgroups). For every group G , both $\{e\}$ and G are normal subgroups.

Indeed, for every $g \in G$,

$$g\{e\}g^{-1} = \{e\}, \quad gGg^{-1} = G.$$

Thus $\{e\} \trianglelefteq G$ and $G \trianglelefteq G$.

Example 2.1.5 (Subgroups of Abelian Groups). Every subgroup of an abelian group is normal.

To see this, let $N \leq G$, where G is abelian. For $g \in G$ and $n \in N$, commutativity gives

$$gng^{-1} = ngg^{-1} = n \in N.$$

Hence $N \trianglelefteq G$.

In particular, $m\mathbb{Z} \trianglelefteq \mathbb{Z}$ for every nonnegative integer m .

Example 2.1.6 (A Non-normal Subgroup). Consider the symmetric group

$$S_3 = \{e, (12), (13), (23), (123), (132)\}$$

and its subgroup $H = \{e, (12)\}$.

Choose $g = (13)$. Since $g^{-1} = g$, we obtain

$$g(12)g^{-1} = (13)(12)(13) = (23).$$

But $(23) \notin H$. Therefore H is not normal in S_3 .

This example shows that a subgroup of a nonabelian group need not be normal.

Example 2.1.7 (The Alternating Subgroup of S_3). Consider the subgroup

$$A_3 = \{e, (123), (132)\}$$

of S_3 .

Since $|S_3| = 6$ and $|A_3| = 3$, Lagrange's theorem gives

$$[S_3 : A_3] = \frac{6}{3} = 2.$$

As every subgroup of index two is normal, a result proved below, we conclude that $A_3 \trianglelefteq S_3$.

2.1.2 Further Properties of Normal Subgroups

The following theorem provides an important sufficient condition for normality.

Theorem 2.1.8 (Subgroups of Index Two). *Every subgroup of index two in a group is normal.*

Proof. Let $H \leq G$ and suppose $[G : H] = 2$. Thus there are exactly two distinct left cosets of H in G , and exactly two distinct right cosets.

Let $g \in G$. If $g \in H$, then $gH = H = Hg$.

If $g \notin H$, the left coset gH is distinct from H . Since the left cosets partition G and there are exactly two of them, we must have

$$gH = G \setminus H.$$

Similarly, the right coset Hg is distinct from H , and hence

$$Hg = G \setminus H.$$

Therefore $gH = Hg$ for every $g \in G$, proving that $H \trianglelefteq G$. ■

Example 2.1.9. The subgroup A_n of even permutations in S_n has index two for $n \geq 2$, since $|S_n| = n!$ and $|A_n| = n!/2$.

Consequently, $A_n \trianglelefteq S_n$.

Theorem 2.1.10 (Intersection of Normal Subgroups). *Let $\{N_i : i \in I\}$ be a nonempty family of normal subgroups of G . Then*

$$N = \bigcap_{i \in I} N_i$$

is a normal subgroup of G .

Proof. Since every N_i is a subgroup, their intersection N is also a subgroup of G .

Let $g \in G$ and $n \in N$. Then $n \in N_i$ for every $i \in I$. Since each N_i is normal,

$$gng^{-1} \in N_i$$

for every $i \in I$. Hence $gng^{-1} \in N$.

By the normal subgroup criterion, $N \trianglelefteq G$. ■

Proposition 2.1.11. *If $N \trianglelefteq G$ and $H \leq G$, then $N \cap H$ is a normal subgroup of H .*

Proof. The intersection $N \cap H$ is a subgroup of H . Let $h \in H$ and $x \in N \cap H$.

Since N is normal in G , we have $h x h^{-1} \in N$. Also, since $h, x \in H$ and H is a subgroup, $h x h^{-1} \in H$.

Therefore $h x h^{-1} \in N \cap H$, and hence $N \cap H \trianglelefteq H$. ■

Proposition 2.1.12. Let H and K be normal subgroups of G . Then

$$HK = \{hk : h \in H, k \in K\}$$

is a normal subgroup of G .

Proof. We first show that HK is a subgroup. Clearly $e \in HK$. Let $h_1 k_1, h_2 k_2 \in HK$. Then

$$(h_1 k_1)(h_2 k_2)^{-1} = h_1(k_1 k_2^{-1})h_2^{-1}.$$

Since H is normal, there exists $h_3 \in H$ such that

$$(k_1 k_2^{-1})h_2^{-1} = h_3(k_1 k_2^{-1}).$$

Hence

$$(h_1 k_1)(h_2 k_2)^{-1} = (h_1 h_3)(k_1 k_2^{-1}) \in HK.$$

Therefore HK is a subgroup.

Now let $g \in G$ and $hk \in HK$. Since H and K are normal,

$$g(hk)g^{-1} = (ghg^{-1})(gkg^{-1}) \in HK.$$

Thus $HK \trianglelefteq G$. ■

Remark 2.1.13. The normality of H and K is important here. The product of two arbitrary subgroups need not be a subgroup.

2.2 Quotient Groups

We have seen that the left cosets of any subgroup partition a group into disjoint subsets. It is natural to ask whether these cosets can themselves be regarded as the elements of a new group.

For an arbitrary subgroup, multiplication of cosets may depend on the representatives chosen. However, when the subgroup is normal, multiplication of cosets is well-defined and produces a group.

This construction is called a *quotient group*.

Definition 2.2.1 (Quotient Group). Let N be a normal subgroup of a group G . The set of all left cosets of N in G is denoted by

$$G/N = \{gN : g \in G\}.$$

Define multiplication on G/N by

$$(aN)(bN) = (ab)N$$

for all $a, b \in G$.

The resulting structure is called the *quotient group* or *factor group* of G by N .

The most important point in this definition is that the product must not depend on the particular representatives of the cosets. We establish this in the following theorem.

Theorem 2.2.2 (Quotient Group Theorem). *Let N be a normal subgroup of G . Then the operation*

$$(aN)(bN) = (ab)N$$

is well-defined and makes G/N a group.

The identity element is N , the inverse of aN is $a^{-1}N$, and if G is finite, then

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

Proof. We first show that multiplication of cosets is well-defined.

Suppose $aN = a'N$ and $bN = b'N$. Then there exist $n_1, n_2 \in N$ such that

$$a' = an_1, \quad b' = bn_2.$$

Consequently,

$$a'b' = an_1bn_2 = ab(b^{-1}n_1b)n_2.$$

Since N is normal, $b^{-1}n_1b \in N$. Therefore $(b^{-1}n_1b)n_2 \in N$, giving

$$a'b'N = abN.$$

Hence the product of two cosets is independent of the chosen representatives.

We now verify the group axioms.

Closure. For $aN, bN \in G/N$, their product is $(ab)N$, which is again a coset of N and hence belongs to G/N .

Associativity. For $aN, bN, cN \in G/N$,

$$\begin{aligned} ((aN)(bN))(cN) &= ((ab)N)(cN) \\ &= ((ab)c)N \\ &= (a(bc))N \\ &= (aN)((bc)N) \\ &= (aN)((bN)(cN)). \end{aligned}$$

Thus multiplication is associative.

Identity. Since $eN = N$, we have

$$(aN)N = (ae)N = aN, \quad N(aN) = (ea)N = aN.$$

Therefore N is the identity element of G/N .

Inverses. For every $aN \in G/N$,

$$(aN)(a^{-1}N) = (aa^{-1})N = N,$$

and similarly $(a^{-1}N)(aN) = N$. Thus the inverse of aN is $a^{-1}N$.

Therefore G/N is a group.

Finally, if G is finite, the elements of G/N are precisely the distinct cosets of N . Hence $|G/N| = [G : N]$. By Lagrange's theorem,

$$|G/N| = \frac{|G|}{|N|}.$$



Remark 2.2.3. The identity element of G/N is the coset N , not the identity element e of G . Although $e \in N$, the elements of G/N are cosets rather than individual elements of G .

2.2.1 Normality and Well-defined Coset Multiplication

The requirement that N be normal is not merely sufficient for defining multiplication of cosets; it is also necessary.

Theorem 2.2.4. *Let H be a subgroup of G . The rule*

$$(aH)(bH) = (ab)H$$

defines a well-defined binary operation on the set of left cosets of H if and only if H is normal in G .

Proof. If H is normal, the quotient group theorem shows that the operation is well-defined. Conversely, suppose the rule is well-defined. Let $g \in G$ and $h \in H$. Since $hH = H = eH$, well-definedness gives

$$(gH)(hH) = (gH)(eH).$$

Therefore $(gh)H = gH$, which implies $g^{-1}(gh) \in H$ but yields no new information. To obtain the required condition, use the equality $hH = eH$ in the first factor:

$$(hH)(gH) = (eH)(gH).$$

Thus $(hg)H = gH$, so $g^{-1}hg \in H$.

Since $g \in G$ and $h \in H$ were arbitrary, the normal subgroup criterion gives $H \trianglelefteq G$. ■

2.2.2 Examples of Quotient Groups

Example 2.2.5 (The Quotient $\mathbb{Z}/n\mathbb{Z}$). Let $G = (\mathbb{Z}, +)$ and $N = n\mathbb{Z}$, where $n \geq 2$. Since $\mathbb{Z}$ is abelian, $n\mathbb{Z}$ is normal.

The quotient group $\mathbb{Z}/n\mathbb{Z}$ consists of the n distinct cosets

$$0 + n\mathbb{Z}, \quad 1 + n\mathbb{Z}, \quad \dots, \quad (n-1) + n\mathbb{Z}.$$

Its operation is

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}.$$

For example, in $\mathbb{Z}/5\mathbb{Z}$,

$$(3 + 5\mathbb{Z}) + (4 + 5\mathbb{Z}) = 7 + 5\mathbb{Z} = 2 + 5\mathbb{Z}.$$

Thus $\mathbb{Z}/n\mathbb{Z}$ is the familiar additive group of integers modulo n .

Example 2.2.6 (The Quotient S_3/A_3). Let

$$G = S_3, \quad N = A_3 = \{e, (123), (132)\}.$$

Since $[S_3 : A_3] = 2$, the subgroup A_3 is normal.

The quotient group has two elements:

$$S_3/A_3 = \{A_3, (12)A_3\}.$$

Since $(12)^2 = e$, we have

$$((12)A_3)^2 = A_3.$$

Therefore S_3/A_3 is a cyclic group of order 2, and hence

$$S_3/A_3 \cong \mathbb{Z}_2.$$

Example 2.2.7 (Quotient of a Finite Cyclic Group). Consider the additive group $G = \mathbb{Z}_{12}$ and the subgroup

$$H = \langle [4] \rangle = \{[0], [4], [8]\}.$$

Since G is abelian, H is normal.

We have $|G| = 12$ and $|H| = 3$, so

$$|G/H| = \frac{12}{3} = 4.$$

The four distinct cosets are

$$H, \quad [1] + H, \quad [2] + H, \quad [3] + H.$$

The coset $[1] + H$ generates G/H , since its successive multiples are

$$[1] + H, \quad [2] + H, \quad [3] + H, \quad H.$$

Thus G/H is cyclic of order 4, and hence

$$\mathbb{Z}_{12}/\langle [4] \rangle \cong \mathbb{Z}_4.$$

Example 2.2.8 (A Quotient of a Matrix Group). Consider the determinant-one subgroup $SL_2(\mathbb{R})$ of $GL_2(\mathbb{R})$.

For $A \in GL_2(\mathbb{R})$ and $B \in SL_2(\mathbb{R})$,

$$\det(ABA^{-1}) = \det(A) \det(B) \det(A^{-1}) = 1.$$

Therefore $ABA^{-1} \in SL_2(\mathbb{R})$, showing that

$$SL_2(\mathbb{R}) \trianglelefteq GL_2(\mathbb{R}).$$

The quotient group $GL_2(\mathbb{R})/SL_2(\mathbb{R})$ is thus well-defined. In the study of group homomorphisms, we shall see that this quotient is isomorphic to the multiplicative group $\mathbb{R} \setminus \{0\}$.

Proposition 2.2.9. If G is abelian and $N \trianglelefteq G$, then G/N is abelian.

Proof. Let $aN, bN \in G/N$. Since G is abelian, $ab = ba$. Therefore

$$(aN)(bN) = (ab)N = (ba)N = (bN)(aN).$$

Hence G/N is abelian. ■

Remark 2.2.10. The converse is not true. A quotient group may be abelian even when the original group is nonabelian. For example, S_3 is nonabelian, but S_3/A_3 is cyclic of order 2 and therefore abelian.

Proposition 2.2.11. If G is cyclic and N is a subgroup of G , then G/N is cyclic.

Proof. Let $G = \langle a \rangle$. Since every cyclic group is abelian, N is normal and G/N is defined.

Every coset of N has the form gN for some $g \in G$. Since $G = \langle a \rangle$, we may write $g = a^k$ for some integer k . Hence

$$gN = a^k N = (aN)^k.$$

Therefore every element of G/N is a power of aN , so

$$G/N = \langle aN \rangle.$$

Thus the quotient group is cyclic. ■

2.3 Group Homomorphisms

Having studied subgroups and quotient groups, we now consider mappings between groups that preserve their algebraic structure. Such mappings are called *group homomorphisms*. They provide a means of comparing groups and establishing relationships between algebraic systems that may have entirely different underlying sets.

The essential property of a homomorphism is that combining two elements before applying the mapping gives the same result as applying the mapping to each element and then combining their images.

Definition 2.3.1 (Group Homomorphism). Let $(G, *)$ and $(K, \cdot)$ be groups. A mapping $\varphi : G \rightarrow K$ is called a *group homomorphism* if

$$\varphi(a * b) = \varphi(a) \cdot \varphi(b)$$

for all $a, b \in G$.

When the operations are understood, we simply write $\varphi(ab) = \varphi(a)\varphi(b)$.

A homomorphism need not be injective or surjective. A homomorphism that is injective is called a *monomorphism*, while a surjective homomorphism is called an *epimorphism*. A bijective homomorphism is called an *isomorphism*. If an isomorphism exists between G and K , we write $G \cong K$.

Example 2.3.2 (Reduction Modulo n). Let $n \geq 2$. Define $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}_n, +)$ by $\varphi(k) = [k]$.

For any $a, b \in \mathbb{Z}$,

$$\varphi(a + b) = [a + b] = [a] + [b] = \varphi(a) + \varphi(b).$$

Thus φ is a homomorphism. It is surjective because every element of $\mathbb{Z}_n$ is represented by an integer.

Example 2.3.3 (The Determinant Homomorphism). Consider the mapping

$$\det : GL_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times,$$

where $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ is a group under multiplication.

Since $\det(AB) = \det(A)\det(B)$ for all invertible matrices A, B , the determinant is a group homomorphism.

Example 2.3.4 (A Homomorphism Between Additive Groups). For a fixed integer m , define $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ by $\varphi(k) = mk$. Then

$$\varphi(a + b) = m(a + b) = ma + mb = \varphi(a) + \varphi(b).$$

Hence φ is a homomorphism.

If $m \neq 0$, this homomorphism is injective, while its image is the subgroup $m\mathbb{Z}$. It is surjective precisely when $m = 1$ or $m = -1$.

Example 2.3.5 (A Map Which Is Not a Homomorphism). Consider $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ defined by $f(k) = k + 1$.

For integers a, b , we have

$$f(a + b) = a + b + 1, \quad f(a) + f(b) = a + b + 2.$$

These are not equal. Therefore f is not a homomorphism.

2.3.1 Elementary Properties of Homomorphisms

The defining property of a homomorphism has several immediate consequences. In particular, a homomorphism preserves identity elements, inverses, and integer powers.

Theorem 2.3.6 (Basic Properties of Homomorphisms). Let $\varphi : G \rightarrow K$ be a group homomorphism, and let e_G and e_K denote the identity elements of G and K , respectively. Then, for every $a \in G$ and $m \in \mathbb{Z}$,

- (i) $\varphi(e_G) = e_K$;
- (ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$;
- (iii) $\varphi(a^m) = \varphi(a)^m$.

Proof. Since $e_G e_G = e_G$, the homomorphism property gives

$$\varphi(e_G) = \varphi(e_G)\varphi(e_G).$$

Multiplying by $\varphi(e_G)^{-1}$ in K , we obtain $\varphi(e_G) = e_K$.

Next, since $aa^{-1} = e_G$, we have

$$\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_K.$$

Similarly, $\varphi(a^{-1})\varphi(a) = e_K$. By uniqueness of inverses, $\varphi(a^{-1}) = \varphi(a)^{-1}$.

For positive integers m , repeated application of the homomorphism property gives

$$\varphi(a^m) = \varphi(a)^m.$$

The result is immediate for $m = 0$ and follows for negative integers from the inverse formula. ■

Proposition 2.3.7. Let $\varphi : G \rightarrow K$ be a homomorphism. If $a \in G$ has finite order, then the order of $\varphi(a)$ divides the order of a .

Proof. Suppose $\text{ord}(a) = n$. Then $a^n = e_G$, so

$$\varphi(a)^n = \varphi(a^n) = \varphi(e_G) = e_K.$$

Therefore $\varphi(a)$ has finite order, say m , and the criterion for the order of an element gives $m \mid n$. ■

Remark 2.3.8. The order of an element may decrease under a homomorphism. For example, under reduction modulo 6, the integer 2 has infinite order in $(\mathbb{Z}, +)$, whereas its image $[2]$ has order 3 in $\mathbb{Z}_6$.

2.3.2 Kernel and Image of a Homomorphism

Two subsets associated with a homomorphism are particularly important: its kernel and its image.

Definition 2.3.9 (Kernel and Image). Let $\varphi : G \rightarrow K$ be a group homomorphism. The *kernel* of φ is

$$\ker \varphi = \{g \in G : \varphi(g) = e_K\}.$$

The *image* of φ is

$$\text{Im } \varphi = \{\varphi(g) : g \in G\}.$$

The kernel consists of those elements of G that are mapped to the identity of K . The image consists of all elements of K that actually occur as values of φ .

The following theorem explains their algebraic significance.

Theorem 2.3.10. *Let $\varphi : G \rightarrow K$ be a group homomorphism. Then*

- (i) $\ker \varphi$ is a normal subgroup of G ;
- (ii) $\text{Im } \varphi$ is a subgroup of K .

Proof. Let $N = \ker \varphi$. Since $\varphi(e_G) = e_K$, we have $e_G \in N$, so N is nonempty.

If $a, b \in N$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_K e_K^{-1} = e_K.$$

Hence $ab^{-1} \in N$, and the subgroup test shows that $N \leq G$.

To establish normality, let $g \in G$ and $n \in N$. Then

$$\begin{aligned} \varphi(gng^{-1}) &= \varphi(g)\varphi(n)\varphi(g)^{-1} \\ &= \varphi(g)e_K\varphi(g)^{-1} \\ &= e_K. \end{aligned}$$

Thus $gng^{-1} \in N$, proving $N \trianglelefteq G$.

Now let $H = \text{Im } \varphi$. Since $\varphi(e_G) = e_K$, the image is nonempty.

Let $x, y \in H$. Then $x = \varphi(a)$ and $y = \varphi(b)$ for some $a, b \in G$. Consequently,

$$xy^{-1} = \varphi(a)\varphi(b)^{-1} = \varphi(ab^{-1}) \in H.$$

Therefore $H \leq K$ by the subgroup test. ■

The kernel determines whether a homomorphism is injective.

Theorem 2.3.11 (Injectivity Criterion). *A group homomorphism $\varphi : G \rightarrow K$ is injective if and only if*

$$\ker \varphi = \{e_G\}.$$

Proof. Suppose φ is injective. If $a \in \ker \varphi$, then

$$\varphi(a) = e_K = \varphi(e_G).$$

Injectivity gives $a = e_G$. Hence $\ker \varphi = \{e_G\}$.

Conversely, suppose $\ker \varphi = \{e_G\}$. If $\varphi(a) = \varphi(b)$, then

$$\varphi(b^{-1}a) = \varphi(b)^{-1}\varphi(a) = e_K.$$

Thus $b^{-1}a \in \ker \varphi$, so $b^{-1}a = e_G$ and hence $a = b$. Therefore φ is injective. ■

Example 2.3.12 (Kernel and Image of Reduction Modulo 6). Let $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_6$ be defined by $\varphi(k) = [k]$.

An integer k belongs to the kernel precisely when $[k] = [0]$, that is, when $6 \mid k$. Therefore

$$\ker \varphi = 6\mathbb{Z}.$$

Every residue class modulo 6 has an integer representative, so $\text{Im } \varphi = \mathbb{Z}_6$.

Thus φ is surjective but not injective.

Example 2.3.13 (The Sign Homomorphism). Define $\text{sgn} : S_n \rightarrow \{1, -1\}$ by assigning 1 to every even permutation and -1 to every odd permutation.

The parity law for products of permutations gives

$$\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau).$$

Thus sgn is a homomorphism. Its kernel is the alternating group A_n , and for $n \geq 2$ its image is $\{1, -1\}$.

2.3.3 Images and Inverse Images of Subgroups

Homomorphisms also preserve subgroup structure in the following sense.

Theorem 2.3.14. *Let $\varphi : G \rightarrow K$ be a group homomorphism.*

(i) *If $H \leq G$, then $\varphi(H) \leq K$.*

(ii) *If $L \leq K$, then*

$$\varphi^{-1}(L) = \{g \in G : \varphi(g) \in L\}$$

is a subgroup of G .

Furthermore, if $L \trianglelefteq K$, then $\varphi^{-1}(L) \trianglelefteq G$.

Proof. Let $H \leq G$. Since $e_G \in H$, the set $\varphi(H)$ contains e_K . If $x = \varphi(a)$ and $y = \varphi(b)$ belong to $\varphi(H)$, where $a, b \in H$, then

$$xy^{-1} = \varphi(ab^{-1}) \in \varphi(H).$$

Thus $\varphi(H)$ is a subgroup.

Next, let $L \leq K$ and put $M = \varphi^{-1}(L)$. Since $\varphi(e_G) = e_K \in L$, the set M is nonempty. If $a, b \in M$, then $\varphi(a), \varphi(b) \in L$, so

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L.$$

Therefore $ab^{-1} \in M$, showing that M is a subgroup.

Finally, suppose L is normal in K . For $g \in G$ and $m \in M$,

$$\varphi(gmg^{-1}) = \varphi(g)\varphi(m)\varphi(g)^{-1} \in L.$$

Hence $gmg^{-1} \in M$, proving $M \trianglelefteq G$. ■

2.4 The Isomorphism Theorems

The isomorphism theorems are fundamental results connecting homomorphisms, normal subgroups, and quotient groups. They explain how quotient groups arise naturally from homomorphisms and provide methods for comparing different quotient constructions.

2.4.1 First Isomorphism Theorem

The kernel of a homomorphism identifies elements that have the same image. The First Isomorphism Theorem states that, after factoring out the kernel, the resulting quotient group is isomorphic to the image.

Theorem 2.4.1 (First Isomorphism Theorem). *Let $\varphi : G \rightarrow K$ be a group homomorphism. Then*

$$G/\ker \varphi \cong \text{Im } \varphi.$$

Proof. Put $N = \ker \varphi$. Since $N \trianglelefteq G$, the quotient group G/N is well-defined.

Define $\tilde{\varphi} : G/N \rightarrow \text{Im } \varphi$ by

$$\tilde{\varphi}(gN) = \varphi(g).$$

We first show that this mapping is well-defined. Suppose $gN = hN$. Then $h^{-1}g \in N$, so

$$\varphi(h^{-1}g) = e_K.$$

Consequently,

$$\varphi(h)^{-1}\varphi(g) = e_K,$$

which implies $\varphi(g) = \varphi(h)$. Hence $\tilde{\varphi}(gN) = \tilde{\varphi}(hN)$.

Next, for $gN, hN \in G/N$,

$$\begin{aligned}\tilde{\varphi}((gN)(hN)) &= \tilde{\varphi}(ghN) \\ &= \varphi(gh) \\ &= \varphi(g)\varphi(h) \\ &= \tilde{\varphi}(gN)\tilde{\varphi}(hN).\end{aligned}$$

Thus $\tilde{\varphi}$ is a homomorphism.

It is surjective because every element of $\text{Im } \varphi$ has the form $\varphi(g)$ for some $g \in G$.

Finally, suppose $\tilde{\varphi}(gN) = \tilde{\varphi}(hN)$. Then $\varphi(g) = \varphi(h)$, so $\varphi(h^{-1}g) = e_K$. Therefore $h^{-1}g \in N$, giving $gN = hN$. Hence $\tilde{\varphi}$ is injective.

Thus $\tilde{\varphi}$ is an isomorphism, and

$$G/\ker \varphi \cong \text{Im } \varphi.$$

■

Example 2.4.2 (The Quotient $\mathbb{Z}/n\mathbb{Z}$). Consider the homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $\varphi(k) = [k]$. Since $\ker \varphi = n\mathbb{Z}$ and $\text{Im } \varphi = \mathbb{Z}_n$, the First Isomorphism Theorem gives

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n.$$

Example 2.4.3 (The Determinant Homomorphism). The determinant map

$$\det : GL_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times$$

is a homomorphism with kernel

$$SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det A = 1\}.$$

It is surjective because, for every $t \in \mathbb{R}^\times$, the diagonal matrix $\text{diag}(t, 1, \dots, 1)$ has determinant t . Therefore

$$GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^\times.$$

Corollary 2.4.4. A normal subgroup N of G is the kernel of a group homomorphism from G .

Proof. Define the natural projection $\pi : G \rightarrow G/N$ by $\pi(g) = gN$.

For $a, b \in G$,

$$\pi(ab) = abN = (aN)(bN) = \pi(a)\pi(b),$$

so π is a homomorphism.

Moreover,

$$\ker \pi = \{g \in G : gN = N\} = N.$$

Thus N is the kernel of a homomorphism. ■

2.4.2 Second Isomorphism Theorem

The Second Isomorphism Theorem describes the relationship between a subgroup, a normal subgroup, and their product.

Theorem 2.4.5 (Second Isomorphism Theorem). *Let $H \leq G$ and $N \trianglelefteq G$. Then HN is a subgroup of G , $N \trianglelefteq HN$, and $H \cap N \trianglelefteq H$. Moreover,*

$$H/(H \cap N) \cong HN/N.$$

Proof. We first show that HN is a subgroup. Clearly $e \in HN$.

Let $h_1n_1, h_2n_2 \in HN$. Then

$$(h_1n_1)(h_2n_2)^{-1} = h_1n_1n_2^{-1}h_2^{-1}.$$

Rearranging, we obtain

$$(h_1n_1)(h_2n_2)^{-1} = (h_1h_2^{-1})(h_2n_1n_2^{-1}h_2^{-1}).$$

Since H is a subgroup, $h_1h_2^{-1} \in H$. Since N is normal in G , the second factor belongs to N . Hence the product belongs to HN , proving that $HN \leq G$.

Since $N \trianglelefteq G$ and $N \leq HN$, we have $N \trianglelefteq HN$.

Also, $H \cap N$ is a subgroup of H . For $h \in H$ and $x \in H \cap N$, the element $h x h^{-1}$ belongs to both H and N . Hence $H \cap N \trianglelefteq H$.

Now define $\varphi : H \rightarrow HN/N$ by

$$\varphi(h) = hN.$$

For $h_1, h_2 \in H$,

$$\varphi(h_1h_2) = (h_1h_2)N = (h_1N)(h_2N) = \varphi(h_1)\varphi(h_2).$$

Thus φ is a homomorphism.

Every element of HN/N has the form $(hn)N = hN$ for some $h \in H$ and $n \in N$, so φ is surjective.

Finally,

$$\ker \varphi = \{h \in H : hN = N\} = H \cap N.$$

By the First Isomorphism Theorem,

$$H/(H \cap N) \cong HN/N.$$

■

Example 2.4.6. Let $G = \mathbb{Z}$, $H = 4\mathbb{Z}$, and $N = 6\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, N is normal.

We have

$$H + N = 4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}$$

and

$$H \cap N = 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}.$$

The Second Isomorphism Theorem gives

$$4\mathbb{Z}/12\mathbb{Z} \cong 2\mathbb{Z}/6\mathbb{Z}.$$

Both quotient groups are cyclic of order 3.

2.4.3 Third Isomorphism Theorem

The Third Isomorphism Theorem explains how successive quotient constructions can be simplified.

Theorem 2.4.7 (Third Isomorphism Theorem). *Let N and K be normal subgroups of G such that $N \subseteq K$. Then K/N is a normal subgroup of G/N , and*

$$(G/N)/(K/N) \cong G/K.$$

Proof. Define $\psi : G/N \rightarrow G/K$ by

$$\psi(gN) = gK.$$

This mapping is well-defined. Indeed, if $gN = hN$, then $h^{-1}g \in N \subseteq K$, so $gK = hK$.
For $gN, hN \in G/N$,

$$\begin{aligned}\psi((gN)(hN)) &= \psi(ghN) \\ &= ghK \\ &= (gK)(hK) \\ &= \psi(gN)\psi(hN).\end{aligned}$$

Thus ψ is a homomorphism. It is surjective because every coset gK is the image of gN .
Its kernel is

$$\ker \psi = \{gN : gK = K\} = \{gN : g \in K\} = K/N.$$

Thus K/N is normal in G/N , and the First Isomorphism Theorem gives

$$(G/N)/(K/N) \cong G/K.$$

■

Example 2.4.8. Consider the normal subgroups

$$12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}.$$

The Third Isomorphism Theorem gives

$$(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}.$$

Thus taking the quotient by $12\mathbb{Z}$ and then by $4\mathbb{Z}/12\mathbb{Z}$ has the same effect, up to isomorphism, as taking the quotient by $4\mathbb{Z}$ directly.

2.5 Products of Subgroups, Centralizers and the Centre

The structure of a group can also be investigated through products of subgroups and through elements that commute with other elements. These ideas lead naturally to centralizers, the centre, and normalizers.

2.5.1 Products of Subgroups

Definition 2.5.1 (Product of Subgroups). Let H and K be subgroups of G . Their *set product* is

$$HK = \{hk : h \in H, k \in K\}.$$

Although H and K are subgroups, their set product need not be a subgroup. The precise condition is given below.

Theorem 2.5.2. Let $H, K \leq G$. Then HK is a subgroup of G if and only if

$$HK = KH.$$

Proof. Suppose HK is a subgroup. Then it is closed under inverses, so

$$HK = (HK)^{-1} = K^{-1}H^{-1} = KH,$$

where $H^{-1} = H$ and $K^{-1} = K$ because both are subgroups.

Conversely, suppose $HK = KH$. Clearly $e \in HK$.

Let $h_1k_1, h_2k_2 \in HK$. Since $k_1h_2 \in KH = HK$, there exist $h_3 \in H$ and $k_3 \in K$ such that $k_1h_2 = h_3k_3$. Hence

$$(h_1k_1)(h_2k_2) = h_1h_3k_3k_2 \in HK.$$

Moreover, if $hk \in HK$, then

$$(hk)^{-1} = k^{-1}h^{-1} \in KH = HK.$$

Thus HK contains the identity and is closed under multiplication and inverses. Therefore it is a subgroup. ■

Corollary 2.5.3. If $H, K \leq G$ and either H or K is normal in G , then HK is a subgroup of G .

Proof. Suppose $H \trianglelefteq G$. Then $gH = Hg$ for every $g \in G$, and in particular $kH = Hk$ for every $k \in K$.

Consequently, $KH = HK$, so the preceding theorem shows that HK is a subgroup. The case in which K is normal is analogous. ■

Proposition 2.5.4. If H and K are finite subgroups of G , then

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Consider the surjective mapping $f : H \times K \rightarrow HK$ defined by $f(h, k) = hk$.

For $x \in HK$, choose $h_0 \in H$ and $k_0 \in K$ such that $x = h_0k_0$. The equality $hk = x$ holds precisely when

$$h_0^{-1}h = k_0k^{-1} \in H \cap K.$$

Thus the pairs mapping to x are exactly

$$(h_0t, t^{-1}k_0), \quad t \in H \cap K.$$

Therefore every element of HK has exactly $|H \cap K|$ preimages under f .

Since $|H \times K| = |H||K|$, counting these preimages gives

$$|H||K| = |HK| |H \cap K|,$$

which proves the formula. ■

2.5.2 Centralizers

Definition 2.5.5 (Centralizer of an Element). Let $a \in G$. The *centralizer* of a in G is

$$C_G(a) = \{g \in G : ga = ag\}.$$

Thus $C_G(a)$ consists of all elements of G that commute with a .

Theorem 2.5.6. For every $a \in G$, the centralizer $C_G(a)$ is a subgroup of G .

Proof. Since $ea = ae = a$, the identity belongs to $C_G(a)$.

Let $g, h \in C_G(a)$. Then $ga = ag$ and $ha = ah$. The second equality implies $h^{-1}a = ah^{-1}$. Conse-

quently,

$$(gh^{-1})a = g(h^{-1}a) = g(ah^{-1}) = (ga)h^{-1} = a(gh^{-1}).$$

Thus $gh^{-1} \in C_G(a)$, and the subgroup test shows that $C_G(a) \leq G$. ■

Example 2.5.7. In S_3 , consider the transposition $a = (12)$. The only permutations commuting with (12) are e and (12) . Therefore

$$C_{S_3}((12)) = \{e, (12)\}.$$

2.5.3 The Centre of a Group

Definition 2.5.8 (Centre of a Group). The *centre* of a group G is the subset

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

The centre consists of all elements that commute with every element of the group.

Theorem 2.5.9. The centre $Z(G)$ is a normal subgroup of G .

Proof. The identity commutes with every element, so $e \in Z(G)$.

Let $a, b \in Z(G)$. For any $g \in G$, both a and b^{-1} commute with g , so

$$(ab^{-1})g = a(b^{-1}g) = a(gb^{-1}) = (ag)b^{-1} = g(ab^{-1}).$$

Thus $ab^{-1} \in Z(G)$. By the subgroup test, $Z(G) \leq G$.

Now let $z \in Z(G)$ and $g \in G$. Since $zg = gz$,

$$gzg^{-1} = zgg^{-1} = z \in Z(G).$$

Hence $Z(G) \trianglelefteq G$. ■

Proposition 2.5.10. For every group G ,

$$Z(G) = \bigcap_{a \in G} C_G(a).$$

Proof. An element $z \in G$ belongs to the intersection on the right if and only if it belongs to $C_G(a)$ for every $a \in G$. This is equivalent to $za = az$ for every $a \in G$, which is precisely the condition that $z \in Z(G)$. ■

Proposition 2.5.11. A group G is abelian if and only if $Z(G) = G$.

Proof. If G is abelian, every element commutes with every other element, so every element belongs to the centre and $Z(G) = G$.

Conversely, if $Z(G) = G$, every element of G commutes with every other element. Hence G is abelian. ■

Example 2.5.12. The centre of S_3 is $\{e\}$. Indeed, no nonidentity permutation in S_3 commutes with every element.

In contrast, if G is abelian, then $Z(G) = G$.

Example 2.5.13 (Centre of a General Linear Group). For $n \geq 2$, the centre of $GL_n(\mathbb{R})$ is

$$Z(GL_n(\mathbb{R})) = \{\lambda I_n : \lambda \in \mathbb{R}^\times\}.$$

Every nonzero scalar matrix commutes with all matrices.

Conversely, let $A \in Z(GL_n(\mathbb{R}))$. Then A commutes with every elementary matrix $I_n + E_{ij}$, where $i \neq j$ and E_{ij} has a single nonzero entry 1 in position (i, j) .

Hence $AE_{ij} = E_{ij}A$ for all $i \neq j$. Comparing matrix entries shows that all off-diagonal entries of A vanish and all diagonal entries are equal. Therefore $A = \lambda I_n$. Since A is invertible, $\lambda \neq 0$.

2.5.4 Normalizers

Definition 2.5.14 (Normalizer of a Subgroup). Let H be a subgroup of G . The *normalizer* of H in G is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

The normalizer consists of those elements of G whose conjugation preserves H as a set.

Theorem 2.5.15. Let $H \leq G$. Then $N_G(H)$ is a subgroup of G containing H , and

$$H \trianglelefteq N_G(H).$$

Moreover, $N_G(H)$ is the largest subgroup of G in which H is normal.

Proof. Since $eHe^{-1} = H$, we have $e \in N_G(H)$.

Let $g, k \in N_G(H)$. Since $gHg^{-1} = H$ and $kHk^{-1} = H$, the latter equality also gives $k^{-1}Hk = H$. Therefore

$$(gk^{-1})H(gk^{-1})^{-1} = g(k^{-1}Hk)g^{-1} = gHg^{-1} = H.$$

Thus $gk^{-1} \in N_G(H)$, so $N_G(H)$ is a subgroup.

If $h \in H$, then $hHh^{-1} = H$, since H is itself a group. Hence $H \subseteq N_G(H)$.

By the definition of the normalizer, every element of $N_G(H)$ conjugates H onto itself. Therefore $H \trianglelefteq N_G(H)$.

Finally, suppose $L \leq G$ and $H \trianglelefteq L$. Then for every $l \in L$, we have $lHl^{-1} = H$, so $l \in N_G(H)$. Hence $L \subseteq N_G(H)$.

Thus $N_G(H)$ is the largest subgroup of G in which H is normal. ■

Corollary 2.5.16. For a subgroup H of G ,

$$H \trianglelefteq G \iff N_G(H) = G.$$

Proof. If $H \trianglelefteq G$, then $gHg^{-1} = H$ for all $g \in G$, so every element belongs to $N_G(H)$.

Conversely, if $N_G(H) = G$, then every $g \in G$ satisfies $gHg^{-1} = H$, proving that H is normal. ■

Example 2.5.17. Let $G = S_3$ and $H = \{e, (12)\}$. Since H has order 2, it is normal in itself. However, it is not normal in S_3 .

The only elements of S_3 that conjugate (12) to itself are e and (12) . Therefore

$$N_{S_3}(H) = H.$$

Thus H is its own normalizer in S_3 .

Remark 2.5.18. The centralizer and normalizer are related but distinct concepts. The centralizer of an element consists of elements commuting with it, while the normalizer of a subgroup consists of elements whose conjugation preserves that subgroup. The centre is the intersection of all element centralizers, and a subgroup is normal in G precisely when its normalizer is all of G .

2.6 Permutation Groups

Permutation groups provide an important class of examples in group theory. They arise naturally in the study of symmetries, arrangements, and transformations of finite sets. More importantly, Cayley's theorem shows that every abstract group can be represented as a group of permutations.

Definition 2.6.1 (Permutation). Let X be a nonempty set. A *permutation* of X is a bijective mapping from X onto itself.

The set of all permutations of X , denoted by $\text{Sym}(X)$, forms a group under composition of mappings. Indeed, the composition of two bijections is a bijection, composition is associative, the identity mapping is an identity element, and every bijection possesses an inverse.

When $X = \{1, 2, \dots, n\}$, this group is called the *symmetric group of degree n* , denoted by S_n .

Since the images of $1, 2, \dots, n$ must be distinct, there are $n!$ possible permutations. Consequently,

$$|S_n| = n!.$$

2.6.1 Representation of Permutations

A permutation $\sigma \in S_n$ may be represented by listing the images of the elements of $\{1, 2, \dots, n\}$.

For example,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$$

means that $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, and $\sigma(4) = 2$.

This representation is called *two-line notation*. Although convenient for describing an arbitrary permutation, it is often more useful to express permutations as products of cycles.

Definition 2.6.2 (Cycle). Let $a_1, a_2, \dots, a_r$ be distinct elements of $\{1, 2, \dots, n\}$. The permutation

$$(a_1 \ a_2 \ \cdots \ a_r)$$

that sends a_1 to a_2 , a_2 to a_3 , and so on, sends a_r to a_1 , and fixes every other element is called a *cycle of length r* , or an *r -cycle*.

A cycle of length two is called a *transposition*. A cycle of length one fixes its only entry and may be omitted from the cycle notation.

Example 2.6.3. The cycle $(1 \ 3 \ 5 \ 2)$ sends

$$1 \mapsto 3, \quad 3 \mapsto 5, \quad 5 \mapsto 2, \quad 2 \mapsto 1,$$

and fixes every other element.

The cycles $(1 \ 3 \ 5 \ 2)$, $(3 \ 5 \ 2 \ 1)$, and $(5 \ 2 \ 1 \ 3)$ represent the same permutation, since each describes the same cyclic movement of elements.

Throughout this chapter, products of permutations are evaluated from right to left. Thus, in the product $\sigma\tau$, the permutation τ is applied first, followed by σ .

Example 2.6.4 (Composition of Permutations). In S_3 , let $\sigma = (12)$ and $\tau = (13)$. Applying τ first and then σ , we obtain

$$\sigma\tau = (132).$$

On the other hand,

$$\tau\sigma = (123).$$

Thus $\sigma\tau \neq \tau\sigma$, proving that S_3 is nonabelian.

2.6.2 Disjoint Cycles

Definition 2.6.5 (Disjoint Cycles). Two cycles are said to be *disjoint* if they have no symbols in common.

For example, $(1\ 4\ 3)$ and $(2\ 5)$ are disjoint, whereas $(1\ 2\ 3)$ and $(3\ 4)$ are not.

A fundamental property of permutations is that every permutation of a finite set can be expressed as a product of disjoint cycles.

Theorem 2.6.6 (Disjoint-Cycle Decomposition). *Every permutation of a finite set can be expressed as a product of pairwise disjoint cycles. This decomposition is unique up to the order of the cycles and cyclic rearrangement of the entries within each cycle, with cycles of length one omitted if desired.*

Proof. Let σ be a permutation of the finite set X .

Choose an element $a \in X$ and consider the sequence

$$a, \sigma(a), \sigma^2(a), \dots$$

Since X is finite, some two terms must be equal. Because σ is bijective, applying a suitable inverse power shows that there exists a positive integer r such that $\sigma^r(a) = a$.

Choose r to be the least such positive integer. Then

$$a, \sigma(a), \dots, \sigma^{r-1}(a)$$

are distinct and form the cycle

$$(a\ \sigma(a)\ \dots\ \sigma^{r-1}(a)).$$

If these elements exhaust X , the required decomposition is obtained. Otherwise, choose an element b not occurring in this cycle and repeat the construction.

The new cycle is disjoint from the first. Indeed, if an iterate of b belonged to the first cycle, applying suitable powers of σ and σ^{-1} would imply that b itself belonged to the first cycle, a contradiction.

Continuing in this way, and using the finiteness of X , we eventually obtain a collection of pairwise disjoint cycles whose union contains every element of X . Their product agrees with σ on all elements of X .

For uniqueness, observe that the set

$$\{\sigma^k(a) : k \in \mathbb{Z}\}$$

is uniquely determined by σ and a . Thus the elements occurring in each cycle are uniquely determined. Their cyclic order is determined by the action of σ , up to the choice of the first entry. Hence the decomposition is unique in the stated sense. ■

Example 2.6.7 (Expressing a Permutation as Disjoint Cycles). Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 1 & 6 & 2 & 4 \end{pmatrix}.$$

Starting with 1, we have $1 \mapsto 3 \mapsto 1$, giving the cycle (13).

Next, $2 \mapsto 5 \mapsto 2$, giving (25). Finally, $4 \mapsto 6 \mapsto 4$, giving (46).

Therefore

$$\sigma = (13)(25)(46).$$

Since these cycles are disjoint transpositions, $\sigma^2 = e$, and hence $\text{ord}(\sigma) = 2$.

Proposition 2.6.8. Disjoint cycles commute.

Proof. Let α and β be disjoint cycles, and let A and B denote the sets of symbols moved by them. If $x \in A$, then $\beta(x) = x$ and $\alpha(x) \in A$. Since β fixes every element of A ,

$$(\alpha\beta)(x) = \alpha(x) = (\beta\alpha)(x).$$

Similarly, if $x \in B$, then both products send x to $\beta(x)$.

If $x \notin A \cup B$, both cycles fix x . Thus $\alpha\beta$ and $\beta\alpha$ agree on every element, proving that $\alpha\beta = \beta\alpha$. ■

2.6.3 Order of a Permutation

The disjoint-cycle decomposition provides a convenient method for determining the order of a permutation.

Theorem 2.6.9 (Order of a Permutation). Let $\sigma \in S_n$ be the product of pairwise disjoint cycles of lengths $r_1, r_2, \dots, r_t$. Then

$$\text{ord}(\sigma) = \text{lcm}(r_1, r_2, \dots, r_t).$$

Proof. Write $\sigma = \sigma_1 \sigma_2 \cdots \sigma_t$, where the σ_i are pairwise disjoint cycles of lengths r_i .

Since disjoint cycles commute,

$$\sigma^m = \sigma_1^m \sigma_2^m \cdots \sigma_t^m.$$

Because the cycles act on disjoint sets, this product is the identity precisely when every σ_i^m is the identity.

An r_i -cycle has order r_i , so $\sigma_i^m = e$ if and only if $r_i \mid m$.

Therefore $\sigma^m = e$ precisely when m is divisible by every r_i . The least positive such integer is their least common multiple, proving the result. ■

Example 2.6.10. Find the order of

$$\tau = (1\ 4\ 3)(2\ 6\ 5\ 7)$$

in S_7 .

The two cycles are disjoint and have lengths 3 and 4. Hence

$$\text{ord}(\tau) = \text{lcm}(3, 4) = 12.$$

Example 2.6.11. Consider $\sigma = (123)(45)(6789)$ in S_9 . Its disjoint cycles have lengths 3, 2, and 4. Consequently,

$$\text{ord}(\sigma) = \text{lcm}(3, 2, 4) = 12.$$

2.6.4 Transpositions and Parity of Permutations

Every permutation can be expressed as a product of transpositions. This observation leads to the classification of permutations as even or odd.

Theorem 2.6.12. Every permutation in S_n can be expressed as a product of transpositions.

Proof. By the disjoint-cycle decomposition theorem, every permutation is a product of cycles. It is therefore sufficient to show that every cycle is a product of transpositions.

Let $\sigma = (a_1 a_2 \cdots a_r)$. Then

$$\sigma = (a_1 a_r)(a_1 a_{r-1}) \cdots (a_1 a_2).$$

Indeed, applying the transpositions from right to left sends a_i to a_{i+1} for $1 \leq i < r$, sends a_r to a_1 , and fixes all other elements.

Thus every cycle, and hence every permutation, is a product of transpositions. ■

Example 2.6.13. The cycle $(1\ 2\ 3\ 4)$ can be written as

$$(1\ 2\ 3\ 4) = (14)(13)(12).$$

Similarly, $(1\ 2\ 3) = (13)(12)$.

Definition 2.6.14 (Even and Odd Permutations). A permutation is called *even* if it can be expressed as a product of an even number of transpositions, and *odd* if it can be expressed as a product of an odd number of transpositions.

This definition requires justification: a permutation may admit many different expressions as products of transpositions. We must show that their parity is always the same.

Theorem 2.6.15 (Parity Theorem). *If a permutation is expressed as a product of transpositions in two different ways, then the numbers of transpositions occurring in the two expressions have the same parity.*

Proof. For $\sigma \in S_n$, define its inversion number by

$$I(\sigma) = |\{(i, j) : 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}|.$$

We show that composing a permutation with a transposition changes the parity of its inversion number.

First consider a transposition that interchanges two adjacent positions in the one-line representation of σ . Only the relative order of the two interchanged entries changes; their combined contribution to inversions with any other entry remains unchanged. Thus the inversion number changes by exactly 1 or -1 , and hence its parity is reversed.

Every transposition can be expressed as a product of an odd number of adjacent transpositions. Indeed, interchanging positions i and j , where $i < j$, can be accomplished using $2(j - i) - 1$ adjacent transpositions.

Consequently, multiplication by any transposition reverses the parity of the inversion number.

The identity permutation has inversion number 0. Therefore, if σ is a product of r transpositions, then

$$I(\sigma) \equiv r \pmod{2}.$$

Hence the parity of r depends only on σ and not on its representation as a product of transpositions. This proves the theorem. ■

Definition 2.6.16 (Sign of a Permutation). The *sign* of a permutation σ is

$$\text{sgn}(\sigma) = (-1)^{I(\sigma)}.$$

Thus $\text{sgn}(\sigma) = 1$ for an even permutation and $\text{sgn}(\sigma) = -1$ for an odd permutation.

Proposition 2.6.17. For $\sigma, \tau \in S_n$,

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau).$$

Proof. Suppose σ and τ are products of r and s transpositions, respectively. Then $\sigma\tau$ is a product of $r + s$ transpositions.

By the parity theorem,

$$\operatorname{sgn}(\sigma\tau) = (-1)^{r+s} = (-1)^r (-1)^s = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau).$$

■

Corollary 2.6.18. An r -cycle has sign $(-1)^{r-1}$.

Proof. Every r -cycle is a product of $r - 1$ transpositions. The result follows from the definition of sign. ■

2.6.5 The Alternating Group

Definition 2.6.19 (Alternating Group). The set of all even permutations in S_n is called the *alternating group of degree n* , denoted by A_n .

Theorem 2.6.20. For every $n \geq 2$, A_n is a normal subgroup of S_n , and

$$|A_n| = \frac{n!}{2}.$$

Proof. The sign mapping

$$\operatorname{sgn} : S_n \longrightarrow \{1, -1\}$$

is a group homomorphism. It is surjective for $n \geq 2$, since the identity has sign 1 and every transposition has sign -1 .

Its kernel is precisely the set of even permutations:

$$\ker(\operatorname{sgn}) = A_n.$$

Since the kernel of a homomorphism is a normal subgroup,

$$A_n \trianglelefteq S_n.$$

By the First Isomorphism Theorem,

$$S_n/A_n \cong \{1, -1\}.$$

Thus $[S_n : A_n] = 2$. Since $|S_n| = n!$, Lagrange's theorem gives $|A_n| = n!/2$. ■

Example 2.6.21. The alternating group A_3 consists of

$$A_3 = \{e, (123), (132)\}.$$

It is cyclic of order 3, generated by (123) .

2.6.6 Cayley's Theorem

Cayley's theorem establishes a fundamental connection between abstract groups and permutation groups. It shows that every group can be represented faithfully as permutations of an appropriate set.

Theorem 2.6.22 (Cayley's Theorem). *Every group G is isomorphic to a subgroup of the permutation group $\text{Sym}(G)$.*

In particular, every finite group of order n is isomorphic to a subgroup of S_n .

Proof. For each $g \in G$, define a mapping $L_g : G \rightarrow G$ by

$$L_g(x) = gx.$$

This is called left translation by g .

Since

$$L_{g^{-1}}(L_g(x)) = g^{-1}(gx) = x,$$

and similarly $L_g(L_{g^{-1}}(x)) = x$, the mapping L_g is bijective, with inverse $L_{g^{-1}}$. Thus $L_g \in \text{Sym}(G)$.

Now define $\Phi : G \rightarrow \text{Sym}(G)$ by $\Phi(g) = L_g$.

For $g, h, x \in G$,

$$(L_g \circ L_h)(x) = g(hx) = (gh)x = L_{gh}(x).$$

Thus

$$\Phi(gh) = L_{gh} = L_g \circ L_h = \Phi(g)\Phi(h).$$

Hence Φ is a homomorphism.

To prove injectivity, suppose $\Phi(g) = \Phi(h)$. Then $L_g = L_h$. Evaluating both maps at the identity e gives

$$g = L_g(e) = L_h(e) = h.$$

Therefore Φ is injective.

Consequently, G is isomorphic to its image $\Phi(G)$, which is a subgroup of $\text{Sym}(G)$.

If $|G| = n$, then $\text{Sym}(G) \cong S_n$, and the final assertion follows. ■

Example 2.6.23 (Cayley's Representation of $\mathbb{Z}_3$). Let $G = \mathbb{Z}_3 = \{[0], [1], [2]\}$ under addition.

Left translation by $[1]$ sends

$$[0] \mapsto [1], \quad [1] \mapsto [2], \quad [2] \mapsto [0].$$

Labeling $[0], [1], [2]$ by $1, 2, 3$, respectively, this translation corresponds to the cycle (123) .

Similarly, translation by $[2]$ corresponds to (132) , while translation by $[0]$ is the identity.

Thus Cayley's theorem realizes $\mathbb{Z}_3$ as the subgroup

$$\{e, (123), (132)\} \leq S_3.$$

2.7 The Alternating Group A_4

The alternating group A_4 provides an important example in finite group theory. It consists of the even permutations of four symbols and has order

$$|A_4| = \frac{4!}{2} = 12.$$

Its elements are the identity, eight 3-cycles, and three products of two disjoint transpositions. More explicitly,

$$\begin{aligned} A_4 = \{ & e, (123), (132), (124), (142), \\ & (134), (143), (234), (243), \\ & (12)(34), (13)(24), (14)(23) \}. \end{aligned}$$

The group A_4 is particularly useful for showing that the converse of Lagrange's theorem does not hold in general.

Theorem 2.7.1. *The alternating group A_4 has no subgroup of order 6.*

Proof. Suppose, to the contrary, that H is a subgroup of A_4 with $|H| = 6$. Since $|A_4| = 12$, Lagrange's theorem gives

$$[A_4 : H] = \frac{12}{6} = 2.$$

Every subgroup of index two is normal, so $H \trianglelefteq A_4$.

The quotient group A_4/H therefore exists and has order 2. Every element of a group of order 2 has square equal to the identity. Consequently, for every $g \in A_4$,

$$(gH)^2 = H,$$

which implies $g^2H = H$, and hence $g^2 \in H$.

Now consider the eight 3-cycles in A_4 . The square of a 3-cycle is its inverse, which is again a 3-cycle. As the squaring operation permutes these eight 3-cycles, the condition $g^2 \in H$ forces all eight 3-cycles to belong to H .

But H has only six elements, giving a contradiction.

Therefore A_4 has no subgroup of order 6. ■

Remark 2.7.2. Lagrange's theorem states that the order of a subgroup divides the order of the group. Its converse is false: although 6 divides 12, the group A_4 has no subgroup of order 6.

2.8 Dihedral Groups

Dihedral groups arise from the rigid symmetries of regular polygons. They provide important geometric examples of nonabelian groups and illustrate the interaction between rotations and reflections.

Definition 2.8.1 (Dihedral Group). For $n \geq 3$, the *dihedral group* D_n is the group of all rigid symmetries of a regular n -gon under composition.

It consists of n rotations and n reflections, so

$$|D_n| = 2n.$$

Let r denote counterclockwise rotation through an angle $2\pi/n$, and let s denote reflection in a fixed symmetry axis.

The elements r and s satisfy

$$r^n = e, \quad s^2 = e, \quad srs = r^{-1}.$$

Equivalently, $sr = r^{-1}s$.

Every symmetry can be written uniquely in one of the forms r^k or sr^k , where $0 \leq k < n$. Thus

$$D_n = \{e, r, \dots, r^{n-1}, s, sr, \dots, sr^{n-1}\}.$$

These relations are summarized by the presentation

$$D_n = \langle r, s \mid r^n = e, s^2 = e, srs = r^{-1} \rangle.$$

Remark 2.8.2. We use D_n for the dihedral group of order $2n$. Some textbooks use the notation D_{2n} for the same group.

2.8.1 Algebraic Properties of Dihedral Groups

The relation $sr = r^{-1}s$ is fundamental for simplifying products in a dihedral group.

Proposition 2.8.3. In D_n , for all integers i, j ,

- (i) $sr^i = r^{-i}s$;
- (ii) $(sr^i)(sr^j) = r^{j-i}$;
- (iii) $(sr^i)^2 = e$.

Proof. The defining relation $sr = r^{-1}s$ implies, by induction and the inverse laws, that $sr^i = r^{-i}s$ for every integer i .

Consequently,

$$(sr^i)(sr^j) = sr^i sr^j = r^{-i} sr^j = r^{j-i}.$$

Taking $j = i$ gives

$$(sr^i)^2 = r^{i-i} = e.$$

Thus every reflection has order 2. ■

Proposition 2.8.4. For every $n \geq 3$, the dihedral group D_n is nonabelian.

Proof. From the defining relations, $sr = r^{-1}s$. If D_n were abelian, we would also have $sr = rs$. Therefore $r^{-1}s = rs$, and right cancellation would give $r^{-1} = r$, so $r^2 = e$. But r has order $n \geq 3$, a contradiction. Thus D_n is nonabelian. ■

2.8.2 The Dihedral Group D_3

The group D_3 is the symmetry group of an equilateral triangle. It has six elements: three rotations and three reflections.

Let r be rotation through 120° and s a reflection. Then

$$D_3 = \{e, r, r^2, s, sr, sr^2\},$$

with defining relations

$$r^3 = e, \quad s^2 = e, \quad srs = r^{-1}.$$

Theorem 2.8.5. The dihedral group D_3 is isomorphic to S_3 .

Proof. Label the vertices of an equilateral triangle by 1, 2, 3. Every symmetry permutes these vertices, giving a mapping

$$\varphi : D_3 \rightarrow S_3.$$

Since the action of a composition of symmetries on the vertices is the composition of their corresponding permutations, φ is a homomorphism.

If a symmetry fixes all three vertices, it must be the identity symmetry. Hence $\ker \varphi = \{e\}$, so φ is injective.

Since $|D_3| = 6 = |S_3|$, an injective mapping between these finite sets is also surjective. Therefore φ is an isomorphism, and $D_3 \cong S_3$. ■

2.8.3 The Dihedral Group D_4

The group D_4 is the symmetry group of a square. It has eight elements:

$$D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\},$$

where r denotes rotation through 90° and s is a reflection.

The defining relations are

$$r^4 = e, \quad s^2 = e, \quad srs = r^{-1}.$$

Example 2.8.6 (Multiplication in D_4). Compute $(sr^2)(sr^3)$.

Using the relation $sr^i s = r^{-i}$, we obtain

$$(sr^2)(sr^3) = sr^2 sr^3 = r^{-2} r^3 = r.$$

Thus the composition of these two reflections is rotation through 90° .

Example 2.8.7. Find the order of sr^3 in D_4 .

Since sr^3 is a reflection, the preceding proposition gives

$$(sr^3)^2 = e.$$

As $sr^3 \neq e$, its order is 2.

2.8.4 The Rotation Subgroup

The rotations of a regular polygon form a cyclic subgroup of the corresponding dihedral group.

Theorem 2.8.8. Let D_n be the dihedral group of order $2n$, with rotation generator r . Then

$$\langle r \rangle \trianglelefteq D_n$$

and

$$D_n / \langle r \rangle \cong \mathbb{Z}_2.$$

Proof. The subgroup

$$\langle r \rangle = \{e, r, \dots, r^{n-1}\}$$

has order n .

Since $|D_n| = 2n$, its index is

$$[D_n : \langle r \rangle] = \frac{2n}{n} = 2.$$

Every subgroup of index two is normal, so $\langle r \rangle \trianglelefteq D_n$.

The quotient group has exactly two elements,

$$D_n / \langle r \rangle = \{\langle r \rangle, s\langle r \rangle\}.$$

Every group of order two is cyclic and isomorphic to $\mathbb{Z}_2$. Hence

$$D_n / \langle r \rangle \cong \mathbb{Z}_2.$$

Remark 2.8.9. The rotation subgroup consists of all orientation-preserving symmetries of the polygon, while the remaining coset consists of reflections, which reverse orientation. Thus the quotient $D_n/\langle r \rangle$ distinguishes the two types of symmetries and records whether a symmetry preserves or reverses orientation.

2.9 Exercises

The following exercises are arranged to progress from routine computations to proofs. Unless stated otherwise, all groups are written multiplicatively.

1. Show directly that every subgroup of $(\mathbb{Z}, +)$ is normal.
2. Determine whether $\{e, (123), (132)\}$ is normal in S_3 .
3. Prove that the intersection of any family of normal subgroups is normal.
4. Give an example of a subgroup of S_3 that is not normal, and exhibit an explicit conjugate outside it.
5. List all cosets of $3\mathbb{Z}$ in $\mathbb{Z}$ and construct the addition table of $\mathbb{Z}/3\mathbb{Z}$.
6. Find the order of $(\mathbb{Z}/18\mathbb{Z})/\langle \bar{6} \rangle$.
7. Prove that a quotient group G/N is abelian if and only if $aba^{-1}b^{-1} \in N$ for every $a, b \in G$.
8. Determine the kernel and image of $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(k) = 5k$.
9. Determine the kernel and image of $f : \mathbb{Z} \rightarrow \mathbb{Z}/15\mathbb{Z}, f(k) = \bar{6}k$.
10. Show that a homomorphism carries inverses to inverses and that its kernel is trivial exactly when it is injective.
11. Use the first isomorphism theorem to identify $\mathbb{Z}/8\mathbb{Z}$ as the image of a homomorphism from $\mathbb{Z}$.
12. Prove that $\text{SL}_2(\mathbb{R})$ is normal in $\text{GL}_2(\mathbb{R})$.
13. Apply the third isomorphism theorem to $(\mathbb{Z}/24\mathbb{Z})/(6\mathbb{Z}/24\mathbb{Z})$.
14. Prove that HK is a subgroup whenever $H \trianglelefteq G$ and $K \leq G$.
15. Compute $C_{S_3}((123))$ and compare it with $Z(S_3)$.
16. Determine the normalizer of $\langle (12) \rangle$ in S_3 .
17. Write $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 5 & 3 \end{pmatrix}$ in disjoint cycles and find its order.
18. Express (12345) as a product of transpositions and determine its parity.
19. Find the order and sign of $(135)(2467)$ as a permutation of seven symbols.
20. Prove that disjoint cycles commute, and give two non-disjoint cycles that do not commute.
21. Use Cayley's construction to embed $\mathbb{Z}/4\mathbb{Z}$ into S_4 .
22. List the elements of A_4 by their cycle types and verify that it has order 12.
23. Explain why A_4 disproves the converse of Lagrange's theorem.

24. In D_4 , simplify sr^3s , r^2sr , and $(sr)^2$.
25. Prove that the centre of D_3 is trivial and that r^2 belongs to the centre of D_4 .
26. Show that D_n has exactly n rotations and n reflections.

Chapter 3

Rings, Homomorphisms and Ideals

In the preceding chapters, we studied groups, which are algebraic structures equipped with a single binary operation. We now turn to rings, in which two binary operations are connected by the distributive laws.

The ring of integers is the most familiar example. Other important examples include polynomial rings, matrix rings, and rings of integers modulo a positive integer. Although these structures differ in their elements and operations, they share many fundamental algebraic properties.

Ring theory investigates these common properties and the ways in which they differ from ordinary integer arithmetic. In particular, the existence of zero divisors, the possibility of multiplicative inverses, and the role of commutativity lead to several important classes of rings.

An especially significant concept in ring theory is that of an *ideal*. Ideals provide the appropriate framework for constructing quotient rings, just as normal subgroups permit the construction of quotient groups. They also play a central role in understanding ring homomorphisms and the structure of commutative rings.

3.1 Rings: Definitions and First Examples

Definition 3.1.1 (Ring). A *ring* is a nonempty set R equipped with two binary operations, called *addition* and *multiplication*, satisfying the following conditions:

(R1) $(R, +)$ is an abelian group.

(R2) Multiplication is associative; that is,

$$(ab)c = a(bc)$$

for all $a, b, c \in R$.

(R3) Multiplication is distributive over addition on both sides; that is,

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc$$

for all $a, b, c \in R$.

Thus, in a ring, addition is associative and commutative, there is an additive identity 0 , and every element a has an additive inverse $-a$. Multiplication is associative, but it need not be commutative, and multiplicative inverses need not exist.

The additive identity and additive inverses are unique because $(R, +)$ is a group.

Definition 3.1.2. A ring R is called *commutative* if $ab = ba$ for all $a, b \in R$.

A ring R is said to have a *multiplicative identity*, or *unity*, if there exists an element $1_R \in R$ such that

$$1_R a = a 1_R = a$$

for every $a \in R$.

Unless otherwise stated, rings in the initial sections of this chapter are not assumed to be commutative or to possess a multiplicative identity. In the study of principal, prime, and maximal ideals, the relevant hypotheses will be stated explicitly.

It is important to distinguish the additive identity 0 from a multiplicative identity 1_R . Every ring has an additive identity, whereas not every ring has a multiplicative identity.

Example 3.1.3 (Standard Examples of Rings). (i) The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are commutative rings with identity under ordinary addition and multiplication.

(ii) For $n \geq 2$, the set $\mathbb{Z}_n$ of residue classes modulo n is a commutative ring with identity $[1]$, where

$$[a] + [b] = [a + b], \quad [a][b] = [ab].$$

Both operations are well-defined because congruence modulo n is compatible with addition and multiplication.

(iii) The set $M_n(\mathbb{R})$ of all $n \times n$ real matrices is a ring under matrix addition and multiplication. Its multiplicative identity is I_n . For $n \geq 2$, matrix multiplication is not commutative in general, so $M_n(\mathbb{R})$ is a noncommutative ring.

(iv) The set $2\mathbb{Z}$ of even integers forms a commutative ring under ordinary addition and multiplication. It has no multiplicative identity.

(v) The polynomial ring $\mathbb{Z}[x]$ is a commutative ring with identity. Similarly, $\mathbb{Z}[x, y]$ is the ring of polynomials in two commuting indeterminates with integer coefficients.

(vi) The singleton set $\{0\}$ forms a ring under $0 + 0 = 0$ and $0 \cdot 0 = 0$. It is called the *zero ring*.

Remark 3.1.4. If a ring has an identity and $1_R = 0$, then for every $a \in R$,

$$a = 1_R a = 0a = 0.$$

Consequently, $R = \{0\}$. Thus the condition $1_R \neq 0$ excludes the zero ring.

3.1.1 Examples of Specially Defined Operations

A ring need not arise from the usual multiplication of numbers. However, its operations must satisfy all the ring axioms.

Example 3.1.5 (A Ring with Zero Multiplication). Let $R = \mathbb{Z}$ with ordinary addition, and define multiplication by

$$a * b = 0$$

for all $a, b \in \mathbb{Z}$.

Since ordinary addition makes $\mathbb{Z}$ an abelian group, only the multiplicative properties need to be verified.

For $a, b, c \in \mathbb{Z}$,

$$(a * b) * c = 0 = a * (b * c),$$

so multiplication is associative.

Also,

$$a * (b + c) = 0 = a * b + a * c,$$

and similarly

$$(a + b) * c = 0 = a * c + b * c.$$

Thus both distributive laws hold.

Therefore $\mathbb{Z}$, with ordinary addition and zero multiplication, is a ring. It has no multiplicative identity.

Example 3.1.6 (An Operation That Does Not Define a Ring). Let $R = \mathbb{Z}$ with ordinary addition, and define $a * b = a + b$.

Although this operation is associative and commutative, it does not satisfy distributivity. Indeed,

$$a * (b + c) = a + b + c,$$

whereas

$$a * b + a * c = (a + b) + (a + c) = 2a + b + c.$$

These expressions are not equal for arbitrary integers a, b, c .

Hence the given operations do not make $\mathbb{Z}$ into a ring.

3.1.2 Elementary Properties of Rings

The ring axioms imply several identities familiar from elementary arithmetic. These hold in every ring, including noncommutative rings and rings without unity.

Theorem 3.1.7 (Basic Identities in a Ring). *Let R be a ring. For all $a, b, c \in R$, the following identities hold:*

(i) $a0 = 0a = 0$;

(ii) $a(-b) = (-a)b = -(ab)$;

(iii) $(-a)(-b) = ab$;

(iv) $a(b - c) = ab - ac$;

(v) $(a - b)c = ac - bc$.

If R has a multiplicative identity, then $(-1_R)a = a(-1_R) = -a$ for every $a \in R$.

Proof. For (i), distributivity gives

$$a0 = a(0 + 0) = a0 + a0.$$

Adding the additive inverse of $a0$ to both sides yields $a0 = 0$. Similarly,

$$0a = (0 + 0)a = 0a + 0a$$

implies $0a = 0$.

For (ii), observe that

$$ab + a(-b) = a(b + (-b)) = a0 = 0.$$

Thus $a(-b)$ is the additive inverse of ab , giving $a(-b) = -(ab)$.

Similarly,

$$ab + (-a)b = (a + (-a))b = 0b = 0,$$

so $(-a)b = -(ab)$.

Applying (ii) twice gives

$$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab,$$

proving (iii).

For (iv), distributivity and (ii) yield

$$a(b - c) = ab + a(-c) = ab - ac.$$

Similarly,

$$(a - b)c = ac + (-b)c = ac - bc,$$

proving (v).

Finally, suppose R has identity 1_R . Then

$$a + (-1_R)a = (1_R + (-1_R))a = 0a = 0.$$

Hence $(-1_R)a = -a$. The identity $a(-1_R) = -a$ follows analogously. ■

Proposition 3.1.8 (Uniqueness of Unity). If a ring has a multiplicative identity, then it is unique.

Proof. Suppose u and v are multiplicative identities of R . Since u is an identity, $uv = v$. Since v is also an identity, $uv = u$. Therefore $u = v$. ■

Definition 3.1.9 (Idempotent and Nilpotent Elements). Let R be a ring. An element $a \in R$ is called *idempotent* if $a^2 = a$.

An element $a \in R$ is called *nilpotent* if $a^n = 0$ for some positive integer n .

Example 3.1.10. In $\mathbb{Z}_6$, the elements $[3]$ and $[4]$ are nontrivial idempotents because

$$[3]^2 = [3], \quad [4]^2 = [4].$$

In $\mathbb{Z}_8$, the element $[2]$ is nilpotent, since

$$[2]^3 = [0],$$

although $[2] \neq [0]$.

Proposition 3.1.11. Every nonzero nilpotent element of a ring is a zero divisor.

Proof. Let $a \neq 0$ be nilpotent, and let $n \geq 2$ be the least positive integer such that $a^n = 0$. Then $a^{n-1} \neq 0$, but

$$a \cdot a^{n-1} = a^n = 0.$$

Thus a is a zero divisor. ■

3.2 Zero Divisors, Integral Domains and Fields

The existence of multiplicative inverses and zero divisors distinguishes several important classes of rings. In ordinary integer arithmetic, the product of two nonzero integers is nonzero, and cancellation by a nonzero factor is valid. These properties need not hold in an arbitrary ring.

We begin by examining zero divisors and units.

3.2.1 Zero Divisors and Units

Definition 3.2.1 (Zero Divisor). Let R be a ring. A nonzero element $a \in R$ is called a *left zero divisor* if there exists a nonzero element $b \in R$ such that $ab = 0$.

A nonzero element $a \in R$ is called a *right zero divisor* if there exists a nonzero element $b \in R$ such that $ba = 0$.

An element that is either a left or a right zero divisor is called a *zero divisor*.

In a commutative ring, the distinction between left and right zero divisors disappears.

Example 3.2.2 (Zero Divisors in $\mathbb{Z}_6$). In the ring $\mathbb{Z}_6$, we have

$$[2][3] = [6] = [0].$$

Since $[2]$ and $[3]$ are nonzero, both are zero divisors.

Thus, unlike in $\mathbb{Z}$, the product of two nonzero elements of $\mathbb{Z}_6$ can be zero.

Example 3.2.3 (Zero Divisors in a Matrix Ring). Consider the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both are nonzero, but

$$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore A and B are zero divisors in $M_2(\mathbb{R})$.

Definition 3.2.4 (Unit). Let R be a ring with identity $1_R \neq 0$. An element $u \in R$ is called a *unit* if there exists $v \in R$ such that

$$uv = vu = 1_R.$$

The element v is called the *multiplicative inverse* of u , denoted by u^{-1} .

The collection of all units of R is denoted by $U(R)$ or $R^\times$.

Proposition 3.2.5. The set $U(R)$ of units of a ring with identity forms a group under multiplication.

Proof. Let $u, v \in U(R)$. Since u and v possess multiplicative inverses,

$$(uv)(v^{-1}u^{-1}) = u(vv^{-1})u^{-1} = 1_R,$$

and similarly

$$(v^{-1}u^{-1})(uv) = 1_R.$$

Thus uv is a unit, with inverse $v^{-1}u^{-1}$.

Multiplication is associative, 1_R is the identity, and the inverse of every unit is again a unit. Therefore $U(R)$ is a group. ■

Example 3.2.6. The only units of $\mathbb{Z}$ are 1 and -1 , whereas every nonzero element of $\mathbb{Q}$ is a unit.

In $\mathbb{Z}_{10}$, the element $[3]$ is a unit because

$$[3][7] = [21] = [1].$$

Thus $[3]^{-1} = [7]$. By contrast, $[2]$ is a zero divisor because $[2][5] = [0]$.

Proposition 3.2.7. In a nonzero ring with identity, no unit is a left or right zero divisor.

Proof. Let u be a unit with inverse u^{-1} . If $ub = 0$, then multiplying on the left by u^{-1} gives

$$b = u^{-1}(ub) = u^{-1}0 = 0.$$

Thus u is not a left zero divisor.

Similarly, if $bu = 0$, multiplication on the right by u^{-1} gives $b = 0$. Hence u is not a right zero divisor. ■

3.2.2 Integral Domains

Definition 3.2.8 (Integral Domain). An *integral domain* is a commutative ring R with identity $1_R \neq 0$ that has no zero divisors.

Equivalently, a commutative ring with identity $1_R \neq 0$ is an integral domain if

$$ab = 0 \implies a = 0 \text{ or } b = 0$$

for all $a, b \in R$.

Example 3.2.9. The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are integral domains.

The polynomial ring $\mathbb{Z}[x]$ is also an integral domain, whereas $\mathbb{Z}_6$ is not, because $[2][3] = [0]$ with both factors nonzero.

An important consequence of the absence of zero divisors is the cancellation law.

Theorem 3.2.10 (Cancellation Law). Let R be an integral domain. If $a, b, c \in R$, $a \neq 0$, and $ab = ac$, then $b = c$.

Proof. Since $ab = ac$, subtracting ac from both sides gives

$$ab - ac = 0.$$

By distributivity, $a(b - c) = 0$.

As R is an integral domain and $a \neq 0$, we must have $b - c = 0$. Therefore $b = c$. ■

Remark 3.2.11. Cancellation can fail in rings with zero divisors. For instance, in $\mathbb{Z}_6$,

$$[2][1] = [2] = [2][4],$$

although $[1] \neq [4]$.

Thus, the familiar cancellation law cannot be used in an arbitrary ring without appropriate hypotheses.

Theorem 3.2.12 (Cancellation Characterization of Domains). Let R be a nonzero commutative ring with identity. Then R is an integral domain if and only if multiplication by every nonzero element satisfies the cancellation law.

Proof. If R is an integral domain, cancellation follows from the preceding theorem.

Conversely, suppose cancellation holds whenever the common factor is nonzero. Let $ab = 0$ with

$a \neq 0$.

Since $a0 = 0$, we have $ab = a0$. Cancellation by a gives $b = 0$. Thus R has no zero divisors and is an integral domain. ■

Theorem 3.2.13 (Polynomial Rings over Domains). *If R is an integral domain, then the polynomial ring $R[x]$ is also an integral domain.*

Proof. The ring $R[x]$ is commutative and has identity because R has these properties.

Let $f(x), g(x) \in R[x]$ be nonzero polynomials of degrees m and n , with leading coefficients a_m and b_n , respectively.

The coefficient of x^{m+n} in $f(x)g(x)$ is a_mb_n . Since R has no zero divisors and both leading coefficients are nonzero,

$$a_mb_n \neq 0.$$

Therefore $f(x)g(x) \neq 0$.

Hence $R[x]$ has no zero divisors and is an integral domain. ■

3.2.3 Division Rings and Fields

Definition 3.2.14 (Division Ring). A *division ring*, also called a *skew field*, is a nonzero ring with identity in which every nonzero element has a multiplicative inverse. Multiplication in a division ring need not be commutative.

Definition 3.2.15 (Field). A *field* is a commutative division ring. Thus a field is a nonzero commutative ring with identity in which every nonzero element is a unit.

The relationship among these structures may be summarized as follows: every field is a commutative division ring and hence an integral domain, but an integral domain need not be a field.

Example 3.2.16. The rings $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields. The ring $\mathbb{Z}$ is an integral domain but not a field because 2 has no multiplicative inverse in $\mathbb{Z}$.

The real quaternion algebra $\mathbb{H}$ is a noncommutative division ring. Thus a division ring need not be a field.

Theorem 3.2.17. *Every division ring has no zero divisors.*

Proof. Let R be a division ring and suppose $ab = 0$ for some $a, b \in R$.

If $a = 0$, there is nothing to prove. Suppose $a \neq 0$. Since a is invertible, multiplying $ab = 0$ on the left by a^{-1} gives

$$b = a^{-1}(ab) = a^{-1}0 = 0.$$

Therefore $ab = 0$ implies $a = 0$ or $b = 0$, so R has no zero divisors. ■

Corollary 3.2.18. *Every field is an integral domain.*

Proof. A field is a commutative division ring with identity $1 \neq 0$. By the preceding theorem, it has no zero divisors. Hence it is an integral domain. ■

The converse is false in general, as the ring $\mathbb{Z}$ demonstrates. However, finiteness makes a significant difference.

Theorem 3.2.19 (Finite Integral Domains Are Fields). *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain, and let $a \in R$ be nonzero.

Define $f : R \rightarrow R$ by $f(r) = ar$.

Suppose $f(r) = f(s)$. Then $ar = as$. Since $a \neq 0$ and R is an integral domain, the cancellation law gives $r = s$. Thus f is injective.

Since R is finite, every injective mapping from R to itself is surjective. Hence f is surjective.

In particular, there exists $b \in R$ such that

$$f(b) = ab = 1_R.$$

Since multiplication is commutative, $ba = 1_R$ as well. Therefore a is invertible.

As a was an arbitrary nonzero element of R , every nonzero element is a unit. Hence R is a field. ■

Remark 3.2.20. The finiteness assumption is essential. The ring $\mathbb{Z}$ is an infinite integral domain but is not a field.

3.2.4 Fields of Integers Modulo n

We now investigate when the ring $\mathbb{Z}_n$ is an integral domain or a field.

Theorem 3.2.21. *Let $n \geq 2$. The following statements are equivalent:*

- (i) n is a prime number;
- (ii) $\mathbb{Z}_n$ is an integral domain;
- (iii) $\mathbb{Z}_n$ is a field.

Proof. Suppose first that $n = p$ is prime. Let $[a], [b] \in \mathbb{Z}_p$ satisfy $[a][b] = [0]$. Then $p \mid ab$.

By Euclid's lemma, $p \mid a$ or $p \mid b$. Hence $[a] = [0]$ or $[b] = [0]$, showing that $\mathbb{Z}_p$ has no zero divisors.

Thus $\mathbb{Z}_p$ is an integral domain.

Since $\mathbb{Z}_p$ is finite, the preceding theorem implies that it is a field.

Every field is an integral domain, so (iii) implies (ii).

Finally, suppose $\mathbb{Z}_n$ is an integral domain. If n were composite, we could write $n = rs$ with $1 < r, s < n$. Then $[r]$ and $[s]$ would be nonzero elements of $\mathbb{Z}_n$ satisfying

$$[r][s] = [rs] = [n] = [0].$$

This contradicts the absence of zero divisors. Therefore n must be prime.

Hence all three statements are equivalent. ■

Example 3.2.22. The rings $\mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$, and $\mathbb{Z}_7$ are fields because 2, 3, 5, and 7 are prime numbers.

In contrast, $\mathbb{Z}_4, \mathbb{Z}_6, \mathbb{Z}_8$, and $\mathbb{Z}_9$ are not fields because their moduli are composite.

3.2.5 Units and Zero Divisors in $\mathbb{Z}_n$

The units of $\mathbb{Z}_n$ admit a simple characterization using the greatest common divisor.

Theorem 3.2.23 (Units in $\mathbb{Z}_n$). *Let $n \geq 2$. A residue class $[a] \in \mathbb{Z}_n$ is a unit if and only if*

$$\gcd(a, n) = 1.$$

Proof. Suppose $[a]$ is a unit. Then there exists $[b] \in \mathbb{Z}_n$ such that $[a][b] = [1]$. Hence $ab \equiv 1 \pmod{n}$. Therefore $ab - 1 = kn$ for some integer k , giving

$$ab - kn = 1.$$

Every common divisor of a and n must divide the left side, and hence must divide 1. Thus $\gcd(a, n) = 1$.

Conversely, suppose $\gcd(a, n) = 1$. By Bézout's identity, there exist integers u, v such that

$$au + nv = 1.$$

Reducing modulo n gives $[a][u] = [1]$. Thus $[u]$ is a multiplicative inverse of $[a]$, proving that $[a]$ is a unit. ■

Theorem 3.2.24 (Zero Divisors in $\mathbb{Z}_n$). *Let $n \geq 2$. A nonzero element $[a] \in \mathbb{Z}_n$ is a zero divisor if and only if*

$$\gcd(a, n) > 1.$$

Proof. Suppose $[a] \neq [0]$ and put $d = \gcd(a, n) > 1$.

Write $a = da_1$ and $n = dn_1$. Since $d > 1$, we have $1 \leq n_1 < n$, so $[n_1] \neq [0]$ in $\mathbb{Z}_n$.

Moreover,

$$[a][n_1] = [an_1] = [da_1n_1] = [a_1n] = [0].$$

Thus $[a]$ is a zero divisor.

Conversely, suppose $[a]$ is a zero divisor. Then there exists $[b] \neq [0]$ such that $[a][b] = [0]$.

If $\gcd(a, n) = 1$, the preceding theorem would imply that $[a]$ is a unit. Multiplying $[a][b] = [0]$ by its inverse would then give $[b] = [0]$, a contradiction.

Therefore $\gcd(a, n) > 1$. ■

Corollary 3.2.25. *Every nonzero element of $\mathbb{Z}_n$ is either a unit or a zero divisor, but not both.*

Proof. For any nonzero $[a] \in \mathbb{Z}_n$, either $\gcd(a, n) = 1$ or $\gcd(a, n) > 1$. The preceding two theorems show that $[a]$ is respectively a unit or a zero divisor.

The two possibilities are mutually exclusive. ■

Example 3.2.26 (Determining Units and Zero Divisors in $\mathbb{Z}_{12}$). We determine all units and nonzero zero divisors of $\mathbb{Z}_{12}$.

By the unit criterion, $[a]$ is invertible precisely when $\gcd(a, 12) = 1$. The corresponding integers are 1, 5, 7, 11. Hence

$$U(\mathbb{Z}_{12}) = \{[1], [5], [7], [11]\}.$$

Indeed, the multiplicative inverses are

$$[1]^{-1} = [1], \quad [5]^{-1} = [5], \quad [7]^{-1} = [7], \quad [11]^{-1} = [11].$$

The remaining nonzero residue classes are

$$[2], [3], [4], [6], [8], [9], [10].$$

Each has greatest common divisor with 12 greater than 1, so these are precisely the nonzero zero divisors of $\mathbb{Z}_{12}$.

For example,

$$[2][6] = [0], \quad [3][4] = [0], \quad [8][3] = [0].$$

Corollary 3.2.27. *For every $n \geq 2$, the group of units $U(\mathbb{Z}_n)$ has order $\varphi(n)$, where φ denotes Euler's totient function.*

Proof. By the unit criterion, $U(\mathbb{Z}_n)$ consists precisely of the residue classes $[a]$ for which $\gcd(a, n) = 1$.

The number of such residue classes is $\varphi(n)$. Hence $|U(\mathbb{Z}_n)| = \varphi(n)$. ■

3.3 Subrings and Ideals

A subset of a ring may itself form a ring under the same addition and multiplication. Such a subset is called a *subring*. However, the construction of quotient rings requires a stronger concept, namely that of an *ideal*.

Ideals are distinguished from arbitrary subrings by an additional property: multiplying an element of an ideal by any element of the ambient ring produces another element of the ideal. This property, called *absorption*, is fundamental in the theory of quotient rings and ring homomorphisms.

3.3.1 Subrings

Definition 3.3.1 (Subring). Let R be a ring. A nonempty subset S of R is called a *subring* of R if S is itself a ring under the operations of addition and multiplication inherited from R .

A subring necessarily contains the additive identity 0 of R , but it need not contain the multiplicative identity 1_R , even when R possesses one.

Since associativity and distributivity are inherited from the ambient ring, it is unnecessary to verify these properties again. The following result gives a convenient criterion for identifying subrings.

Theorem 3.3.2 (Subring Test). *A nonempty subset S of a ring R is a subring of R if and only if*

$$a - b \in S, \quad ab \in S$$

for all $a, b \in S$.

Proof. Suppose S is a subring of R . Then $(S, +)$ is an additive subgroup of $(R, +)$, so $a - b \in S$ whenever $a, b \in S$. Also, S is closed under multiplication, giving $ab \in S$.

Conversely, suppose S is nonempty and satisfies the stated conditions.

Since $a - b \in S$ for all $a, b \in S$, the subgroup test shows that $(S, +)$ is a subgroup of $(R, +)$. In particular, $0 \in S$, every element of S has its additive inverse in S , and S is closed under addition. By assumption, S is also closed under multiplication. Associativity of multiplication and both distributive laws hold in S because they hold in R .

Therefore S is a ring under the inherited operations, and hence a subring of R . ■

Example 3.3.3 (Standard Examples of Subrings). (i) The integers $\mathbb{Z}$ form a subring of $\mathbb{Q}$, and $\mathbb{Q}$ is a subring of $\mathbb{R}$.

(ii) The even integers $2\mathbb{Z}$ form a subring of $\mathbb{Z}$. Indeed, the difference and product of any two even integers are even.

(iii) The Gaussian integers

$$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$$

form a subring of $\mathbb{C}$. For $a + bi, c + di \in \mathbb{Z}[i]$, their difference is

$$(a - c) + (b - d)i \in \mathbb{Z}[i],$$

and their product is

$$(ac - bd) + (ad + bc)i \in \mathbb{Z}[i].$$

(iv) The set of diagonal matrices

$$D_n(\mathbb{R}) = \{\text{diag}(a_1, \dots, a_n) : a_i \in \mathbb{R}\}$$

is a subring of $M_n(\mathbb{R})$.

(v) For every ring R , the subsets $\{0\}$ and R are subrings of R .

Remark 3.3.4. Some authors require a subring of a ring with identity to contain the same multiplicative identity. Under that convention, $2\mathbb{Z}$ is not a unital subring of $\mathbb{Z}$. Throughout this chapter, we use the more general definition, under which a subring need not contain 1_R .

3.3.2 Ideals

A subring is closed under multiplication of its own elements. An ideal satisfies the stronger requirement that it absorbs multiplication by arbitrary elements of the ambient ring.

Definition 3.3.5 (Ideal). Let R be a ring. A nonempty subset I of R is called a *two-sided ideal*, or simply an *ideal*, if

- (i) $a - b \in I$ for all $a, b \in I$;
- (ii) $ra \in I$ and $ar \in I$ for every $a \in I$ and $r \in R$.

The first condition says that $(I, +)$ is a subgroup of $(R, +)$. The second condition is called the *absorption property*.

If R is commutative, the conditions $ra \in I$ and $ar \in I$ are identical.

In a noncommutative ring, it is also possible to consider left ideals and right ideals, where only one of the absorption conditions is imposed. Unless stated otherwise, the word *ideal* will mean a two-sided ideal.

Example 3.3.6 (Ideals in the Integers). For any integer n , the subset

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is an ideal of $\mathbb{Z}$.

Indeed, if $na, nb \in n\mathbb{Z}$, then

$$na - nb = n(a - b) \in n\mathbb{Z}.$$

Furthermore, if $r \in \mathbb{Z}$, then

$$r(na) = n(ra) \in n\mathbb{Z}.$$

Thus $n\mathbb{Z}$ satisfies the ideal conditions.

Example 3.3.7 (An Ideal in a Polynomial Ring). Let $R = \mathbb{R}[x]$ and consider

$$I = x\mathbb{R}[x] = \{xf(x) : f(x) \in \mathbb{R}[x]\}.$$

The set I consists precisely of the polynomials whose constant term is zero.

If $xf(x), xg(x) \in I$, then

$$xf(x) - xg(x) = x(f(x) - g(x)) \in I.$$

Also, for $h(x) \in R$,

$$h(x) \cdot xf(x) = x(h(x)f(x)) \in I.$$

Therefore I is an ideal of $\mathbb{R}[x]$.

Example 3.3.8 (A Subring Which Is Not an Ideal). The ring $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it is not an ideal of $\mathbb{Q}$.

Indeed, $1 \in \mathbb{Z}$ and $\frac{1}{2} \in \mathbb{Q}$, yet

$$\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}.$$

Thus $\mathbb{Z}$ fails the absorption property.

Proposition 3.3.9. Every ideal of a ring is a subring, but the converse need not hold.

Proof. Let I be an ideal of R . By definition, I is nonempty and closed under subtraction.

If $a, b \in I$, then $a \in I$ and $b \in R$, so the absorption property gives $ab \in I$. Hence I is also closed under multiplication.

By the subring test, I is a subring of R .

The converse fails, as shown by the subring $\mathbb{Z}$ of $\mathbb{Q}$, which is not an ideal. ■

3.3.3 Elementary Properties of Ideals

Every ring has at least two ideals: the zero ideal and the entire ring.

Definition 3.3.10. The ideals $\{0\}$ and R are called the *trivial ideals* of R . An ideal I is called a *proper ideal* if $I \neq R$.

In a ring with identity, proper ideals are characterized by the absence of the identity element.

Theorem 3.3.11 (Unit Criterion for Ideals). Let R be a ring with identity 1_R , and let I be an ideal of R . Then

$$I = R \iff 1_R \in I.$$

Consequently, if I contains a unit of R , then $I = R$.

Proof. If $I = R$, then clearly $1_R \in I$.

Conversely, suppose $1_R \in I$. For any $r \in R$, the absorption property gives

$$r = r1_R \in I.$$

Hence every element of R belongs to I , and therefore $I = R$.

Now suppose I contains a unit u . Since $u^{-1} \in R$ and $u \in I$, the absorption property gives

$$1_R = u^{-1}u \in I.$$

By the first part, $I = R$. ■

Corollary 3.3.12. *In a field F , the only ideals are $\{0\}$ and F .*

Proof. Let I be a nonzero ideal of F . Choose $0 \neq a \in I$. Since F is a field, a is a unit. By the unit criterion, $I = F$. Thus the only ideals of F are $\{0\}$ and F . ■

3.3.4 Intersections, Sums and Products of Ideals

Ideals can be combined to construct new ideals. The most important operations are intersection, sum, and product.

Theorem 3.3.13 (Intersection of Ideals). *Let $\{I_\lambda : \lambda \in \Lambda\}$ be a nonempty family of ideals of a ring R . Then*

$$\bigcap_{\lambda \in \Lambda} I_\lambda$$

is an ideal of R .

Proof. Every ideal contains 0, so the intersection is nonempty. Let a, b belong to the intersection. Then $a, b \in I_\lambda$ for every $\lambda \in \Lambda$. Since each I_λ is an ideal, $a - b \in I_\lambda$ for every λ . Hence $a - b$ belongs to the intersection. Similarly, for any $r \in R$, both ra and ar belong to every I_λ , and therefore to their intersection. Thus the intersection is an ideal. ■

Definition 3.3.14 (Sum of Ideals). Let I and J be ideals of R . Their *sum* is

$$I + J = \{i + j : i \in I, j \in J\}.$$

Theorem 3.3.15. *If I and J are ideals of R , then $I + J$ is an ideal of R . Moreover, it is the smallest ideal containing both I and J .*

Proof. Since $0 = 0 + 0 \in I + J$, the sum is nonempty. Let $a = i_1 + j_1$ and $b = i_2 + j_2$ belong to $I + J$. Then

$$a - b = (i_1 - i_2) + (j_1 - j_2) \in I + J,$$

because $i_1 - i_2 \in I$ and $j_1 - j_2 \in J$.

For $r \in R$,

$$ra = ri_1 + rj_1 \in I + J,$$

and similarly

$$ar = i_1r + j_1r \in I + J.$$

Therefore $I + J$ is an ideal.

Clearly $I \subseteq I + J$ and $J \subseteq I + J$, because both ideals contain 0.

If K is any ideal containing both I and J , then $i + j \in K$ for every $i \in I$ and $j \in J$. Hence $I + J \subseteq K$.

Thus $I + J$ is the smallest ideal containing both I and J . ■

Definition 3.3.16 (Product of Ideals). Let I and J be ideals of R . Their *product* is

$$IJ = \left\{ \sum_{k=1}^m i_k j_k : m \geq 1, i_k \in I, j_k \in J \right\}.$$

Thus IJ consists of all finite sums of products of elements of I and J .

Remark 3.3.17. The set $\{ij : i \in I, j \in J\}$ need not be closed under addition. For this reason, the definition of the product ideal requires finite sums of such products.

Theorem 3.3.18. If I and J are two-sided ideals of R , then IJ is a two-sided ideal. Moreover,

$$IJ \subseteq I \cap J.$$

Proof. The set IJ is nonempty because $0 = 0 \cdot 0 \in IJ$.

Let

$$a = \sum_{k=1}^m i_k j_k, \quad b = \sum_{\ell=1}^t i'_\ell j'_\ell$$

be elements of IJ .

Then

$$a - b = \sum_{k=1}^m i_k j_k + \sum_{\ell=1}^t (-i'_\ell) j'_\ell.$$

Since $-i'_\ell \in I$, this expression belongs to IJ .

For $r \in R$,

$$ra = \sum_{k=1}^m (ri_k) j_k \in IJ,$$

because $ri_k \in I$. Similarly,

$$ar = \sum_{k=1}^m i_k (j_k r) \in IJ,$$

because $j_k r \in J$.

Thus IJ is an ideal.

Finally, every product ij , with $i \in I$ and $j \in J$, belongs to I because I absorbs multiplication on the right, and belongs to J because J absorbs multiplication on the left.

Hence $ij \in I \cap J$. Since $I \cap J$ is closed under addition, every finite sum of such products also belongs to $I \cap J$. Therefore

$$IJ \subseteq I \cap J.$$

■

Example 3.3.19. In $\mathbb{Z}$, let $I = 4\mathbb{Z}$ and $J = 6\mathbb{Z}$. Then

$$I + J = 2\mathbb{Z}, \quad I \cap J = 12\mathbb{Z}, \quad IJ = 24\mathbb{Z}.$$

The sum consists of multiples of $\gcd(4, 6) = 2$, while the intersection consists of multiples of $\text{lcm}(4, 6) = 12$.

The product is $24\mathbb{Z}$, since every finite sum of products of multiples of 4 and 6 is a multiple of 24, and 24 itself is such a product.

3.4 Principal Ideals and Principal Ideal Domains

We now turn to ideals generated by individual elements. These are called *principal ideals*. They are especially important because divisibility of elements can be expressed through containment of principal ideals.

Throughout this section, R denotes a commutative ring with identity $1_R \neq 0$, unless otherwise stated.

3.4.1 Ideals Generated by Elements

Definition 3.4.1 (Generated Ideal). Let $a_1, \dots, a_t \in R$. The ideal generated by $a_1, \dots, a_t$ is

$$(a_1, \dots, a_t) = \{r_1 a_1 + \dots + r_t a_t : r_1, \dots, r_t \in R\}.$$

The elements $a_1, \dots, a_t$ are called *generators* of the ideal.

Definition 3.4.2 (Principal Ideal). An ideal I of R is called *principal* if there exists $a \in R$ such that

$$I = (a) = \{ra : r \in R\}.$$

The element a is called a generator of the principal ideal (a) .

The zero ideal is principal, since $(0) = \{0\}$, and the whole ring is principal, since $(1_R) = R$.

Theorem 3.4.3 (Smallest Ideal Containing Given Elements). Let $a_1, \dots, a_t \in R$. Then $(a_1, \dots, a_t)$ is the smallest ideal of R containing all the elements $a_1, \dots, a_t$. In particular, (a) is the smallest ideal containing a .

Proof. Let

$$I = (a_1, \dots, a_t) = \left\{ \sum_{i=1}^t r_i a_i : r_i \in R \right\}.$$

Clearly $0 \in I$. If $x = \sum r_i a_i$ and $y = \sum s_i a_i$ belong to I , then

$$x - y = \sum_{i=1}^t (r_i - s_i) a_i \in I.$$

If $r \in R$, then

$$rx = \sum_{i=1}^t (rr_i) a_i \in I.$$

Since R is commutative, the absorption property holds on both sides. Hence I is an ideal.

For each j , taking $r_j = 1_R$ and all other coefficients equal to 0 gives $a_j \in I$.

Now let J be any ideal containing all $a_1, \dots, a_t$. Since J absorbs multiplication by arbitrary elements of R , it contains every $r_i a_i$, and therefore every finite sum $\sum r_i a_i$.

Thus $I \subseteq J$, proving that I is the smallest ideal containing the specified elements. ■

Proposition 3.4.4 (Divisibility and Principal Ideals). Let $a, b \in R$. Then

$$(a) \subseteq (b) \iff b \mid a.$$

Proof. Suppose $(a) \subseteq (b)$. Since $a \in (a)$, we have $a \in (b)$. Therefore $a = br$ for some $r \in R$, showing that $b \mid a$.

Conversely, suppose $b \mid a$. Then $a = br$ for some $r \in R$. Every element of (a) has the form $sa = sbr$, which belongs to (b) . Thus $(a) \subseteq (b)$. ■

Remark 3.4.5. Containment of principal ideals reverses the direction of divisibility. If b divides a , then every multiple of a is a multiple of b , so $(a) \subseteq (b)$.

For example,

$$(12) \subseteq (6) \subseteq (3)$$

in $\mathbb{Z}$, because $3 \mid 6 \mid 12$.

Proposition 3.4.6. If R is an integral domain and $a, b \in R$ are nonzero, then

$$(a) = (b) \iff a = ub$$

for some unit $u \in R$.

Proof. If $a = ub$ with u a unit, then $(a) \subseteq (b)$. Also $b = u^{-1}a$, so $(b) \subseteq (a)$. Hence $(a) = (b)$.

Conversely, suppose $(a) = (b)$. Then $a = rb$ and $b = sa$ for some $r, s \in R$.

Substituting gives $a = rsa$, so

$$(1_R - rs)a = 0.$$

Since R is an integral domain and $a \neq 0$, we obtain $rs = 1_R$.

Thus r is a unit, and $a = rb$ gives the required relation. ■

3.4.2 Principal Ideal Domains

Definition 3.4.7 (Principal Ideal Domain). An integral domain R is called a *principal ideal domain*, abbreviated *PID*, if every ideal of R is principal.

Thus, in a PID, every ideal can be generated by a single element. The rings $\mathbb{Z}$ and $F[x]$, where F is a field, are the most important elementary examples.

Theorem 3.4.8 (Every Ideal of $\mathbb{Z}$ Is Principal). Every ideal I of $\mathbb{Z}$ has the form

$$I = n\mathbb{Z} = (n)$$

for a unique integer $n \geq 0$. Consequently, $\mathbb{Z}$ is a PID.

Proof. If $I = \{0\}$, then $I = 0\mathbb{Z}$, and the conclusion holds with $n = 0$.

Suppose $I \neq \{0\}$. Since I contains a nonzero integer and is closed under additive inverses, it contains a positive integer.

By the well-ordering principle, there exists a least positive integer $n \in I$.

Since I is an ideal and $n \in I$, every multiple of n belongs to I . Therefore $n\mathbb{Z} \subseteq I$.

Conversely, let $a \in I$. By the division algorithm, there exist integers q, r such that

$$a = qn + r, \quad 0 \leq r < n.$$

Since $a, n \in I$ and I is an ideal, we have

$$r = a - qn \in I.$$

By the minimality of n , the only possibility is $r = 0$. Hence $a = qn \in n\mathbb{Z}$.

Thus $I \subseteq n\mathbb{Z}$, and therefore $I = n\mathbb{Z}$.

The integer n is unique because, for a nonzero ideal, it is the least positive element of I . ■

Example 3.4.9. The ideals of $\mathbb{Z}$ include

$$(0) = \{0\}, \quad (1) = \mathbb{Z}, \quad (2) = 2\mathbb{Z}, \quad (3) = 3\mathbb{Z},$$

and so on.

Every ideal of $\mathbb{Z}$ is obtained in this manner. In particular, there are no nonprincipal ideals in $\mathbb{Z}$.

3.4.3 Ideals in Polynomial Rings

The polynomial ring over a field is another important example of a principal ideal domain.

Theorem 3.4.10 (Polynomial Rings over Fields). *If F is a field, then $F[x]$ is a principal ideal domain.*

Proof. Let I be an ideal of $F[x]$.

If $I = \{0\}$, then $I = (0)$ is principal.

Suppose $I \neq \{0\}$. Choose a nonzero polynomial $f(x) \in I$ of least degree. Such a polynomial exists because the degrees of nonzero polynomials in I form a nonempty set of nonnegative integers.

Since $f(x) \in I$, every multiple of $f(x)$ belongs to I , so $(f) \subseteq I$.

Conversely, let $g(x) \in I$. By the polynomial division algorithm over the field F , there exist $q(x), r(x) \in F[x]$ such that

$$g(x) = q(x)f(x) + r(x),$$

where either $r(x) = 0$ or $\deg r < \deg f$.

Since $g(x) \in I$ and $q(x)f(x) \in I$, their difference $r(x)$ belongs to I .

If $r(x) \neq 0$, its degree would be smaller than that of $f(x)$, contradicting the choice of $f(x)$. Therefore $r(x) = 0$.

Thus $g(x) = q(x)f(x) \in (f)$, proving $I \subseteq (f)$.

Consequently $I = (f)$, so every ideal of $F[x]$ is principal. ■

Example 3.4.11. In $\mathbb{R}[x]$, the set

$$I = \{f(x) \in \mathbb{R}[x] : f(2) = 0\}$$

is the principal ideal $(x - 2)$.

Indeed, the factor theorem states that $f(2) = 0$ if and only if $x - 2$ divides $f(x)$.

Remark 3.4.12. The assumption that F is a field is important. Although $\mathbb{Z}$ is a PID, the polynomial ring $\mathbb{Z}[x]$ is not a PID, as we shall show below.

3.4.4 Sums and Intersections of Principal Ideals

In a principal ideal domain, sums and intersections of principal ideals have a direct interpretation in terms of greatest common divisors and least common multiples.

Theorem 3.4.13 (Greatest Common Divisor and Ideal Sum). *Let R be a PID, and let $a, b \in R$ be nonzero. Then*

$$(a) + (b) = (d),$$

where d is a greatest common divisor of a and b .

Proof. Since R is a PID, the ideal $(a) + (b)$ is principal. Write

$$(a) + (b) = (d).$$

Because $a, b \in (d)$, the element d divides both a and b .

Now let c be any common divisor of a and b . Then c divides every linear combination $ra + sb$, where $r, s \in R$.

Since $d \in (a) + (b)$, there exist $r, s \in R$ such that

$$d = ra + sb.$$

Hence $c \mid d$.

Thus d is a greatest common divisor of a and b . ■

Corollary 3.4.14 (Bézout's Identity in a PID). *If R is a PID and d is a greatest common divisor of $a, b \in R$, then there exist $r, s \in R$ such that*

$$d = ra + sb.$$

Proof. By the preceding theorem, $(a) + (b) = (d)$. Since $d \in (a) + (b)$, it has the required representation. ■

Theorem 3.4.15 (Least Common Multiple and Ideal Intersection). *Let R be a PID, and let $a, b \in R$ be nonzero. Then*

$$(a) \cap (b) = (m),$$

where m is a least common multiple of a and b .

Proof. Since R is a PID, the ideal $(a) \cap (b)$ is principal. Write

$$(a) \cap (b) = (m).$$

Because $m \in (a)$ and $m \in (b)$, both a and b divide m . Thus m is a common multiple.

Now let c be any common multiple of a and b . Then $c \in (a)$ and $c \in (b)$, so $c \in (a) \cap (b) = (m)$.

Therefore $m \mid c$. Hence m is a least common multiple of a and b . ■

Example 3.4.16 (Operations on Ideals of $\mathbb{Z}$). Find the sum, intersection, and product of the ideals (18) and (30) in $\mathbb{Z}$.

Since

$$\gcd(18, 30) = 6, \quad \text{lcm}(18, 30) = 90,$$

we have

$$(18) + (30) = (6)$$

and

$$(18) \cap (30) = (90).$$

The product ideal is

$$(18)(30) = (540).$$

Indeed, every product of a multiple of 18 and a multiple of 30 is divisible by 540, and $540 = 18 \cdot 30$ belongs to the product ideal.

Thus

$$(18) + (30) = (6), \quad (18) \cap (30) = (90), \quad (18)(30) = (540).$$

Proposition 3.4.17. If R is a commutative ring with identity and $a, b \in R$, then

$$(a)(b) = (ab).$$

Proof. Every element of $(a)(b)$ is a finite sum of terms $(ra)(sb)$, where $r, s \in R$. By commutativity,

$$(ra)(sb) = (rs)(ab) \in (ab).$$

Hence $(a)(b) \subseteq (ab)$.

Conversely, ab is the product of $a \in (a)$ and $b \in (b)$, so $ab \in (a)(b)$. Since $(a)(b)$ is an ideal, it contains the smallest ideal generated by ab , namely (ab) .

Thus $(ab) \subseteq (a)(b)$, proving equality. ■

3.4.5 Nonprincipal Ideals

Not every integral domain is a principal ideal domain. The following examples illustrate why certain ideals cannot be generated by a single element.

Theorem 3.4.18. The ideal $(2, x)$ of $\mathbb{Z}[x]$ is not principal. Consequently, $\mathbb{Z}[x]$ is not a PID.

Proof. Consider

$$I = (2, x) = \{2f(x) + xg(x) : f(x), g(x) \in \mathbb{Z}[x]\}.$$

Suppose, for a contradiction, that $I = (h(x))$ for some $h(x) \in \mathbb{Z}[x]$.

Since $2 \in I$, the polynomial $h(x)$ divides the constant polynomial 2. Since $\mathbb{Z}[x]$ is an integral domain, degree considerations show that $h(x)$ must be a nonzero constant polynomial.

Moreover, the constant integer $h(x)$ must divide 2 in $\mathbb{Z}$. Hence

$$h(x) \in \{1, -1, 2, -2\}.$$

If $h(x) = \pm 1$, then $I = \mathbb{Z}[x]$. This is impossible because every polynomial in I has an even constant term, so $1 \notin I$.

If $h(x) = \pm 2$, then $I = (2)$. This is also impossible because $x \in I$ but $x \notin (2)$.

Both possibilities lead to contradictions. Therefore I is not principal.

Hence $\mathbb{Z}[x]$ is not a principal ideal domain. ■

Example 3.4.19 (A Nonprincipal Ideal in $F[x, y]$.) Let F be a field and consider the ideal

$$I = (x, y)$$

of $F[x, y]$.

Suppose $I = (f)$ for some polynomial $f \in F[x, y]$. Since both x and y belong to I , the polynomial f must divide both x and y .

But x and y have no common nonunit polynomial factor. Consequently, f must be a unit, which would imply $I = F[x, y]$.

This is impossible because every polynomial in (x, y) has zero constant term, whereas the constant polynomial 1 does not.

Therefore (x, y) is not principal, and hence $F[x, y]$ is not a PID.

Remark 3.4.20. The preceding examples show that the principal ideal property is sensitive to the structure of the ring.

The rings $\mathbb{Z}$ and $F[x]$ are PIDs, but $\mathbb{Z}[x]$ and $F[x, y]$ are not. Thus the property that every ideal is generated by one element does not automatically extend to polynomial rings in additional

indeterminates.

3.5 Quotient Rings and Ring Homomorphisms

In group theory, normal subgroups allow us to construct quotient groups. A similar construction exists in ring theory, where ideals play the role of normal subgroups.

Every ideal of a ring is an additive subgroup, so its additive cosets partition the ring. The additional absorption property ensures that these cosets can be multiplied consistently. The resulting algebraic structure is called a quotient ring.

Ring homomorphisms provide another fundamental tool. They describe mappings that preserve both addition and multiplication and establish important connections between ideals and quotient rings.

3.5.1 Construction of Quotient Rings

Let R be a ring and let I be a two-sided ideal of R . Since $(I, +)$ is a subgroup of the abelian group $(R, +)$, we may consider the additive cosets

$$a + I = \{a + i : i \in I\}, \quad a \in R.$$

Two elements $a, b \in R$ determine the same coset precisely when $a - b \in I$. Thus

$$a + I = b + I \iff a - b \in I.$$

The collection of all these cosets is denoted by

$$R/I = \{a + I : a \in R\}.$$

We define addition and multiplication of cosets by

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Before treating these operations as ring operations, it is necessary to establish that their values do not depend on the representatives chosen for the cosets.

Theorem 3.5.1 (Quotient Ring Theorem). *Let I be a two-sided ideal of a ring R . Then R/I is a ring under the operations*

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Its additive identity is $I = 0 + I$, and the additive inverse of $a + I$ is $-a + I$.

If R is commutative, then R/I is commutative. If R has a multiplicative identity 1_R , then $1_R + I$ is a multiplicative identity of R/I .

Proof. We first verify that the operations are well-defined.

Suppose

$$a + I = a' + I, \quad b + I = b' + I.$$

Then $a' - a \in I$ and $b' - b \in I$. Write $a' = a + i$ and $b' = b + j$, where $i, j \in I$.

For addition,

$$(a' + b') - (a + b) = i + j \in I.$$

Hence $(a' + b') + I = (a + b) + I$, proving that addition is independent of representatives.

For multiplication, we have

$$\begin{aligned} a'b' - ab &= (a + i)(b + j) - ab \\ &= aj + ib + ij. \end{aligned}$$

Since I is a two-sided ideal, all three terms aj , ib , and ij belong to I . Thus $a'b' - ab \in I$, so

$$a'b' + I = ab + I.$$

Therefore multiplication is also well-defined.

The additive group axioms follow from those of $(R, +)$. In particular, addition is associative and commutative, I is the additive identity, and

$$(a + I) + (-a + I) = I.$$

For multiplication, associativity follows from

$$\begin{aligned} ((a + I)(b + I))(c + I) &= ((ab)c) + I \\ &= (a(bc)) + I \\ &= (a + I)((b + I)(c + I)). \end{aligned}$$

The distributive laws follow from the corresponding laws in R . For example,

$$\begin{aligned} (a + I)((b + I) + (c + I)) &= a(b + c) + I \\ &= (ab + ac) + I \\ &= (ab + I) + (ac + I). \end{aligned}$$

The right distributive law is verified similarly.

Therefore R/I is a ring.

If R is commutative, then

$$(a + I)(b + I) = ab + I = ba + I = (b + I)(a + I).$$

Finally, if R has identity 1_R , then

$$(1_R + I)(a + I) = a + I = (a + I)(1_R + I),$$

so $1_R + I$ is the multiplicative identity of R/I . ■

Remark 3.5.2. The requirement that I be a two-sided ideal is essential for multiplication of cosets to be well-defined. An arbitrary additive subgroup need not satisfy the absorption property. Notice also that if R has identity and $I = R$, then R/I is the zero ring, whose multiplicative identity coincides with its additive identity.

If I is proper, then $1_R + I \neq I$, so the quotient has distinct additive and multiplicative identities.

Example 3.5.3 (Integers Modulo n). Let $R = \mathbb{Z}$ and $I = n\mathbb{Z}$, where $n \geq 2$. Then

$$\mathbb{Z}/n\mathbb{Z} = \{0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \dots, (n-1) + n\mathbb{Z}\}.$$

The quotient operations are

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}$$

and

$$(a + n\mathbb{Z})(b + n\mathbb{Z}) = ab + n\mathbb{Z}.$$

Thus $\mathbb{Z}/n\mathbb{Z}$ is precisely the familiar ring $\mathbb{Z}_n$ of integers modulo n .

For example, in $\mathbb{Z}/7\mathbb{Z}$,

$$(5 + 7\mathbb{Z}) + (6 + 7\mathbb{Z}) = 4 + 7\mathbb{Z}$$

and

$$(5 + 7\mathbb{Z})(6 + 7\mathbb{Z}) = 2 + 7\mathbb{Z}.$$

Example 3.5.4 (A Quotient of a Polynomial Ring). Consider the ideal (x) of $\mathbb{Z}[x]$.

Two polynomials belong to the same coset modulo (x) if and only if they have the same constant term. Indeed, $f(x) - g(x) \in (x)$ precisely when $f(0) = g(0)$.

Thus every coset contains a unique constant polynomial, and the quotient $\mathbb{Z}[x]/(x)$ has the same ring structure as $\mathbb{Z}$.

We shall establish this formally using a homomorphism below.

3.5.2 Ring Homomorphisms

A ring homomorphism is a mapping that preserves both of the operations defining a ring.

Definition 3.5.5 (Ring Homomorphism). Let R and S be rings. A mapping $\varphi : R \rightarrow S$ is called a *ring homomorphism* if, for all $a, b \in R$,

$$\varphi(a + b) = \varphi(a) + \varphi(b)$$

and

$$\varphi(ab) = \varphi(a)\varphi(b).$$

If R and S have multiplicative identities, the homomorphism is called *unital* if

$$\varphi(1_R) = 1_S.$$

A ring homomorphism need not be injective or surjective. A bijective ring homomorphism is called a *ring isomorphism*. If an isomorphism exists, we write $R \cong S$.

Throughout this section, homomorphisms between rings with identity are not automatically assumed to preserve the identity; this additional condition will be stated whenever needed.

Example 3.5.6 (Reduction Modulo n). Define $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ by $\varphi(a) = [a]$.

For $a, b \in \mathbb{Z}$,

$$\varphi(a + b) = [a + b] = [a] + [b]$$

and

$$\varphi(ab) = [ab] = [a][b].$$

Hence φ is a ring homomorphism. It is unital and surjective.

Example 3.5.7 (Evaluation Homomorphism). Let F be a field and let $a \in F$. Define $\text{ev}_a : F[x] \rightarrow F$ by

$$\text{ev}_a(f(x)) = f(a).$$

For any $f, g \in F[x]$,

$$(f + g)(a) = f(a) + g(a)$$

and

$$(fg)(a) = f(a)g(a).$$

Therefore ev_a is a ring homomorphism.

It is surjective because every element of F is the image of the corresponding constant polynomial. Moreover, $\text{ev}_a(1) = 1$, so it is unital.

Example 3.5.8 (A Map That Is Not a Homomorphism). Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(a) = a + 1$.

Then

$$f(a + b) = a + b + 1,$$

whereas

$$f(a) + f(b) = a + b + 2.$$

Thus f does not preserve addition and is not a ring homomorphism.

3.5.3 Elementary Properties of Ring Homomorphisms

The defining conditions of a ring homomorphism imply several useful identities.

Proposition 3.5.9. Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then, for every $a, b \in R$ and every positive integer n ,

- (i) $\varphi(0_R) = 0_S$;
- (ii) $\varphi(-a) = -\varphi(a)$;
- (iii) $\varphi(a - b) = \varphi(a) - \varphi(b)$;
- (iv) $\varphi(a^n) = \varphi(a)^n$.

Proof. Since $0_R + 0_R = 0_R$,

$$\varphi(0_R) = \varphi(0_R) + \varphi(0_R).$$

Additive cancellation in S gives $\varphi(0_R) = 0_S$.

Next,

$$\varphi(a) + \varphi(-a) = \varphi(a - a) = \varphi(0_R) = 0_S.$$

Therefore $\varphi(-a) = -\varphi(a)$, proving (ii).

Using (ii),

$$\varphi(a - b) = \varphi(a + (-b)) = \varphi(a) - \varphi(b),$$

which proves (iii).

Finally, repeated application of multiplicativity gives

$$\varphi(a^n) = \underbrace{\varphi(a) \cdots \varphi(a)}_{n \text{ factors}} = \varphi(a)^n.$$

■

Remark 3.5.10. If φ is unital, the power formula also holds for $n = 0$. Without the unital condition, one cannot generally assert that $\varphi(1_R) = 1_S$.

3.5.4 Kernel and Image

As in group theory, the kernel and image of a homomorphism provide fundamental information about the mapping.

Definition 3.5.11 (Kernel and Image). Let $\varphi : R \rightarrow S$ be a ring homomorphism. Its *kernel* and *image* are defined by

$$\ker \varphi = \{r \in R : \varphi(r) = 0_S\}$$

and

$$\text{Im } \varphi = \{\varphi(r) : r \in R\},$$

respectively.

The kernel consists of the elements mapped to zero, whereas the image consists of all elements actually attained by the homomorphism.

Theorem 3.5.12 (Kernel and Image Theorem). *Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then*

- (i) $\ker \varphi$ is a two-sided ideal of R ;
- (ii) $\text{Im } \varphi$ is a subring of S .

Proof. Put $K = \ker \varphi$.

Since $\varphi(0_R) = 0_S$, we have $0_R \in K$.

If $a, b \in K$, then

$$\varphi(a - b) = \varphi(a) - \varphi(b) = 0_S,$$

so $a - b \in K$.

Let $r \in R$ and $a \in K$. Then

$$\varphi(ra) = \varphi(r)\varphi(a) = 0_S$$

and

$$\varphi(ar) = \varphi(a)\varphi(r) = 0_S.$$

Hence $ra, ar \in K$, proving that K is a two-sided ideal of R .

Now let $T = \text{Im } \varphi$. Since $0_S = \varphi(0_R)$, the set T is nonempty.

If $x, y \in T$, write $x = \varphi(a)$ and $y = \varphi(b)$. Then

$$x - y = \varphi(a - b) \in T$$

and

$$xy = \varphi(ab) \in T.$$

By the subring test, T is a subring of S . ■

Theorem 3.5.13 (Injectivity Criterion). *A ring homomorphism $\varphi : R \rightarrow S$ is injective if and only if*

$$\ker \varphi = \{0_R\}.$$

Proof. Suppose φ is injective. If $a \in \ker \varphi$, then

$$\varphi(a) = 0_S = \varphi(0_R).$$

Injectivity gives $a = 0_R$. Thus $\ker \varphi = \{0_R\}$.

Conversely, suppose $\ker \varphi = \{0_R\}$. If $\varphi(a) = \varphi(b)$, then

$$\varphi(a - b) = 0_S.$$

Hence $a - b \in \ker \varphi$, so $a - b = 0_R$ and therefore $a = b$.

Thus φ is injective. ■

Example 3.5.14 (Kernel and Image of Reduction Modulo n). Consider $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$, defined by $\varphi(a) = [a]$.

An integer a belongs to the kernel precisely when $[a] = [0]$, that is, when $n \mid a$. Therefore

$$\ker \varphi = n\mathbb{Z}.$$

Every residue class has an integer representative, so $\text{Im } \varphi = \mathbb{Z}_n$. Thus the mapping is surjective but not injective.

Example 3.5.15 (Kernel of Evaluation). Let $\text{ev}_a : F[x] \rightarrow F$ be evaluation at $a \in F$.

Then

$$\ker(\text{ev}_a) = \{f(x) \in F[x] : f(a) = 0\}.$$

By the factor theorem, $f(a) = 0$ if and only if $(x - a) \mid f(x)$. Consequently,

$$\ker(\text{ev}_a) = (x - a).$$

3.5.5 The Natural Projection

Every quotient ring comes equipped with a particularly important homomorphism.

Theorem 3.5.16 (Natural Projection). Let I be a two-sided ideal of a ring R . The mapping $\pi : R \rightarrow R/I$, defined by

$$\pi(r) = r + I,$$

is a surjective ring homomorphism with

$$\ker \pi = I.$$

If R has identity, then π is unital.

Proof. For $a, b \in R$,

$$\pi(a + b) = (a + b) + I = (a + I) + (b + I)$$

and

$$\pi(ab) = ab + I = (a + I)(b + I).$$

Thus π is a ring homomorphism.

It is surjective because every element of R/I has the form $r + I = \pi(r)$ for some $r \in R$.

Furthermore,

$$\ker \pi = \{r \in R : r + I = I\} = I.$$

If R has identity, then $\pi(1_R) = 1_R + I$, the identity of R/I . Hence π is unital. ■

Corollary 3.5.17. A subset I of a ring R is a two-sided ideal if and only if it is the kernel of some ring homomorphism defined on R .

Proof. Every kernel is a two-sided ideal by the kernel and image theorem.

Conversely, if I is a two-sided ideal, the natural projection $\pi : R \rightarrow R/I$ is a ring homomorphism with kernel I . ■

3.5.6 The First Isomorphism Theorem

The First Isomorphism Theorem explains the precise relationship between the kernel and image of a ring homomorphism.

Theorem 3.5.18 (First Isomorphism Theorem for Rings). Let $\varphi : R \rightarrow S$ be a ring homomorphism.

Then

$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

Proof. Put $K = \ker \varphi$. Since K is a two-sided ideal of R , the quotient ring R/K exists.

Define

$$\bar{\varphi} : R/K \longrightarrow \operatorname{Im} \varphi$$

by

$$\bar{\varphi}(r + K) = \varphi(r).$$

We first show that $\bar{\varphi}$ is well-defined. Suppose $r + K = s + K$. Then $r - s \in K$, and hence

$$\varphi(r) - \varphi(s) = \varphi(r - s) = 0_S.$$

Therefore $\varphi(r) = \varphi(s)$.

For $r, s \in R$,

$$\begin{aligned} \bar{\varphi}((r + K) + (s + K)) &= \bar{\varphi}((r + s) + K) \\ &= \varphi(r + s) \\ &= \varphi(r) + \varphi(s), \end{aligned}$$

and

$$\begin{aligned} \bar{\varphi}((r + K)(s + K)) &= \bar{\varphi}(rs + K) \\ &= \varphi(rs) \\ &= \varphi(r)\varphi(s). \end{aligned}$$

Thus $\bar{\varphi}$ preserves addition and multiplication.

It is surjective onto $\operatorname{Im} \varphi$ by definition.

Finally, suppose $\bar{\varphi}(r + K) = 0_S$. Then $\varphi(r) = 0_S$, so $r \in K$. Consequently $r + K = K$, the zero element of R/K .

Thus the kernel of $\bar{\varphi}$ is the zero ideal, so $\bar{\varphi}$ is injective.

Therefore $\bar{\varphi}$ is a ring isomorphism, and

$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

■

Example 3.5.19 (A Quotient of $F[x]$).] Let F be a field and let $a \in F$. The evaluation homomorphism $\operatorname{ev}_a : F[x] \rightarrow F$ is surjective, and

$$\ker(\operatorname{ev}_a) = (x - a).$$

By the First Isomorphism Theorem,

$$F[x]/(x - a) \cong F.$$

In particular, taking $a = 0$ gives

$$F[x]/(x) \cong F.$$

Example 3.5.20 (A Quotient of $\mathbb{Z}[x]$).] Define $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}$ by $\varphi(f(x)) = f(0)$.

This is a surjective ring homomorphism. Its kernel consists precisely of polynomials with zero constant term, so

$$\ker \varphi = (x).$$

Hence

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

Example 3.5.21. Define $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_2$ by

$$\varphi(f(x)) = [f(0)].$$

The mapping evaluates at 0 and reduces the result modulo 2, so it is a surjective unital ring homomorphism.

Its kernel consists of polynomials whose constant terms are even. Every such polynomial can be written as

$$f(x) = 2m + xg(x)$$

for some $m \in \mathbb{Z}$ and $g(x) \in \mathbb{Z}[x]$. Therefore

$$\ker \varphi = (2, x).$$

By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}_2.$$

3.5.7 The Second Isomorphism Theorem

The Second Isomorphism Theorem relates a subring, an ideal, and their sum.

Theorem 3.5.22 (Second Isomorphism Theorem for Rings). *Let S be a subring of a ring R , and let I be a two-sided ideal of R . Then $S + I$ is a subring of R , $S \cap I$ is a two-sided ideal of S , and I is a two-sided ideal of $S + I$.*

Moreover,

$$S/(S \cap I) \cong (S + I)/I.$$

Proof. We first show that $S + I$ is a subring of R .

Let $s_1 + i_1, s_2 + i_2 \in S + I$. Their difference is

$$(s_1 + i_1) - (s_2 + i_2) = (s_1 - s_2) + (i_1 - i_2) \in S + I.$$

Also,

$$(s_1 + i_1)(s_2 + i_2) = s_1s_2 + s_1i_2 + i_1s_2 + i_1i_2.$$

Since $s_1s_2 \in S$ and the remaining terms belong to I , the product lies in $S + I$.

Thus $S + I$ is a subring.

The intersection $S \cap I$ is an additive subgroup of S . If $s \in S$ and $a \in S \cap I$, then sa, as belong to both S and I . Hence $S \cap I$ is an ideal of S .

Since $I \subseteq S + I$ and I is an ideal of R , it is also an ideal of $S + I$.

Now define

$$\varphi : S \longrightarrow (S + I)/I$$

by $\varphi(s) = s + I$.

This is a ring homomorphism. Every coset of I in $S + I$ has the form

$$(s + i) + I = s + I$$

for some $s \in S$ and $i \in I$. Hence φ is surjective.

Moreover,

$$\begin{aligned} \ker \varphi &= \{s \in S : s + I = I\} \\ &= \{s \in S : s \in I\} \\ &= S \cap I. \end{aligned}$$

By the First Isomorphism Theorem,

$$S/(S \cap I) \cong (S + I)/I.$$



Example 3.5.23. Let $R = \mathbb{Z}$, $S = 4\mathbb{Z}$, and $I = 6\mathbb{Z}$.

Then

$$S + I = 4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}$$

and

$$S \cap I = 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}.$$

The Second Isomorphism Theorem gives

$$4\mathbb{Z}/12\mathbb{Z} \cong 2\mathbb{Z}/6\mathbb{Z}.$$

Both quotient rings have three elements. The isomorphism sends

$$4k + 12\mathbb{Z} \mapsto 4k + 6\mathbb{Z}.$$

3.5.8 The Third Isomorphism Theorem

Successive quotient constructions can often be combined into a single quotient.

Theorem 3.5.24 (Third Isomorphism Theorem for Rings). *Let I and J be two-sided ideals of a ring R such that $I \subseteq J$. Then J/I is a two-sided ideal of R/I , and*

$$(R/I)/(J/I) \cong R/J.$$

Proof. Define

$$\psi : R/I \longrightarrow R/J$$

by

$$\psi(r + I) = r + J.$$

To show that ψ is well-defined, suppose $r + I = s + I$. Then $r - s \in I \subseteq J$, so $r + J = s + J$. For $r, s \in R$,

$$\psi((r + s) + I) = (r + s) + J = (r + J) + (s + J)$$

and

$$\psi(rs + I) = rs + J = (r + J)(s + J).$$

Thus ψ is a ring homomorphism.

It is surjective because every coset $r + J$ is the image of $r + I$.

Its kernel is

$$\begin{aligned} \ker \psi &= \{r + I : r + J = J\} \\ &= \{r + I : r \in J\} \\ &= J/I. \end{aligned}$$

Therefore J/I is an ideal of R/I . By the First Isomorphism Theorem,

$$(R/I)/(J/I) \cong R/J.$$

■

Example 3.5.25. Consider the ideals

$$12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}.$$

By the Third Isomorphism Theorem,

$$(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}.$$

Thus quotienting first by $12\mathbb{Z}$ and then by $4\mathbb{Z}/12\mathbb{Z}$ has the same effect, up to isomorphism, as quotienting directly by $4\mathbb{Z}$.

3.5.9 Correspondence of Ideals

The following theorem describes all ideals of a quotient ring in terms of ideals of the original ring.

Theorem 3.5.26 (Correspondence Theorem for Ideals). *Let I be a two-sided ideal of a ring R . There is a one-to-one correspondence between the two-sided ideals of R/I and the two-sided ideals J of R satisfying $I \subseteq J$.*

The correspondence is given by

$$J \longmapsto J/I,$$

with inverse

$$L \longmapsto \pi^{-1}(L),$$

where $\pi : R \rightarrow R/I$ is the natural projection.

Proof. Let J be an ideal of R containing I . Then J/I is an ideal of R/I by the Third Isomorphism Theorem.

Conversely, let L be an ideal of R/I . The inverse image

$$\pi^{-1}(L) = \{r \in R : r + I \in L\}$$

is an ideal of R . Moreover, since $\pi(i) = I$ for every $i \in I$ and every ideal contains zero, we have $I \subseteq \pi^{-1}(L)$.

These constructions are inverse to one another.

Indeed, if J contains I , then

$$\pi^{-1}(J/I) = J.$$

Also, since π is surjective, for every ideal L of R/I ,

$$\pi(\pi^{-1}(L)) = L.$$

Hence the stated correspondence is bijective. ■

Remark 3.5.27. The correspondence theorem preserves inclusions. Consequently, the ideals of a quotient ring can be studied by examining the ideals of the original ring that contain the ideal used in forming the quotient.

This theorem is especially useful in the study of prime and maximal ideals.

3.6 Prime Ideals

We now study a distinguished class of ideals in commutative ring theory. Prime ideals extend the fundamental divisibility property of prime numbers and are closely connected with integral domains.

Throughout this section, R denotes a commutative ring with multiplicative identity $1_R \neq 0$.

3.6.1 Definition and First Examples

Recall that a prime integer p has the property that if p divides a product ab , then p divides at least one of the factors.

Prime ideals generalize this property by replacing divisibility by membership in an ideal.

Definition 3.6.1 (Prime Ideal). A proper ideal P of R is called a *prime ideal* if, for all $a, b \in R$,

$$ab \in P \implies a \in P \text{ or } b \in P.$$

The requirement that P be proper is essential. The whole ring cannot be a prime ideal.

Example 3.6.2. In $\mathbb{Z}$, the ideal (5) is prime. Indeed, if $ab \in (5)$, then $5 \mid ab$. Since 5 is a prime integer, Euclid's lemma gives $5 \mid a$ or $5 \mid b$. Hence $a \in (5)$ or $b \in (5)$.

In contrast, (6) is not prime because

$$2 \cdot 3 = 6 \in (6),$$

although neither 2 nor 3 belongs to (6) .

Example 3.6.3. The zero ideal (0) of $\mathbb{Z}$ is prime. Indeed, if $ab \in (0)$, then $ab = 0$, which implies $a = 0$ or $b = 0$.

However, the zero ideal of $\mathbb{Z}_6$ is not prime, since $[2][3] = [0]$ while both factors are nonzero.

3.6.2 Prime Ideals and Integral Domains

The most important characterization of prime ideals is expressed in terms of quotient rings.

Theorem 3.6.4 (Quotient Characterization of Prime Ideals). *Let P be an ideal of a commutative ring R with identity $1_R \neq 0$. Then*

$$P \text{ is prime} \iff R/P \text{ is an integral domain.}$$

Proof. Suppose first that P is a prime ideal. Since P is proper, $1_R \notin P$, and hence

$$1_R + P \neq P.$$

Therefore R/P is a nonzero commutative ring with identity.

Suppose

$$(a + P)(b + P) = P.$$

By the definition of quotient multiplication, this means $ab + P = P$, or equivalently, $ab \in P$.

Since P is prime, $a \in P$ or $b \in P$. Hence

$$a + P = P \quad \text{or} \quad b + P = P.$$

Thus R/P has no zero divisors and is an integral domain.

Conversely, suppose R/P is an integral domain. Since an integral domain is nonzero, $P \neq R$, so P is proper.

Let $a, b \in R$ satisfy $ab \in P$. Then

$$(a + P)(b + P) = ab + P = P.$$

Since R/P is an integral domain, at least one of the factors must be zero. Hence

$$a + P = P \quad \text{or} \quad b + P = P.$$

Equivalently, $a \in P$ or $b \in P$.

Therefore P is a prime ideal. ■

Remark 3.6.5. The preceding theorem provides a powerful method for determining whether an ideal is prime. Rather than checking the defining condition directly for arbitrary products, one may identify the quotient ring and determine whether it is an integral domain.

Corollary 3.6.6. *The zero ideal (0) of R is prime if and only if R is an integral domain.*

Proof. Since $R/(0) \cong R$, the result follows immediately from the quotient characterization. ■

Example 3.6.7 (A Prime Ideal in $\mathbb{Z}[x]$).] Consider the ideal (x) in $\mathbb{Z}[x]$.

By evaluation at zero,

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

Since $\mathbb{Z}$ is an integral domain, (x) is a prime ideal of $\mathbb{Z}[x]$.

Example 3.6.8 (The Ideal (2) in $\mathbb{Z}[x]$).] Reducing the coefficients of a polynomial modulo 2 gives a surjective ring homomorphism

$$\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{F}_2[x].$$

Its kernel is (2) , the set of polynomials with even integer coefficients.

By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(2) \cong \mathbb{F}_2[x].$$

Since $\mathbb{F}_2[x]$ is an integral domain, the ideal (2) is prime in $\mathbb{Z}[x]$.

Example 3.6.9 (A Nonprime Ideal). Consider (x^2) in $F[x]$, where F is a field.

We have

$$x \cdot x = x^2 \in (x^2),$$

but $x \notin (x^2)$.

Therefore (x^2) is not prime. Equivalently, the nonzero coset $x + (x^2)$ has square zero in $F[x]/(x^2)$, so the quotient is not an integral domain.

3.6.3 The Ideal-Product Characterization

Prime ideals can also be characterized using products of ideals instead of products of individual elements.

Theorem 3.6.10 (Ideal-Product Criterion). *Let P be a proper ideal of a commutative ring R with identity. Then P is prime if and only if, for every pair of ideals I, J of R ,*

$$IJ \subseteq P \implies I \subseteq P \text{ or } J \subseteq P.$$

Proof. Suppose P is prime. Let I, J be ideals of R such that $IJ \subseteq P$.

Assume that neither I nor J is contained in P . Then there exist

$$a \in I \setminus P, \quad b \in J \setminus P.$$

Since $a \in I$ and $b \in J$, their product belongs to IJ , so

$$ab \in IJ \subseteq P.$$

Primality of P implies $a \in P$ or $b \in P$, contradicting the choice of a and b .

Hence $I \subseteq P$ or $J \subseteq P$.

Conversely, suppose the ideal-product condition holds.

Let $a, b \in R$ satisfy $ab \in P$. Consider the principal ideals (a) and (b) . Since R is commutative,

$$(a)(b) = (ab) \subseteq P.$$

By assumption,

$$(a) \subseteq P \quad \text{or} \quad (b) \subseteq P.$$

Since $a \in (a)$ and $b \in (b)$, we obtain $a \in P$ or $b \in P$.

Thus P is prime. ■

Corollary 3.6.11. *Let P be a prime ideal of R . If $I_1, \dots, I_n$ are ideals satisfying*

$$I_1 I_2 \cdots I_n \subseteq P,$$

then $I_j \subseteq P$ for at least one $j \in \{1, \dots, n\}$.

Proof. The result follows by induction on n . For $n = 2$, it is the preceding theorem.

For $n > 2$, apply the ideal-product criterion to $(I_1 \cdots I_{n-1})I_n \subseteq P$. Either $I_n \subseteq P$ or $I_1 \cdots I_{n-1} \subseteq P$. In the latter case, apply the induction hypothesis. ■

3.6.4 Prime Ideals of the Integers

The prime ideals of $\mathbb{Z}$ can be classified completely.

Theorem 3.6.12 (Prime Ideals of $\mathbb{Z}$). *The prime ideals of $\mathbb{Z}$ are precisely*

$$(0) \quad \text{and} \quad (p),$$

where p is a positive prime integer.

Proof. Since $\mathbb{Z}$ is a principal ideal domain, every ideal has the form (n) for a unique integer $n \geq 0$. The zero ideal (0) is prime because $\mathbb{Z}$ is an integral domain.

The ideal $(1) = \mathbb{Z}$ is not prime because it is not proper.

Now suppose $n \geq 2$. By the quotient characterization, (n) is prime if and only if $\mathbb{Z}/(n)$ is an integral domain.

But

$$\mathbb{Z}/(n) \cong \mathbb{Z}_n,$$

and $\mathbb{Z}_n$ is an integral domain precisely when n is prime.

Therefore the nonzero prime ideals of $\mathbb{Z}$ are exactly the ideals (p) , where p is prime. ■

Example 3.6.13. The ideals (2) , (3) , (5) , and (7) are prime ideals of $\mathbb{Z}$.

The ideals (4) , (6) , and (9) are not prime.

For example, $2 \cdot 2 = 4 \in (4)$, while $2 \notin (4)$.

3.6.5 Prime Ideals in Polynomial Rings

Polynomial rings provide further important examples of prime ideals.

Theorem 3.6.14. *Let F be a field and let $f(x) \in F[x]$ be a nonzero, nonunit polynomial. Then $(f(x))$ is a prime ideal of $F[x]$ if and only if $f(x)$ is irreducible over F .*

Proof. Suppose $f(x)$ is irreducible. Since $F[x]$ is a principal ideal domain, irreducible elements are prime elements.

Hence, whenever $f(x)$ divides $g(x)h(x)$, it divides $g(x)$ or $h(x)$.

Now suppose

$$g(x)h(x) \in (f(x)).$$

Then $f(x)$ divides $g(x)h(x)$, so $f(x)$ divides $g(x)$ or $h(x)$. Equivalently,

$$g(x) \in (f(x)) \quad \text{or} \quad h(x) \in (f(x)).$$

Since $f(x)$ is a nonunit, $(f(x))$ is proper, and therefore prime.

Conversely, suppose $(f(x))$ is prime. If $f(x)$ were reducible, we could write

$$f(x) = g(x)h(x),$$

where both $g(x)$ and $h(x)$ have positive degree strictly less than $\deg f$.

Then $g(x)h(x) \in (f(x))$, so primality gives $g(x) \in (f(x))$ or $h(x) \in (f(x))$.

But either inclusion would imply that $f(x)$ divides a polynomial of smaller positive degree, which is impossible.

Therefore $f(x)$ is irreducible. ■

Example 3.6.15. In $\mathbb{R}[x]$, the polynomial $x^2 + 1$ is irreducible because it has no real root and has degree two.

Therefore the ideal $(x^2 + 1)$ is prime in $\mathbb{R}[x]$.

In contrast, $x^2 - 1$ is reducible because

$$x^2 - 1 = (x - 1)(x + 1).$$

Hence $(x^2 - 1)$ is not a prime ideal.

Remark 3.6.16. The assumption that f be a nonzero nonunit is necessary. For example, the zero ideal of $F[x]$ is prime because $F[x]$ is an integral domain, even though the zero polynomial is not irreducible.

3.6.6 Preimages of Prime Ideals

Prime ideals behave particularly well under inverse images of ring homomorphisms.

Theorem 3.6.17 (Preimage of a Prime Ideal). *Let $\varphi : R \rightarrow S$ be a unital ring homomorphism between commutative rings with identities $1_R \neq 0$ and $1_S \neq 0$.*

If Q is a prime ideal of S , then

$$\varphi^{-1}(Q) = \{r \in R : \varphi(r) \in Q\}$$

is a prime ideal of R .

Proof. Put $P = \varphi^{-1}(Q)$.

Since $\varphi(0_R) = 0_S \in Q$, we have $0_R \in P$.

If $a, b \in P$, then $\varphi(a), \varphi(b) \in Q$. Since Q is an ideal,

$$\varphi(a - b) = \varphi(a) - \varphi(b) \in Q.$$

Thus $a - b \in P$.

If $r \in R$ and $a \in P$, then

$$\varphi(ra) = \varphi(r)\varphi(a) \in Q,$$

so $ra \in P$. Therefore P is an ideal.

Since φ is unital,

$$\varphi(1_R) = 1_S.$$

Because Q is proper, $1_S \notin Q$. Hence $1_R \notin P$, proving that P is proper. Finally, suppose $ab \in P$. Then

$$\varphi(a)\varphi(b) = \varphi(ab) \in Q.$$

Since Q is prime, $\varphi(a) \in Q$ or $\varphi(b) \in Q$. Therefore $a \in P$ or $b \in P$. Thus P is a prime ideal of R . ■

Example 3.6.18. Let $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}$ be evaluation at zero, defined by $\varphi(f(x)) = f(0)$.

The ideal (2) is prime in $\mathbb{Z}$. Its inverse image is

$$\varphi^{-1}((2)) = (2, x),$$

because a polynomial has even constant term precisely when it belongs to $(2, x)$.

Therefore $(2, x)$ is a prime ideal of $\mathbb{Z}[x]$.

In fact,

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}_2,$$

so the quotient characterization also establishes its primality.

Remark 3.6.19. The inverse image of a prime ideal under a unital homomorphism is always prime. By contrast, the image of a prime ideal under an arbitrary ring homomorphism need not be a prime ideal of the codomain.

For example, the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ maps the prime ideal (2) of $\mathbb{Z}$ to $2\mathbb{Z}$, which is not even an ideal of $\mathbb{Q}$.

3.6.7 Prime Ideals and Maximal Ideals

The theory of prime ideals is closely connected with that of maximal ideals, which will be studied in the next section.

The essential distinction is reflected in their quotient rings:

$$\begin{aligned} P \text{ prime} &\iff R/P \text{ is an integral domain,} \\ M \text{ maximal} &\iff R/M \text{ is a field.} \end{aligned}$$

Since every field is an integral domain, every maximal ideal is prime. The converse, however, does not hold in general.

For instance, (x) is prime in $\mathbb{Z}[x]$ because $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but it is not maximal because $\mathbb{Z}$ is not a field.

These observations will be established formally in the next section.

3.7 Maximal Ideals

Prime ideals are characterized by the property that their quotient rings are integral domains. We now study a stronger class of ideals, called *maximal ideals*, whose quotient rings are fields. These ideals occupy an important position in commutative algebra because they describe the largest proper ideals of a ring and provide a systematic way of constructing fields from more general rings.

Throughout this section, R denotes a commutative ring with identity $1_R \neq 0$.

Definition 3.7.1 (Maximal Ideal). A proper ideal M of R is called a *maximal ideal* if there is no ideal J of R satisfying $M \subsetneq J \subsetneq R$.

Thus, a maximal ideal is a proper ideal that cannot be enlarged without obtaining the entire ring. Equivalently, M is maximal if whenever J is an ideal satisfying $M \subseteq J \subseteq R$, then $J = M$ or $J = R$.

Example 3.7.2. The ideal (5) is maximal in $\mathbb{Z}$. Indeed, if $(5) \subseteq (n) \subseteq \mathbb{Z}$, then $n \mid 5$. Since 5 is prime, $n = \pm 1$ or $n = \pm 5$. Hence $(n) = \mathbb{Z}$ or $(n) = (5)$.

On the other hand, (6) is not maximal because $(6) \subsetneq (2) \subsetneq \mathbb{Z}$.

3.7.1 Characterization of Maximal Ideals

The following theorem provides the fundamental characterization of maximal ideals and is one of the most useful results in elementary commutative algebra.

Theorem 3.7.3 (Quotient Characterization of Maximal Ideals). *Let M be an ideal of R . Then M is maximal if and only if R/M is a field.*

Proof. Suppose first that M is maximal. Since M is proper, $1_R \notin M$, and hence R/M is a nonzero commutative ring with identity $1_R + M$.

Let $a + M$ be a nonzero element of R/M . Then $a \notin M$. Consider the ideal $M + (a)$. Since $a \in M + (a)$ but $a \notin M$, we have $M \subsetneq M + (a)$. By maximality, $M + (a) = R$.

In particular, $1_R \in M + (a)$, so there exist $m \in M$ and $r \in R$ such that

$$1_R = m + ra.$$

Passing to the quotient gives

$$1_R + M = (m + M) + (ra + M) = (r + M)(a + M).$$

Thus $r + M$ is a multiplicative inverse of $a + M$. Since every nonzero element of R/M is invertible, R/M is a field.

Conversely, suppose R/M is a field. Since a field is nonzero, M is a proper ideal of R .

Let J be an ideal satisfying $M \subseteq J \subseteq R$. By the ideal correspondence theorem, J/M is an ideal of R/M . Since the only ideals of a field are the zero ideal and the whole field, either $J/M = \{M\}$ or $J/M = R/M$.

In the first case, $J = M$, while in the second case, $J = R$. Therefore there is no ideal properly between M and R , proving that M is maximal. ■

Remark 3.7.4. The quotient characterization provides a convenient method for testing maximality. Rather than examining all ideals containing M , one may determine whether the quotient ring R/M is a field.

Corollary 3.7.5 (Every Maximal Ideal Is Prime). *Every maximal ideal of a commutative ring with identity is prime.*

Proof. Let M be a maximal ideal of R . By the preceding theorem, R/M is a field. Every field is an integral domain, so R/M is an integral domain. By the quotient characterization of prime ideals, M is prime. ■

Remark 3.7.6. The converse is false in general. For example, (0) is a prime ideal of $\mathbb{Z}$ because $\mathbb{Z}$ is an integral domain, but it is not maximal because $\mathbb{Z}$ is not a field. Similarly, (x) is prime in $\mathbb{Z}[x]$, since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but it is not maximal.

Thus every maximal ideal is prime, whereas a prime ideal need not be maximal.

Proposition 3.7.7. A proper ideal M of R is maximal if and only if, for every $a \in R \setminus M$, there exist $m \in M$ and $r \in R$ such that $1_R = m + ra$.

Proof. Suppose M is maximal and let $a \notin M$. Then $M + (a)$ is an ideal properly containing M , so $M + (a) = R$. Consequently, $1_R = m + ra$ for some $m \in M$ and $r \in R$.

Conversely, suppose the stated condition holds. Let J be an ideal satisfying $M \subsetneq J \subseteq R$. Choose $a \in J \setminus M$. By assumption, $1_R = m + ra$ for some $m \in M$ and $r \in R$. Since $m, a \in J$ and J is an ideal, $1_R \in J$. Hence $J = R$, proving that M is maximal. ■

3.7.2 Maximal Ideals of the Integers

The maximal ideals of $\mathbb{Z}$ admit a complete classification.

Theorem 3.7.8 (Maximal Ideals of $\mathbb{Z}$). *The maximal ideals of $\mathbb{Z}$ are precisely the ideals (p) , where p is a positive prime integer.*

Proof. Every ideal of $\mathbb{Z}$ has the form (n) for a unique integer $n \geq 0$.

The ideal $(1) = \mathbb{Z}$ is not proper and hence is not maximal. The zero ideal (0) is not maximal because $\mathbb{Z}/(0) \cong \mathbb{Z}$, which is not a field.

For $n \geq 2$, the quotient ring $\mathbb{Z}/(n)$ is isomorphic to $\mathbb{Z}_n$. By the quotient characterization, (n) is maximal if and only if $\mathbb{Z}_n$ is a field.

We have already established that $\mathbb{Z}_n$ is a field precisely when n is prime. Therefore the maximal ideals of $\mathbb{Z}$ are exactly the ideals (p) , where p is a positive prime integer. ■

Example 3.7.9. The ideals (2) , (3) , (5) and (7) are maximal in $\mathbb{Z}$, whereas (4) , (6) and (9) are not.

For example, (4) is properly contained in (2) , and (2) is properly contained in $\mathbb{Z}$. Therefore (4) cannot be maximal.

3.7.3 Maximal Ideals in Polynomial Rings

The relationship between maximal ideals and irreducible polynomials is especially important in polynomial rings over fields.

Theorem 3.7.10 (Prime and Maximal Ideals in $F[x]$). *Let F be a field and let $f(x) \in F[x]$ be a nonzero nonunit polynomial. Then the following conditions are equivalent:*

- (i) $f(x)$ is irreducible over F ;
- (ii) $(f(x))$ is a maximal ideal of $F[x]$;
- (iii) $(f(x))$ is a prime ideal of $F[x]$.

Proof. Suppose first that $f(x)$ is irreducible. Let J be an ideal of $F[x]$ satisfying $(f) \subseteq J \subseteq F[x]$. Since $F[x]$ is a principal ideal domain, there exists $g(x) \in F[x]$ such that $J = (g)$. The inclusion $(f) \subseteq (g)$ implies that g divides f .

Because f is irreducible, either g is a unit or g is an associate of f . In the first case, $J = F[x]$, while in the second case, $J = (f)$. Therefore (f) is maximal.

Every maximal ideal is prime, so (ii) implies (iii).

Finally, suppose (f) is prime and write $f(x) = g(x)h(x)$ for some $g, h \in F[x]$. Since $gh = f \in (f)$, primality gives $g \in (f)$ or $h \in (f)$.

If $g \in (f)$, then $g = fu$ for some $u \in F[x]$. Substituting into $f = gh$ gives $f = fuh$. Since $F[x]$ is an integral domain and $f \neq 0$, cancellation yields $uh = 1$. Hence h is a unit.

Similarly, if $h \in (f)$, then g is a unit. Thus every factorization of f has a unit factor, proving that f is irreducible. ■

Corollary 3.7.11. *Every nonzero prime ideal of $F[x]$ is maximal.*

Proof. Let P be a nonzero prime ideal of $F[x]$. Since $F[x]$ is a PID, $P = (f)$ for some nonzero polynomial f . Since P is proper, f is not a unit. By the preceding theorem, (f) is maximal. ■

Remark 3.7.12. The zero ideal of $F[x]$ is prime because $F[x]$ is an integral domain, but it is not maximal because $F[x]$ is not a field. Thus the nonzero hypothesis in the preceding corollary cannot be omitted.

Example 3.7.13 (A Maximal Ideal of $\mathbb{R}[x]$.] Consider the ideal $I = (x^2 + 1)$ of $\mathbb{R}[x]$. Since $x^2 + 1$ has no real root and has degree two, it is irreducible over $\mathbb{R}$. Therefore I is maximal and hence prime.

In fact, the quotient is isomorphic to the complex field:

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

To see this, define $\varphi : \mathbb{R}[x] \rightarrow \mathbb{C}$ by $\varphi(f(x)) = f(i)$. Every complex number $a + bi$ is the image of $a + bx$, so φ is surjective. By polynomial division, $\ker \varphi = (x^2 + 1)$. The First Isomorphism Theorem gives the result.

Example 3.7.14 (A Nonmaximal Polynomial Ideal). Consider $(x^2 - 1)$ in $\mathbb{R}[x]$. Since $x^2 - 1 = (x - 1)(x + 1)$, the polynomial is reducible. Therefore $(x^2 - 1)$ is neither prime nor maximal.

Indeed, neither $x - 1$ nor $x + 1$ belongs to $(x^2 - 1)$, although their product does.

3.7.4 Existence of Maximal Ideals

The preceding results identify maximal ideals in familiar rings. We now state a general existence theorem.

Theorem 3.7.15 (Existence of Maximal Ideals). *Let R be a nonzero commutative ring with identity. Then every proper ideal of R is contained in a maximal ideal. In particular, every such ring possesses at least one maximal ideal.*

Proof. Let I be a proper ideal of R , and consider the collection

$$\mathcal{S} = \{J : J \text{ is an ideal of } R, I \subseteq J \subsetneq R\},$$

ordered by inclusion.

The set $\mathcal{S}$ is nonempty because $I \in \mathcal{S}$.

Let $\mathcal{C}$ be a nonempty chain in $\mathcal{S}$ and put $K = \bigcup_{J \in \mathcal{C}} J$. Since the ideals in $\mathcal{C}$ are linearly ordered by inclusion, any two elements of K belong to a common ideal in the chain. Consequently, K is closed under subtraction and absorbs multiplication by elements of R , so K is an ideal.

Moreover, $I \subseteq K$. Since every $J \in \mathcal{C}$ is proper, no such J contains 1_R . Hence $1_R \notin K$, and therefore K is proper. Thus $K \in \mathcal{S}$ and is an upper bound for $\mathcal{C}$.

The empty chain also has an upper bound, namely I . Therefore every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ contains a maximal element M .

If $M \subsetneq J \subsetneq R$ for some ideal J , then $J \in \mathcal{S}$, contradicting the maximality of M in $\mathcal{S}$. Hence M is a maximal ideal of R containing I .

Taking $I = (0)$ proves that every nonzero commutative ring with identity has a maximal ideal. ■

Remark 3.7.16. The preceding theorem uses Zorn's lemma, which is equivalent to the axiom of choice. It is more advanced than the elementary results of this chapter but is included to emphasize that maximal ideals exist in complete generality for nonzero commutative rings with identity.

3.7.5 Maximal Ideals and Quotient Rings

Recall that if I is an ideal of R , the ideal correspondence theorem gives a bijection between ideals of R/I and ideals of R containing I .

This correspondence also preserves primality and maximality.

Proposition 3.7.17. Let $I \subseteq J$ be ideals of R . Then:

- (i) J is prime in R if and only if J/I is prime in R/I ;
- (ii) J is maximal in R if and only if J/I is maximal in R/I .

Proof. By the Third Isomorphism Theorem,

$$(R/I)/(J/I) \cong R/J.$$

Consequently, $(R/I)/(J/I)$ is an integral domain if and only if R/J is an integral domain. By the quotient characterization of prime ideals, this is equivalent to J/I being prime in R/I if and only if J is prime in R .

Similarly, $(R/I)/(J/I)$ is a field if and only if R/J is a field. By the quotient characterization of maximal ideals, this is equivalent to J/I being maximal in R/I if and only if J is maximal in R . ■

Example 3.7.18 (Prime and Maximal Ideals of $\mathbb{Z}_{12}$). We determine all prime and maximal ideals of $\mathbb{Z}_{12}$.

Since $\mathbb{Z}_{12} \cong \mathbb{Z}/12\mathbb{Z}$, the ideal correspondence theorem identifies its prime ideals with prime ideals of $\mathbb{Z}$ containing (12) .

The prime ideals of $\mathbb{Z}$ are (0) and (p) , where p is prime. The containment $(12) \subseteq (p)$ holds precisely when $p \mid 12$. Hence the only possibilities are $p = 2$ and $p = 3$.

Therefore the prime ideals of $\mathbb{Z}_{12}$ are

$$([2]) = \{[0], [2], [4], [6], [8], [10]\}$$

and

$$([3]) = \{[0], [3], [6], [9]\}.$$

Their quotient rings are

$$\mathbb{Z}_{12}/([2]) \cong \mathbb{Z}_2, \quad \mathbb{Z}_{12}/([3]) \cong \mathbb{Z}_3.$$

Since both quotient rings are fields, these ideals are maximal as well. Thus $\mathbb{Z}_{12}$ has exactly two prime ideals, and both are maximal.

Example 3.7.19 (A Maximal Ideal in Two Variables). Let F be a field and let $a, b \in F$. We show that the ideal $I = (x - a, y - b)$ is maximal in $F[x, y]$.

Define $\varphi : F[x, y] \rightarrow F$ by $\varphi(f(x, y)) = f(a, b)$. This is a surjective ring homomorphism because every element of F is the image of a constant polynomial.

Clearly $x - a$ and $y - b$ belong to $\ker \varphi$, so $I \subseteq \ker \varphi$.

Conversely, suppose $f(a, b) = 0$. Then

$$f(x, y) = (f(x, y) - f(a, y)) + (f(a, y) - f(a, b)).$$

Regarding $f(x, y)$ as a polynomial in x with coefficients in $F[y]$, the first difference is divisible by $x - a$. Similarly, the second difference is divisible by $y - b$. Hence $f(x, y) \in (x - a, y - b)$.

Thus $\ker \varphi = I$, and the First Isomorphism Theorem gives

$$F[x, y]/(x - a, y - b) \cong F.$$

Since the quotient is a field, $(x - a, y - b)$ is maximal.

3.8 Boolean Rings

We conclude our discussion of rings with a class of rings in which every element is idempotent. These rings arise naturally in set theory and mathematical logic.

Definition 3.8.1 (Boolean Ring). A ring R is called a *Boolean ring* if

$$a^2 = a$$

for every $a \in R$.

The definition does not require the ring to be commutative or to possess a multiplicative identity. Nevertheless, the Boolean property forces multiplication to be commutative and every element to be its own additive inverse.

Theorem 3.8.2 (Basic Properties of Boolean Rings). *Let R be a Boolean ring. Then:*

(i) $a + a = 0$ for every $a \in R$;

(ii) $ab = ba$ for all $a, b \in R$.

Consequently, every Boolean ring is commutative, and every nonzero Boolean ring has characteristic 2.

Proof. Let $a \in R$. Since R is Boolean, $a^2 = a$ and $(-a)^2 = -a$. On the other hand, the ring identities give $(-a)^2 = a^2 = a$. Therefore $-a = a$, and hence

$$a + a = 0.$$

This proves (i).

Now let $a, b \in R$. By the Boolean property,

$$(a + b)^2 = a + b.$$

Expanding the left side gives

$$a^2 + ab + ba + b^2 = a + b.$$

Since $a^2 = a$ and $b^2 = b$, additive cancellation yields

$$ab + ba = 0.$$

By (i), every element is its own additive inverse, so $ba = -ba$. Hence $ab = ba$, proving (ii).

Thus R is commutative. If R is nonzero, the equality $a + a = 0$ for every $a \in R$ shows that its additive group has exponent 2 and its characteristic is 2. ■

Example 3.8.3 (The Ring $\mathbb{Z}_2$). The ring $\mathbb{Z}_2 = \{[0], [1]\}$ is Boolean, because

$$[0]^2 = [0], \quad [1]^2 = [1].$$

Example 3.8.4 (The Power Set of a Set). Let X be a set and let $\mathcal{P}(X)$ denote its power set. Define addition by symmetric difference and multiplication by intersection:

$$A + B = A \triangle B, \quad AB = A \cap B.$$

Then $\mathcal{P}(X)$ is a commutative ring. Its additive identity is the empty set $\emptyset$, and every element is its own additive inverse because $A \triangle A = \emptyset$. Multiplication is associative and distributive over symmetric difference.

Moreover, $A^2 = A \cap A = A$ for every $A \subseteq X$. Hence $\mathcal{P}(X)$ is a Boolean ring. If X is nonempty, its multiplicative identity is X .

Proposition 3.8.5. A Boolean ring has no nonzero nilpotent elements.

Proof. Let R be a Boolean ring and suppose $a^n = 0$ for some positive integer n . Since $a^2 = a$, induction gives $a^n = a$ for every positive integer n . Therefore $a = 0$. ■

Proposition 3.8.6. Every Boolean integral domain is isomorphic to $\mathbb{Z}_2$.

Proof. Let R be a Boolean integral domain. For every $a \in R$, the Boolean property gives $a^2 = a$, so

$$a(a - 1_R) = 0.$$

Since R is an integral domain, either $a = 0$ or $a = 1_R$.

Thus R consists of exactly two elements, 0 and 1_R . The map $\mathbb{Z}_2 \rightarrow R$ defined by $[0] \mapsto 0$ and $[1] \mapsto 1_R$ is therefore a ring isomorphism. ■

Corollary 3.8.7. Every field that is a Boolean ring is isomorphic to $\mathbb{Z}_2$.

Proof. Every field is an integral domain, so the result follows immediately from the preceding proposition. ■

3.9 Summary and Guided Review

The principal concepts developed in this chapter are rings, integral domains, fields, subrings, ideals, quotient rings, ring homomorphisms, and principal, prime, and maximal ideals. The following table summarizes the most important characterizations for commutative rings with identity $1_R \neq 0$.

Concept	Characterization
Integral domain	A nonzero commutative ring with identity and no nonzero zero divisors.
Field	A nonzero commutative ring with identity in which every nonzero element is invertible.
Principal ideal (a)	$(a) = \{ra : r \in R\}$, the smallest ideal containing a .
Principal ideal domain	An integral domain in which every ideal is principal.
Prime ideal P	$P \neq R$ and $ab \in P$ implies $a \in P$ or $b \in P$; equivalently, R/P is an integral domain.
Maximal ideal M	$M \neq R$ and there is no ideal strictly between M and R ; equivalently, R/M is a field.
Boolean ring	A ring satisfying $a^2 = a$ for every element a .

The relationship between prime and maximal ideals is particularly important. Every maximal ideal is prime, but a prime ideal need not be maximal. In $\mathbb{Z}$ and $F[x]$, every nonzero prime ideal is maximal, whereas in $\mathbb{Z}[x]$ this need not be the case.

When examining a given ideal, it is usually advantageous to identify the corresponding quotient ring. For example, the primality of (x) in $\mathbb{Z}[x]$ follows from $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, while the maximality of $(x^2 + 1)$ in $\mathbb{R}[x]$ follows from $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$. These examples illustrate how quotient rings transform questions about ideals into questions about integral domains and fields.

3.10 Exercises

Basic concepts and computations.

1. Verify directly that $3\mathbb{Z}$ is a subring and an ideal of $\mathbb{Z}$.
2. Find all units and all nonzero zero divisors in $\mathbb{Z}_8$ and $\mathbb{Z}_{15}$.
3. Prove that a unit cannot be a zero divisor in a nonzero commutative ring.
4. Show that an integral domain has no nonzero nilpotent elements.
5. Determine $(24) + (36)$, $(24) \cap (36)$ and $(24)(36)$ in $\mathbb{Z}$.
6. Prove that $(a) = (b)$ in an integral domain if and only if $a = ub$ for some unit u (allow $a = b = 0$).
7. Show that $(x^2 - 4, x - 2) = (x - 2)$ in $\mathbb{Q}[x]$.
8. Use the division algorithm to prove $(x^2 + 1, x + 1) = (1)$ in $\mathbb{Q}[x]$.
9. Describe the elements of $\mathbb{Z}[x]/(2, x)$ and identify this quotient ring.
10. Decide whether (4) is a prime ideal of $\mathbb{Z}$; justify with a product outside the ideal.
11. Decide whether (0) is prime and/or maximal in $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{Z}_6$.
12. List all ideals of $\mathbb{Z}_{18}$. Which are prime? Which are maximal?

Proof-based exercises.

13. Prove that the intersection of an arbitrary family of ideals is an ideal.
14. Prove that the sum of two ideals is the smallest ideal containing both.
15. Prove that $IJ \subseteq I \cap J$, and give an example where the inclusion is strict.
16. Let P be prime. Prove that if $a_1 \cdots a_n \in P$, then some $a_i \in P$.
17. Prove that every prime ideal of a finite commutative ring with identity is maximal.
18. Show that the union of two ideals is an ideal if and only if one is contained in the other.
19. Show that (x) is prime but not maximal in $\mathbb{Z}[x]$.
20. Prove that $(x^2 + 2)$ is maximal in $\mathbb{Q}[x]$.
21. Find the kernel and image of $\mathbb{Z}[x] \rightarrow \mathbb{Z}_5, f \mapsto [f(2)]_5$.
22. Show that $\mathbb{Z}[x]/(x^2 + 1)$ is not a field, although $\mathbb{R}[x]/(x^2 + 1)$ is.
23. Prove that if $I + J = R$, then $I \cap J = IJ$ for commutative R .
24. For $R = \mathbb{Z} \times \mathbb{Z}$, prove that $\mathbb{Z} \times \{0\}$ is prime but not maximal.
25. Explain why the statement “the preimage of every maximal ideal is maximal” requires extra hypotheses, and find a counterexample using an inclusion of rings.

Chapter 4

Tutorial

This tutorial reviews selected concepts from group theory and ring theory through examples and applications. The emphasis is on recognizing algebraic structures, identifying their defining properties, and applying standard results to concrete situations.

Semigroups and Monoids

Definition 4.0.1. A *semigroup* is a nonempty set S equipped with an associative binary operation. Thus, for all $a, b, c \in S$, one has $(ab)c = a(bc)$. A *monoid* is a semigroup possessing an identity element.

Every monoid is a semigroup, but the converse is not necessarily true. The distinction is illustrated by the following examples.

- Example 4.0.2.**
- (i) The set $\mathbb{Z}^+$ of positive integers is a semigroup under addition because addition is associative and closed on $\mathbb{Z}^+$. It is not a monoid because its additive identity 0 does not belong to $\mathbb{Z}^+$.
 - (ii) The set $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ is a monoid under addition, with identity 0. It is not a group because positive integers have no additive inverses in $\mathbb{N}_0$.
 - (iii) The set $(\mathbb{Z}, \cdot)$ is a monoid under ordinary multiplication, with identity 1. It is not a group because, for example, 2 has no multiplicative inverse in $\mathbb{Z}$.
 - (iv) The set $M_n(\mathbb{R})$ of real $n \times n$ matrices is a monoid under matrix multiplication. Its identity is I_n . It is not a group because singular matrices have no multiplicative inverses.
 - (v) Let Σ be a nonempty alphabet. The set Σ^+ of all nonempty finite strings over Σ is a semigroup under concatenation. The set Σ^* of all finite strings, including the empty string ε , is a monoid under concatenation, with identity ε .

Remark 4.0.3. The existence of an identity does not guarantee the existence of inverses. This is precisely what distinguishes a monoid from a group.

Abelian and Nonabelian Groups

Recall that a group G is *abelian* if $ab = ba$ for all $a, b \in G$. A group that does not satisfy this condition is called *nonabelian*.

Example 4.0.4. The following are examples of abelian groups:

- (i) $(\mathbb{Z}, +)$, the additive group of integers;
- (ii) $(\mathbb{Z}_n, +)$, the additive group of residue classes modulo n ;
- (iii) $(\mathbb{Q}^\times, \cdot)$, the multiplicative group of nonzero rational numbers;
- (iv) $\mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group under componentwise addition.

In each case, the group operation is commutative.

Example 4.0.5. The symmetric group S_3 is nonabelian. Indeed, under right-to-left composition,

$$(12)(13) = (132), \quad (13)(12) = (123).$$

Since these products are different, multiplication in S_3 is not commutative.

Example 4.0.6. The group $GL_2(\mathbb{R})$ is nonabelian. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both matrices are invertible, but

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus $AB \neq BA$, proving that $GL_2(\mathbb{R})$ is nonabelian.

Example 4.0.7. The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ is a nonabelian group of order 8. Its multiplication satisfies $i^2 = j^2 = k^2 = -1$ and $ij = k$, whereas $ji = -k$. Hence the group operation is not commutative.

Cyclic and Noncyclic Groups

Definition 4.0.8. A group G is *cyclic* if there exists $a \in G$ such that $G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}$. The element a is called a generator of G .

- Example 4.0.9.**
- (i) The group $(\mathbb{Z}, +)$ is infinite cyclic, with generators 1 and -1 .
 - (ii) The group $(\mathbb{Z}_n, +)$ is cyclic of order n , generated by $[1]$.
 - (iii) The set of n th roots of unity, $\mu_n = \{z \in \mathbb{C} : z^n = 1\}$, forms a cyclic group under multiplication. It is generated by $e^{2\pi i/n}$.
 - (iv) The multiplicative group $U(10) = \{[1], [3], [7], [9]\}$ is cyclic, since

$$[3]^1 = [3], \quad [3]^2 = [9], \quad [3]^3 = [7], \quad [3]^4 = [1].$$

Hence $U(10) = \langle [3] \rangle$.

Example 4.0.10. The following groups are not cyclic:

- (i) The Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic because every nonidentity element has order 2, whereas the group has order 4.
- (ii) The symmetric group S_3 is not cyclic because no element has order 6. Its elements have orders 1, 2, or 3.
- (iii) The additive group $(\mathbb{Q}, +)$ is not cyclic. Indeed, if $\mathbb{Q} = \langle q \rangle$ for some $q \in \mathbb{Q}$, then every rational number would be an integer multiple of q . In particular, both q and $q/2$ would belong to $\langle q \rangle$, which is impossible for $q \neq 0$.

Determining Generators of a Cyclic Group

Recall that if $G = \langle a \rangle$ has finite order n , then a^k generates G if and only if $\gcd(k, n) = 1$. Consequently, the number of generators is $\varphi(n)$, where φ is Euler's totient function.

Example 4.0.11. Determine all generators of $(\mathbb{Z}_{12}, +)$.

An element $[k] \in \mathbb{Z}_{12}$ is a generator precisely when $\gcd(k, 12) = 1$. Among the integers $0, 1, \dots, 11$, the integers relatively prime to 12 are 1, 5, 7, 11.

Therefore the generators are $[1], [5], [7], [11]$, and their number is $\varphi(12) = 4$.

Example 4.0.12. Determine the order of $[8]$ in $\mathbb{Z}_{12}$.

Using the order formula for an element of an additive cyclic group,

$$\text{ord}([8]) = \frac{12}{\gcd(12, 8)} = \frac{12}{4} = 3.$$

Indeed, $\langle [8] \rangle = \{[0], [8], [4]\}$, which contains three elements.

Structure Theorem for Cyclic Groups

Theorem 4.0.13 (Structure Theorem for Cyclic Groups). *Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$, and every finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.*

This theorem completely classifies cyclic groups up to isomorphism. In particular, any two cyclic groups of the same finite order are isomorphic, even when their elements and operations appear different.

Even and Odd Permutations

A *transposition* is a permutation that interchanges two symbols and fixes all others. Every permutation can be expressed as a product of transpositions, although the representation is not unique.

Theorem 4.0.14. *Every permutation in S_n can be expressed as a product of transpositions. Moreover, the parity of the number of transpositions is independent of the representation.*

Definition 4.0.15. A permutation is called *even* if it can be expressed as a product of an even number of transpositions, and *odd* if it can be expressed as a product of an odd number of transpositions.

The identity permutation is even because it can be expressed as a product of zero transpositions.

Example 4.0.16. Consider $\sigma = (132) \in S_3$. Since

$$(132) = (12)(13),$$

the permutation σ is even.

On the other hand, (1234) is odd because

$$(1234) = (14)(13)(12),$$

which is a product of three transpositions.

Proposition 4.0.17. A cycle of length r is even if r is odd, and odd if r is even.

Proof. Every r -cycle can be expressed as

$$(a_1 a_2 \cdots a_r) = (a_1 a_r)(a_1 a_{r-1}) \cdots (a_1 a_2).$$

The right side contains $r - 1$ transpositions. Therefore the cycle is even precisely when $r - 1$ is even, which is equivalent to r being odd. ■

Example 4.0.18. Determine the parity of $\sigma = (123)(45)(6789)$.

The permutation is a product of disjoint cycles of lengths 3, 2, and 4. These cycles require 2, 1, and 3 transpositions, respectively. The total number is $2 + 1 + 3 = 6$, which is even. Hence σ is an even permutation.

Remark 4.0.19. The set of all even permutations in S_n forms the alternating group A_n . For $n \geq 2$, it is a normal subgroup of index two, so $|A_n| = n!/2$.

Commutative and Noncommutative Rings

A ring R is *commutative* if $ab = ba$ for every $a, b \in R$. Otherwise, it is called *noncommutative*.

Example 4.0.20. The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_n$, and $\mathbb{R}[x]$ are commutative under their usual operations. The polynomial ring $\mathbb{R}[x, y]$ is also commutative because the indeterminates commute.

Example 4.0.21. The matrix ring $M_2(\mathbb{R})$ is noncommutative. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$AB = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Since $AB \neq BA$, the ring $M_2(\mathbb{R})$ is noncommutative.

Example 4.0.22. The ring of Hamilton's quaternions $\mathbb{H}$ is noncommutative. Its elements include i, j, k , satisfying $i^2 = j^2 = k^2 = -1$ and $ij = k$, whereas $ji = -k$. Thus $ij \neq ji$.

Left, Right and Two-Sided Ideals

In noncommutative rings, it is necessary to distinguish between left ideals, right ideals, and two-sided ideals. The distinction arises because multiplication on the left and on the right need not give the same result.

Definition 4.0.23. Let R be a ring and let I be an additive subgroup of $(R, +)$.

- (i) I is a *left ideal* if $ra \in I$ for all $r \in R$ and $a \in I$.
- (ii) I is a *right ideal* if $ar \in I$ for all $r \in R$ and $a \in I$.
- (iii) I is a *two-sided ideal* if it is both a left ideal and a right ideal.

In a commutative ring, every left ideal is also a right ideal, and hence every one-sided ideal is two-sided.

Example 4.0.24. Let $R = M_2(\mathbb{R})$ and consider

$$I = \left\{ \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

The set I is an additive subgroup of R . We show that it is a left ideal but not a right ideal.

Let

$$A = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \in R, \quad B = \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} \in I.$$

Then

$$AB = \begin{pmatrix} ra + sb & 0 \\ ta + ub & 0 \end{pmatrix} \in I.$$

Thus I is a left ideal.

However, choosing

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in I, \quad C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in R,$$

we obtain

$$BC = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \notin I.$$

Therefore I is not a right ideal, and hence is not a two-sided ideal.

Relation Between Subrings and Ideals

Every two-sided ideal is a subring, but every subring need not be an ideal. A subring is closed under subtraction and multiplication of its own elements, whereas an ideal must also absorb multiplication by arbitrary elements of the ambient ring.

Example 4.0.25. The ring $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it is not an ideal of $\mathbb{Q}$. Indeed, $1 \in \mathbb{Z}$ and $\frac{1}{2} \in \mathbb{Q}$, whereas $\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}$. Therefore the absorption property fails.

Example 4.0.26. The subset $2\mathbb{Z}$ is both a subring and an ideal of $\mathbb{Z}$. Indeed, the difference of two even integers is even, and the product of an even integer with any integer is again even.

Sums, Intersections and Products of Ideals

Let I and J be two-sided ideals of a ring R . Their sum, intersection, and product are again ideals, but these operations generally produce different subsets.

Definition 4.0.27. The *sum* of I and J is

$$I + J = \{i + j : i \in I, j \in J\}.$$

It is the smallest ideal of R containing both I and J .

Definition 4.0.28. The *product* of I and J is

$$IJ = \left\{ \sum_{k=1}^m i_k j_k : m \geq 1, i_k \in I, j_k \in J \right\}.$$

It consists of all finite sums of products of elements of I and J .

Recall that $IJ \subseteq I \cap J$. In general, equality need not hold.

Example 4.0.29. Let $I = 4\mathbb{Z}$ and $J = 6\mathbb{Z}$ be ideals of $\mathbb{Z}$. We determine their sum, intersection, and product.

Since $\gcd(4, 6) = 2$, the ideal sum is

$$I + J = 4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}.$$

Since $\text{lcm}(4, 6) = 12$, their intersection is

$$I \cap J = 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}.$$

Finally, the product ideal is

$$IJ = (4\mathbb{Z})(6\mathbb{Z}) = 24\mathbb{Z}.$$

Thus

$$I + J = 2\mathbb{Z}, \quad I \cap J = 12\mathbb{Z}, \quad IJ = 24\mathbb{Z}.$$

In particular, $IJ \subsetneq I \cap J$.

Example 4.0.30. Let $I = 6\mathbb{Z}$ and $J = 15\mathbb{Z}$. Determine $I + J$, $I \cap J$, and IJ .

Since $\gcd(6, 15) = 3$, $\text{lcm}(6, 15) = 30$, and $6 \cdot 15 = 90$, we obtain

$$I + J = 3\mathbb{Z}, \quad I \cap J = 30\mathbb{Z}, \quad IJ = 90\mathbb{Z}.$$

Remark 4.0.31. For principal ideals of $\mathbb{Z}$, the sum is generated by the greatest common divisor, the intersection by the least common multiple, and the product by the product of the generators. These relationships are particularly useful for computations involving ideals of the integers.